Physics-Informed Neural Quantum States for Quantum Dots: Tangent-Space Geometry, Message Passing, and Wigner Molecules

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Abstract

A neural network used as a variational wavefunction is judged by its energy. This thesis asks what the energy cannot see: two networks landing on the same number need not be the same object, a preconditioner decisive in one training regime does nothing in another, and an objective that reproduces a Markov chain's answer in one corner of the phase diagram departs from it in the next. The system is the two-dimensional harmonic quantum dot, from the kinetic-energy-dominated droplet to the dilute Wigner molecule, for $N \in \{2, 6, 12, 20\}$ electrons and confinements ω from 1 to 10^{-3}.

The ansatz is a Slater determinant dressed by a neural Jastrow correlator, which reshapes the amplitude, and a coordinate backflow, which reshapes the nodes. Each exists in a separable form and a message-passing one. Training is by residual pretraining followed by stochastic reconfiguration, and separately by importance-sampled collocation that places no Markov chain in the gradient. The trained states are read through the geometry of their log-derivatives (the quantum geometric tensor and the neural tangent kernel) and through structural diagnostics of the electron arrangement. The energies agree closely with every available published fixed-node diffusion-Monte-Carlo reference, to 0.0008–0.04%—an agreement, not an independent validation, since the same references guide pretraining. Below $\omega = 0.1$ for $N \geq 6$ no reference exists, and those energies are predictions.

Message passing changes the variational geometry before it changes the energy. At strong confinement, and under the parametrisations used here, a message-passing correlator reaches the same reference-quality solution with its Fisher weight concentrated in about half as many effective directions as a separable network of equal or greater capacity; at every confinement far more of that weight lies inside a compact dictionary of collective operators, and its local-energy variance is lower. In the nodal channel, toward the Wigner regime, a conventional backflow collapses spectrally to rank one and loses about half a percent in energy while a larger message-passing field retains its rank; the two are not capacity-matched, so this is an observation and not yet an ablation. Holding the backflow fixed and swapping only the correlator moves the collapse boundary, which points to one relational bottleneck rather than two deficits.

Conditioning alone does not decide whether natural-gradient preconditioning earns its cost: under $|\Psi|^2$ sampling the empirical metric is badly conditioned on exactly the models where stochastic reconfiguration and Adam agree to four decimals. We propose, as a hypothesis the observed failures support but do not establish, that what matters is whether the metric is still *estimable* from a batch.

The trained states then serve as instruments, and the crossover they resolve is *staged*: the cloud expands and its shells stiffen radially well before the electrons acquire angular order within them. A polygon shell model fitted on held-out frames reproduces the measured pair distribution better than an angle-randomised null, by a margin that grows as the trap weakens. At $\omega = 10^{-3}$ the twelve-electron dot forms a two-shell molecule with a modal $(3, 9)$ split and the twenty-electron dot a three-shell $(1, 7, 12)$ molecule, which by $\omega = 10^{-2}$ the radial classifier reads as one shell about a central electron, with no angular order.

Those measurements return to the solver. A generic collocation proposal loses the Wigner state entirely; a proposal rebuilt from the measured shell geometry lifts the effective sample size by 15–32× with no training at all and carries training with no Markov chain in the gradient to $\omega = 0.0035$, where a different limit takes over—a transfer instability in the Slater–Jastrow cancellation that no change of proposal repairs.

Two limitations qualify these results. Every state is the lowest found within a fixed spin-projection sector and total spin is not scanned, which matters most in the Wigner regime, where spin transitions are plausible. The architecture comparison is seeded at N=6, rests on fewer models at N=12 and on a single production state at N=20, and the two backflow variants are not capacity-matched.

To go wrong in one's own way is better than to go right in someone else's.

Fyodor Dostoevsky
Crime and Punishment

Preface

How easy it is
to look out across the sea
while you rest on the white sand.

Perhaps you never noticed,
but you swam so swiftly
that no one saw you leave.

The waves carried you
while you simply lay there.
You reached for air
as your vision faded.

You gasp to call out
and that was when your chest burned.

When I came to UiO, Morten read my background and said: "You know PINNs, you know many-body quantum systems. I want you to do PINNs on quantum dots." I said "sure." I knew enough about each of those things separately to imagine that putting them together would mostly be a technical problem.

It was not.

The difficult part was rarely producing a result. The difficult part was knowing what a result meant. When something stopped improving, there were always several possible explanations. Perhaps the model was wrong for the problem, perhaps the optimisation was failing, perhaps the physics itself had become more difficult, or perhaps I had simply made a mistake somewhere along the way. Much of this thesis grew out of trying to tell those possibilities apart.

The trouble with having so many explanations was that, as one after another failed to account for what I was seeing, a much simpler one began to acquire weight: perhaps it simply could not be done. At first it was only another possibility. Gradually, it became difficult to see past it.

For a while, I stopped expecting the project to succeed.

I kept working, but for a different reason. If it could not be done, I wanted at least to understand why. In some ways, that became a more difficult problem than the original one. The worst periods were not those in which the method failed, but those in which I knew that it had failed and could not say what the failure meant.

I was fortunate to spend part of that period in Rome. The city gave the days something larger than the problem I happened to be working on, but it was Elida, more than anything, who gave me a steadiness I could not have supplied for myself. Her patience, warmth, and faith in me carried me through more of this work than I know how to put into words.

Eventually, some of the mystery began to give way. Not every obstacle turned out to be profound. Some were almost embarrassingly ordinary. Things I had taken for limitations of the method occasionally turned out to be limitations of my own implementation; other failures, once understood properly, pointed toward more interesting questions than the ones I had originally been asking.

That changed the way I looked at the project. A disappointing result was no longer merely something to overcome. Sometimes it contained information about the model, sometimes about the physics, and sometimes about the question itself. What began as an attempt to obtain a sufficiently good answer gradually became an attempt to understand what the answers, and the failures, were telling me.

I owe a warm and sincere thanks to my family, who have always made sure there was a home for me

to return to, whenever I needed one. And to my friends, for countless conversations in which we tried to work out what we actually understood, and usually discovered that it was less than we thought.

To Morten, again, and properly: thank you for the assignment, for the two years we have had together, and above all for a genuine and undisguised interest in what I was finding at every stage, including the stages where what I found was that I had been wrong. That interest is most of what encouragement actually consists of.

When I look back, it is easy to make a catalogue of what I would now do differently. Most of that knowledge, however, was purchased by the failures and could not have been read off in advance. In a sense, the regret and the result are the same object seen from two sides.

I began this project thinking that the difficult part would be finding an answer to the many-body problem. More often, it was discovering which question a disappointing result was actually answering.

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Part I

Introduction

The world was to me a secret which I desired to divine.

Mary Shelley
Frankenstein

Introduction

A system of many interacting electrons is described completely by its wavefunction. The same object puts the system almost out of reach, because a complete description of N interacting particles is a function of all their coordinates at once, and the cost of representing such a function grows exponentially with every particle added. The equation that fixes this wavefunction is known exactly; the entire difficulty is solving it.

The competition that makes it hard is physical. Kinetic energy tends to delocalise the particles; interactions and external potentials drive localisation and correlation; and the balance between them plays out in a configuration space far too large to represent explicitly. Yet the quantities we need (energies, correlation functions, phase boundaries) are precisely the ones that depend on getting that balance right. This is the *many-body problem*.

Established methods pay for accuracy in one of two currencies: cost, or a prior decision about which correlations may be described. Full configuration interaction pays in cost. It is exact within a finite basis, but its cost grows combinatorially with that basis, so a modest two-dimensional dot of twenty electrons is already far beyond reach. Hartree–Fock and coupled-cluster theory pay in the second currency: the first keeps only an average field, and the second truncates the excitations it includes, which serves well near equilibrium and less well where correlation is strong or states are nearly degenerate. Quantum Monte Carlo represents the wavefunction through stochastic sampling and provides accurate benchmarks for continuum systems, but in its fixed-node form its accuracy is bounded by the nodes of a trial wavefunction, and good trial wavefunctions become harder to construct as correlation grows. In practice, then, much of the many-body problem reduces to a narrower one: choosing an expressive, well-conditioned ansatz and optimising it reliably.

Quantum dots as a controlled testbed. Two-dimensional parabolic quantum dots are a clean and tunable place to study that narrower problem. As the trap frequency ω decreases, the single-particle level spacing collapses while Coulomb repulsion becomes dominant, and the electrons localise into rings and shells: one system spans the route from Fermi-liquid-like behaviour to Wigner-molecule physics. In the lab frame, rotational symmetry and quantum fluctuations smear these patterns into smooth densities; in a co-rotating frame they recover crystalline structures whose topology depends on particle number. For N=6 and N=12, classical and semiclassical studies have identified preferred shellings such as $(1, 5)$ and $(3, 9)$ and mapped parts of the crossover [1–4].

Quantum dots suit this thesis for two reasons. They are genuinely non-trivial: Coulomb singularities, strong correlation and a real crossover appear already at moderate N. And they are simple enough that accurate DMC references exist in many regimes and that detailed structural diagnostics are computationally feasible. For fully quantum dots, quantitative topology-resolved diagnostics and a systematic picture of the Wigner regime remain relatively sparse, and it is not well understood how modern neural ansätze represent these states internally.

Neural wavefunctions. Machine learning has become a standard tool for representing complex, high-dimensional functions, and in physics it has taken two broad forms. Neural networks act as *surrogates*, learning maps from parameters to observables or approximating the solution operators of partial differential equations, as physics-informed neural networks and operator learning do. Or they act as *variational ansätze* for quantum states, where the network outputs the amplitude of a many-body configuration and is trained by variational Monte Carlo. This thesis began in the first tradition and ended in the second:

the physics-informed residual objective it started from survives as the pretraining stage of a variational method, and as the collocation route of Chapter 7, rather than as the method itself.

Representing a quantum state by a neural network and optimising it variationally was introduced by Carleo and Troyer [5]. For continuum fermions the idea has since matured into highly accurate *ab initio* solvers such as FermiNet [6] and PauliNet [7], and into explicit neural *backflow* transformations [8], the direct ancestor of the coordinate backflow used here. Message passing has recently been built into neural quantum states for extended fermion systems [9, 10]. Against that backdrop the present work is deliberately narrow. It does not push a new architecture to ever larger systems. It asks why particular architectural and training choices succeed or fail on a problem where the answer can be checked, and what the trained state can then say about the physics. Accuracy is the precondition for that, not its point.

Universal approximation establishes representational capacity, not trainability. The theorems guarantee that a sufficiently large network can *represent* any suitably regular function to arbitrary accuracy; they say nothing about whether gradient-based optimisation will *find* that representation, nor at what sample cost. Representability is the easier question; trainability is the harder one. The case for a neural ansatz therefore cannot rest on expressiveness, and has to be made on cost. A variational neural ansatz pays in neither of the two currencies above: it requires no enumeration of a determinant space, and it fixes no prior truncation of which correlations may be described. That is not freedom from scaling, since capacity can still have to grow with N or with the accuracy demanded (Chapter 7), but it is freedom to concentrate capacity wherever the solution actually varies. Whether it does so in practice is an empirical question, and this thesis treats it as one.

In practice, several issues arise. Objectives built on local energies involve second derivatives and can be numerically delicate; coordinates that do not respect known structure produce large gradients and Laplacian spikes near electron coalescence; and generic architectures can waste capacity on degrees of freedom that are physically irrelevant. The ansatz used here therefore builds in as much known structure as possible (antisymmetry, the electron–electron cusp, trap scaling and permutation symmetry) and leaves the network to learn the remaining correlation. On top of a Slater determinant of harmonic-oscillator orbitals, two learnable objects carry that correlation: a Jastrow correlator, which reshapes the amplitude, and a coordinate backflow, which reshapes the nodes. Chapter 3 gives the construction.

What the energy cannot see. A further choice concerns not *what* structure is imposed but *how* the network represents the electrons. Correlation is intrinsically relational, since what one electron should do depends on where the others are, so architectures that let particles exchange information are a natural candidate for both the correlator and the backflow. Each can be built either way: as a separable network that encodes and pools per-particle and pairwise features in a single pass, as in a DeepSet, or as a *message-passing* network in which the particles exchange and update information through explicit edge states. Holding everything else fixed, the two can be trained to the same energy. The total energy then cannot say what the relational channel buys, or whether it buys anything, and the difference has to be looked for elsewhere.

That suggests a different use for a solver. Established methods are, for the most part, constructed outward from physical principles: an approximation is chosen, justified physically, and then implemented. A trained network reverses part of that order. Physical structure is still imposed on it, but the specific correlations it develops within those constraints are not prescribed, and can be interrogated afterwards. Unlike an implicit stochastic representation, it is a concrete parameterised object whose internal structure can be examined directly. A twenty-electron dot is a function of forty coordinates, and any basis expansion of it grows combinatorially, but it does not follow that the solution itself is forty-dimensional in any useful sense. How many directions does a given parameterisation actually use to represent the solved state? That is a question about the solution and its representation, not about the equation, and the answer depends on how the state is parameterised (§2.2.3).

We make that question quantitative through the geometry of the wavefunction's log-derivatives. Collecting the parameter sensitivities of a trained ansatz over an ensemble of configurations into a covariance matrix gives, in the simplest reading, a principal-component analysis of the state: the eigenvalue distribution counts how many directions are doing genuine work. The physically appropriate version of that covariance is the quantum geometric tensor, and the corresponding count is the participation ratio of its spectrum. The same tensor is the metric a natural-gradient optimiser inverts, and it is estimated from whatever samples define the loss, so it connects the architecture to the optimiser and to the sampling measure. Chapter 7 develops the instrument and applies it to all three.

Questions. When a variational calculation stalls, or two models reach the same energy by different routes, several explanations remain open: the representation may be wrong for the problem, the optimisation may be failing, the samples that define the loss may no longer describe the state, or the physics itself may have become harder. This thesis is organised as an attempt to tell those explanations apart, and asks four questions, one for each.

1. **Architecture.** What does a message-passing network learn that a separable one cannot? We compare the two forms in the correlator and in the backflow, holding everything else fixed.
2. **Optimiser.** When does natural-gradient (stochastic reconfiguration) preconditioning change the result, and when is it unnecessary or harmful?
3. **Paradigm.** Can an accurate wavefunction be trained with no Markov chain in the gradient, using importance-sampled collocation instead, and how far into the correlated regime does that reach before it fails?
4. **Physics.** Can the trained wavefunction serve as an instrument for Wigner-molecule physics (shell topologies, angular order, melting), and what does it show about the crossover?

The results take these in a different order, because a state has to be shown accurate enough to trust before either its physics or the solver that produced it can be interpreted. Chapter 6 establishes the energies and reads the physics from the states; Chapter 7 turns to the solver.

Main contributions.

1. A compact Slater–Jastrow–backflow ansatz that agrees closely with every available fixed-node reference for $N \in \{2, 6, 12, 20\}$ and extends smoothly into the Wigner regime, where its energies are predictions.
2. Evidence that message passing changes the variational geometry before it changes the energy. Under the parametrisations used here, a message-passing correlator uses fewer effective tangent directions and reaches a lower local-energy variance than a separable one at equal energy, and a conventional backflow collapses to a single displacement mode at weak confinement where a larger message-passing one does not.
3. A hypothesis for when natural-gradient preconditioning earns its cost. Poor conditioning alone does not decide it, and that much is measured. The failures suggest that the metric must also be estimable from the batch.
4. Markov-chain-free training by importance-sampled collocation, mapped to its limits: within about a tenth of a percent of DMC at strong confinement, and carried to $\omega = 0.0035$ once the proposal is built from the measured shell geometry, where an instability of the ansatz, not of the sampler, takes over.
5. Topology-resolved diagnostics of the Wigner crossover, read against random-angle baselines and a held-out polygon shell model, which show that radial stiffening completes well before angular order appears, complementing classical and quantum studies of parabolic dots [1–3, 11].

Thesis structure. Part II sets out the physics and machine-learning background (Chapters 1 and 2). Part III gives the ansatz, the optimisation pipeline and the diagnostics (Chapters 3 to 5). Part IV reports the results: Chapter 6 the energies and the Wigner-molecule structure, and Chapter 7 the analysis of architecture, optimisation and sampling together with the Markov-chain-free training programme. Part V discusses and concludes. The appendices in Part VI develop the conditioning theory, the collocation–backflow obstruction that motivated the weak-form objective, the full diagnostic tables and the reproducibility record.

Part II

Theory

Men do not know how what is at variance agrees with itself. It is an attunement of opposite tensions, like that of the bow and the lyre.

HERACLITUS
Fragments

Chapter 1

The many-body problem

The system is N electrons in a two-dimensional harmonic trap, repelling one another through the Coulomb interaction. Its Hamiltonian is elementary, and three of its properties make the numerical solution difficult.

The electrons are identical fermions, so the wavefunction changes sign under exchange, and that sign structure is the expensive part of the problem. The interaction diverges where two electrons meet, so the local energy $\hat{H}\Psi/\Psi$ diverges with it unless the wavefunction is constructed to cancel the divergence. And the kinetic term is a second derivative, which exposes any method that evaluates the energy pointwise to the curvature of the function it is given.

1.1 The many-body problem

A quantum state is a normalised vector in a complex Hilbert space, and in the coordinate representation $\psi(x) = \langle x|\psi\rangle$ is the wavefunction. Which basis one expands in is not a formality: choosing it to match the physics (harmonic-oscillator orbitals for confined charges, plane waves for periodic systems) is what makes a representation compact, and it is why the orbitals of Chapter 3 are those of the trap itself.

For N particles the space is the N-fold tensor product $\mathcal{H}_1^{\otimes N}$, and a fermionic state expands exactly in Slater determinants built from N distinct spin-orbitals, $|\Psi\rangle = \sum_I C_I\,|\Phi_I\rangle$. That expansion is exact and, as §1.2 makes precise, unaffordable.

We work in atomic units ($\hbar = m_e = e = 1$) with the many-body Hamiltonian

$$H = -\frac{1}{2}\sum_{i=1}^{N}\nabla_{\mathbf{r}_i}^2 \;+\; \sum_{i=1}^{N}V_{\text{ext}}(\mathbf{r}_i) \;+\; \sum_{i<j}V_{\text{int}}(|\mathbf{r}_i-\mathbf{r}_j|), \qquad H\Psi(R) = E\Psi(R), \tag{1.1}$$

where $R = (\mathbf{r}_1,\dots,\mathbf{r}_N)$ and Ψ is normalised. Identical particles restrict Ψ to a symmetry sector, antisymmetric for the fermions studied here. For these quantum dots $V_{\text{ext}}(\mathbf{r}) = \frac{1}{2}\omega^2 r^2$ and $V_{\text{int}}(r) = 1/r$ in two dimensions. Everything that follows approximates the lowest eigenpair of Eq. (1.1).

1.2 Correlation, closed shells, and the cost of exactness

Three ideas from electronic structure are used throughout this thesis and computed nowhere in it. They are what the accuracy claims of later chapters are calibrated against.

The first is *correlation energy*. Hartree–Fock approximates the ground state by a single determinant whose orbitals are themselves optimised, each electron moving in the mean field of the others; the correlation energy $\Delta E = E_{\text{exact}} - E_{\text{HF}}$ is whatever that cannot reach. It is a small fraction of the total

and essentially all of the difficulty, which is why a method can look converged in total energy while representing the correlation badly—a distinction Chapter 7 makes quantitative.

The second is *closed-shell structure*. When the occupied orbitals fill complete shells the determinant is a spin singlet and is unique, with no degenerate manifold to choose from. Those particle numbers, $N \in \{2, 6, 12, 20\}$ in a two-dimensional parabolic trap, are the ones studied here.

The third is *exactness within a basis*, and its price. Expanding in all N-electron determinants of a given one-particle basis and diagonalising gives the exact ground state within that basis; what cannot be avoided is the $\binom{M}{N}$ growth of the space itself, which for the twenty-electron dot puts the determinant count far beyond reach. That combinatorial wall is what motivates variational methods. The occupation-number formalism of second quantisation, in which antisymmetry is carried by anticommuting creation and annihilation operators, not by the wavefunction, is the usual language for all three; our own ansatz is first-quantised throughout, so it appears here as vocabulary rather than machinery.

Our determinant is built from harmonic-oscillator orbitals, not self-consistent Hartree–Fock ones, and our external benchmark is diffusion Monte Carlo [12], which for these systems is both more accurate than an affordable configuration-interaction calculation and available at the particle numbers we care about.

1.3 Quantum dots

1.3.1 Model and units

We consider N spin-$\frac{1}{2}$ fermions in d=2 with harmonic confinement and Coulomb repulsion,

$$H \;=\; \sum_{i=1}^{N}\Big(\,-\tfrac{1}{2}\nabla_{\mathbf{r}_i}^2 + \tfrac{1}{2}\omega^2 r_i^2\Big) \;+\; \sum_{i<j}\frac{1}{|\mathbf{r}_i-\mathbf{r}_j|}, \qquad (\hbar = m_e = e = 1).$$

Spin does not appear in H, so the eigenstates can be classified by S and S_z and the spatial problem solved at fixed spin assignment. For more than two identical fermions a spin-adapted eigenstate of S^2 is in general a sum of spatial–spin products, not a single one; what we use computationally is the weaker and sufficient statement that with the numbers of up- and down-spin electrons held fixed, the antisymmetriser factorises and the spatial problem may be treated on its own.

1.3.2 One–particle structure and shell filling

We use two-dimensional harmonic-oscillator spin–orbitals. In the implementation these are the *Cartesian* 2D-HO eigenstates $\{\phi_{n_x n_y}(\mathbf{r})\chi_\sigma\}$ with energies $E_{n_x n_y} = \omega(n_x + n_y + 1)$ and spin $\sigma \in \{\uparrow, \downarrow\}$. The principal index $k = n_x + n_y$ labels a shell of $(k{+}1)$ degenerate spatial states, each twofold spin-degenerate; for the closed shells studied here this Cartesian basis spans the same filled subspace as any complete basis of those shells, so the reference determinant is basis-independent. Filling from the bottom yields the *closed-shell* electron numbers

$$N_{\text{closed}} = (K{+}1)(K{+}2), \quad K = 0, 1, 2, \ldots \quad (2, 6, 12, 20, 30, \ldots).$$

All other N correspond to a *partially filled* top shell (open shell), with a degenerate noninteracting manifold.

1.3.3 Spin structure and Slater form

Write single-particle spin-orbitals as $\varepsilon_\alpha(x) = \phi_\alpha(\mathbf{r})\,\chi_\sigma(s)$, with $x = (\mathbf{r}, s)$. Because $\chi_\uparrow$ and $\chi_\downarrow$ are orthogonal and H is spin-independent, the Slater determinant simplifies:

Closed shells. For even N with doubly occupied spatial orbitals $\{\phi_1, \ldots, \phi_{N/2}\}$, the determinant factorizes into two identical spatial determinants, one for each spin:

$$\Psi_{\text{closed}}(X) = \frac{1}{(N/2)!}\,\det[\phi_a(\mathbf{r}_i)]_{i\in\uparrow, a=1..N/2}\,\det[\phi_a(\mathbf{r}_j)]_{j\in\downarrow, a=1..N/2}\;\times\;\Xi_{S=0,S_z=0},$$

where Ξ is the singlet spin function. In the computational representation this last factor is not carried: fixing which electrons are spin-up and which spin-down makes the full spin-orbital determinant block

diagonal, and the product of the two spatial determinants *is* the ansatz. This has the *form* of a restricted closed-shell determinant, but its orbitals are the harmonic-oscillator eigenstates of the trap, not orbitals optimised self-consistently in the mean field of the other electrons. It is therefore the closed-shell *non-interacting* reference, not an RHF one, and no self-consistent-field calculation is performed anywhere in this thesis. The distinction matters for how the later results should be read: everything the correlator and the backflow achieve is measured against a determinant that has not been given the benefit of a mean field, so what they recover includes the orbital relaxation an HF calculation would have supplied as well as the correlation beyond it.

1.3.4 Kato cusp conditions in two dimensions

For Coulomb interactions the potential diverges as $1/r$ at coalescence. Finiteness of the local energy

$$E_L(R) = \frac{(H\Psi)(R)}{\Psi(R)}$$

requires that the kinetic singularity cancels the $1/r$ divergence. Kato's theorem, first proved in [13], gives the necessary short–distance behaviour of the (spherically averaged) wavefunction. Let r_{ij} be the interparticle distance, $\mu_{ij} = m_i m_j/(m_i + m_j)$ the reduced mass (in a.u.), Z_i the charge number (electron $Z = -1$, nucleus $Z = +Z_A$), and D the spatial dimension. Then

$$\frac{\partial}{\partial r_{ij}} \ln \overline{\Psi}\bigg|_{r_{ij}\to 0} = \frac{2\,\mu_{ij}\,Z_i Z_j}{D-1} \quad \text{(s-wave / opposite-spin channel).}$$

For *identical fermions* the s-wave is excluded; the leading channel is p-wave (relative angular momentum $k = 1$), and the wavefunction carries a linear node at coalescence. The cusp condition then applies to the *regular part* left after that node is factored out, and this distinction is easy to lose. Writing $\overline{\Psi}(r) = r\,\overline{u}(r)$ with $\overline{u}$ regular and nonzero at contact, the logarithmic derivative of the full wavefunction diverges, $\partial_r \ln \overline{\Psi} = 1/r + \partial_r \ln \overline{u}$, so it is $\overline{u}$ and not $\overline{\Psi}$ that obeys a finite cusp condition:

$$\frac{\partial}{\partial r_{ij}} \ln \overline{u}\bigg|_{r_{ij}\to 0} = \frac{2\,\mu_{ij}\,Z_i Z_j}{2k+D-1} = \frac{2\,\mu_{ij}\,Z_i Z_j}{D+1} \quad \text{(like-spin, } k = 1\text{).}$$

In a Slater–Jastrow form the antisymmetric factor supplies the linear zero and the Jastrow supplies $\overline{u}$, so this is exactly a condition on the Jastrow slope—which is how it is imposed in Chapter 3.

Specializing to 2D electrons (atomic units). With $m_e = 1$ and $\mu_{ee} = 1/2$:

$$\boxed{a_{ee}^{\uparrow\downarrow} = \frac{\partial}{\partial r_{ij}} \ln \overline{\Psi}\bigg|_0 = 1\ , \qquad a_{ee}^{\uparrow\uparrow} = \frac{\partial}{\partial r_{ij}} \ln \overline{u}\bigg|_0 = \tfrac{1}{3}\ .}$$

The two derivatives are taken on different objects: the full spherically averaged wavefunction in the antiparallel case, where it is finite at contact, and its regular part in the parallel case, where the antisymmetric node has been removed. The like-spin value follows from the p-wave (linear-node) analysis carried out in full in Appendix B.2, the canonical derivation used throughout this thesis; it gives the familiar $\frac{1}{4}$ in three dimensions and $\frac{1}{3}$ here in two. The 2D like-spin cusp is therefore *not* simply half the opposite-spin value—a coincidence that holds only in 3D—and the implemented value is $1/(D+1) = \frac{1}{3}$. For electron–nucleus coalescence ($\mu_{eA} \approx 1$, $Z_i Z_j = -Z_A$):

$$\boxed{a_{eA} = \frac{\partial}{\partial r_{iA}} \ln \overline{\Psi}\bigg|_0 = -2Z_A\ .}$$

Why this matters. These conditions ensure the kinetic energy's r^{-1} singularity cancels the Coulomb r^{-1} term so that E_L remains finite as particles coalesce. Enforcing the cusp explicitly improves conditioning.

1.4 Structural diagnostics and Wigner markers

A Wigner molecule is not defined by a single number, so the crossover has to be read from several markers at once. This section says what each one measures and why it responds to Wigner order; the estimators, thresholds and stability protocol are defined once, in the Analysis chapter (§5.1), and the measurements themselves are in Chapter 6.

Radial localisation. The pair distribution $g(r)$ and the radial probability $P(r)$ say how far apart the electrons sit and how tightly that distance is fixed. The informative quantity is not the typical radius, which simply grows as the trap weakens, but the *ratio* of the radial width to it. A distribution that becomes relatively narrower around a growing radius is a shell stiffening; one that merely dilates is a droplet expanding. This is the finite-system analogue of a Lindemann criterion.

Shell topology. In a parabolic trap the classical minimum energy configuration is a set of concentric rings with definite occupancies—$(1, 5)$ at N=6, $(3, 9)$ at N=12 [2, 3]. Detecting those occupancies frame by frame turns a qualitative picture into a distribution over topologies, which is what makes it possible to say that a state is *mostly* $(3, 9)$ with real weight on $(2, 10)$—a rotating molecule fluctuating between near-degenerate shellings rather than a single frozen crystal.

Orientational order. Radial structure alone cannot distinguish a ring of electrons at fixed angular spacing from a ring free to slide. The bond-orientational order parameter Φ_m measures m-fold angular order on a detected ring after removing global rotation, and an angular Lindemann number measures how stiff those spacings are. Both are standard in two-dimensional Coulomb and Yukawa systems [11, 14], and together with the radial markers they separate angular from radial melting—the two need not happen at the same ω.

Classical scaling. Finally, the strong-coupling limit makes a parameter-free prediction: inter-particle distances should approach $r \propto \omega^{-2/3}$. Fitting the measured exponent and watching it approach $-2/3$ from above is an external check on the whole picture, since nothing in the ansatz knows about that number.

How to read them together. As ω falls, the width ratios and angular Lindemann numbers shrink, shell gaps sharpen, the shell-resolved $|\Phi_m|$ rise above their finite-ring random-angle baseline, and the radial exponent approaches its classical value. No one of these is decisive—a narrow $P(r)$ can be an artefact of poor sampling, and a large $|\Phi_m|$ can be a fluctuation on a short chain—but they fail in different ways, which is why Chapter 6 reports them together and Chapter 5 states the stability checks each one is subjected to. Quantum Wigner markers in parabolic dots are reviewed in Refs. [1, 4, 11, 14].

1.5 Quantum Monte Carlo benchmarks

Two stochastic methods supply the numbers this thesis is measured against, and the difference between what they guarantee is the whole of its external evidence. Variational Monte Carlo optimises a trial state under the variational principle. Diffusion Monte Carlo projects toward the ground state in imaginary time. Only the first is what we do; only the second is what we are scored against; and neither is exact for fermions.

1.5.1 Variational Monte Carlo

For any normalised trial state $\Psi_T(\mathbf{R}; \boldsymbol{\alpha})$ the Rayleigh quotient

$$E[\Psi_T] = \frac{\langle \Psi_T | \hat{H} | \Psi_T \rangle}{\langle \Psi_T | \Psi_T \rangle} \;\geq\; E_0$$

bounds the exact ground-state energy from above, so minimising it over $\boldsymbol{\alpha}$ can only improve the estimate. Writing the expectation as an average of the *local energy*

$$E_L(\mathbf{R}) \;=\; \frac{\hat{H}\Psi_T(\mathbf{R})}{\Psi_T(\mathbf{R})} \tag{1.2}$$

over configurations drawn from $P(\mathbf{R}) = |\Psi_T(\mathbf{R})|^2 / \int |\Psi_T|^2$ turns the integral into a sampling problem,

$$E[\Psi_T] \;\approx\; \frac{1}{N_{\mathrm{MC}}} \sum_i E_L(\mathbf{R}_i),$$

with the configurations generated by Metropolis–Hastings [15] at acceptance

$$A(\mathbf{R} \to \mathbf{R}') = \min\!\left(1,\; \frac{|\Psi_T(\mathbf{R}')|^2}{|\Psi_T(\mathbf{R})|^2}\, \frac{T(\mathbf{R}' \to \mathbf{R})}{T(\mathbf{R} \to \mathbf{R}')}\right),$$

optionally drifted by the quantum force $\mathbf{F} = 2\nabla \ln |\Psi_T|$ toward regions of high density [16].

The zero-variance principle. One property of Eq. (1.2) does more work in this thesis than the energy itself. If Ψ_T were an exact eigenstate, $\hat{H}\Psi_T = E\Psi_T$ pointwise, and E_L would be *constant* over the whole of configuration space. Its variance

$$\mathrm{Var}(E_L) \;=\; \big\langle E_L^2 \big\rangle - \big\langle E_L \big\rangle^2 \tag{1.3}$$

therefore vanishes only at an exact eigenstate, and decreases as the trial state improves. This makes it a measure of wavefunction quality that is independent of any external reference—which matters here for two reasons. It keeps saying something after the energy has saturated, when two ansätze agree to four figures and the energy can no longer tell them apart. And it is defined on the state rather than on any component of it, so it survives changes of parameterisation that quantities such as a component's rank do not. Chapter 7 uses it in both roles.

1.5.2 Diffusion Monte Carlo, and the fixed-node constraint

DMC removes the dependence on a trial state by projection. In imaginary time $\tau = it$ the Schrödinger equation becomes a diffusion equation,

$$-\frac{\partial \Psi(\mathbf{R}, \tau)}{\partial \tau} = (\hat{H} - E_T)\, \Psi(\mathbf{R}, \tau),$$

whose long-time limit is the ground state, provided the initial state overlaps it. The evolution is simulated by an ensemble of walkers that diffuse, drift along the quantum force, and are branched or killed with weight $w = \exp[-(E_L(\mathbf{R}) - E_T)\Delta\tau]$, so that the walker density converges on the ground-state density.

That argument works because a diffusion equation propagates a *probability*. For bosons the ground state is nodeless and can be treated as one directly. For fermions it cannot: antisymmetry forces Ψ to change sign, and a signed density is not a probability. Splitting it into positive and negative populations is formally possible but their difference is swamped by their sum at a rate exponential in $N\tau$—the *fermion sign problem* [17].

The standard escape is to impose the sign structure instead of sampling it. The *fixed-node* approximation confines the walkers to the nodal pockets of a trial state Ψ_T, killing any walker that crosses a node. Within each pocket the wavefunction has a definite sign, the projection is well posed, and the method returns the exact ground-state energy *of the problem with those nodes imposed as a boundary condition*. That energy is a rigorous upper bound on E_0, and it is exact only if the nodes are. Otherwise it carries a systematic *nodal bias* that no amount of sampling removes. The bound is a statement about the fixed-node eigenvalue in the limit of zero time step and infinite walker population; a published DMC number is an estimate of it and carries time-step, population-control and finite-statistics errors of its own, which is part of why comparing a variational energy against one at the fifth decimal place requires the matched uncertainty analysis we did not undertake (§8.1).

Three consequences follow. A published DMC energy is not ground truth; it is the best energy attainable inside the nodal surface of the trial state used to produce it. A variational state with better nodes than that surface may therefore lie *below* a fixed-node DMC number without violating anything, since the variational principle bounds the energy from below by E_0 and not by another approximation (§8.1). And improving nodes becomes the thing to attempt, which is what the coordinate backflow of Chapter 3 is for.

1.5.3 Requirements on the trial state

Both methods take a trial state as given. VMC is only as good as the state it optimises, and fixed-node DMC only as good as the nodes it is handed, so the quality of the ansatz stops being an implementation detail and becomes the object of study. Three requirements bear on it at once.

Antisymmetry has to hold exactly and by construction, since the sign structure is what makes the problem hard in the first place. In practice that means a determinant, multiplied by a symmetric correlation factor of the kind Jastrow introduced for strongly interacting systems [18]. The nodes have to be placed well, since that is where the remaining systematic error lives once the amplitude is right. And the state has to be smooth enough that $E_L = \hat{H}\Psi/\Psi$, containing as it does a Laplacian and a potential that diverges at coalescence, can be evaluated without the two singularities amplifying one another. The cusp conditions of §1.3.4 say that those two divergences can be made to cancel exactly, and hard-wiring that cancellation is the first design decision of Chapter 3.

The remaining design choice is the functional form of the smooth correlation factor. The next chapter takes it from a literature that grew up asking different questions.

Chapter 2

Neural quantum states and variational geometry

2.1 The variational learning objective

Machine learning is usually introduced as *empirical risk minimisation*: an unknown joint distribution $P(X, Y)$ generates data, a loss ℓ scores predictions, and one minimises the empirical average $\hat{\mathcal{R}}(\theta) = \frac{1}{n}\sum_i \ell(f_\theta(x_i), y_i)$ over a finite sample as a proxy for the population risk $\mathcal{R}(\theta) = \mathbb{E}_P[\ell(f_\theta(X), Y)]$. The gap between the two is the subject of the field: overfitting, the bias–variance tradeoff, regularisation, held-out validation [19, 20].

The variational problem considered here differs from that setting in two respects. There is no fixed labelled dataset, and so no conventional train/test generalisation problem. What replaces it is not the absence of a finite-sample difficulty but a different one: the objective is a population expectation that must be estimated afresh at every step, and an optimiser can certainly overreact to a particular finite batch—which is exactly what the effective-sample-size and weight-tail diagnostics of Chapter 7 measure. The objective is an expectation,

$$E[\Psi_\theta] \;=\; \mathbb{E}_{\mathbf{R}\sim|\Psi_\theta|^2}\big[E_L(\mathbf{R})\big], \tag{2.1}$$

under a distribution the model itself defines, re-estimated from fresh samples at every step. The optimisation target is the physical state rather than predictive performance on unseen data.

What replaces generalisation as the difficulty is already visible in the form of Eq. (2.1). Its measure $|\Psi_\theta|^2$ moves: it changes with every update, so the gradient, the samples used to estimate it, and the metric used to precondition it are all functions of the current parameters. In supervised learning the data distribution sits still and none of this arises. Meanwhile $E_L = \hat{H}\Psi_\theta/\Psi_\theta$ contains a Laplacian and a potential that diverges, so the objective reads the network's second derivatives, not its values, and reads them near a singularity. Smoothness stops being a stylistic preference and becomes a requirement of the loss.

One term from the usual taxonomy recurs below. The objectives here are *unsupervised*: no labelled targets exist, and the loss comes entirely from the physics the network is asked to satisfy. The exception is the second phase of residual pretraining, where a published reference energy is used as an explicit target, and that is why those runs are later described as trained without a Markov chain rather than as unsupervised solutions of the Schrödinger equation. Reinforcement learning enters obliquely, through the REINFORCE estimator borrowed for the collocation gradient.

2.2 Optimisation

Whatever the objective, minimising it over a nonlinear, high-dimensional hypothesis class admits no closed form, so the parameters $\theta \in \mathbb{R}^p$ are updated iteratively according to the local geometry of the loss.

2.2.1 First-order methods

Gradient descent, $\theta^{(t+1)} = \theta^{(t)} - \eta \nabla_\theta \hat{\mathcal{R}}(\theta^{(t)})$, moves along the direction of steepest local decrease; too small a learning rate η converges slowly and too large a one diverges. Evaluating the full gradient is usually unaffordable, so the estimate is formed on a mini-batch, which is unbiased and whose variance sets the noise scale of the trajectory [21]. Momentum accumulates a running average of past gradients, damping oscillation across narrow directions and building displacement along consistent ones. Adam [22] adds a per-parameter normalisation by a running second moment,

$$m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t, \qquad v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2, \qquad \theta^{(t+1)} = \theta^{(t)} - \eta \, \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}, \tag{2.2}$$

with g_t the mini-batch gradient and $\hat{m}_t, \hat{v}_t$ bias-corrected. That normalisation is a diagonal rescaling of parameter space, and the question of whether a non-diagonal one is worth its cost is what §2.2.2 and, empirically, §7.2 are about. Adam is the default optimiser in this work [23].

2.2.2 Second-order methods and natural gradient

First-order methods only use gradient information. In principle, convergence can be accelerated by incorporating curvature via the Hessian $H(\theta) = \nabla_\theta^2 \hat{\mathcal{R}}(\theta)$. Newton's method updates parameters as

$$\theta^{(t+1)} = \theta^{(t)} - H(\theta^{(t)})^{-1} \nabla_\theta \hat{\mathcal{R}}(\theta^{(t)}). \tag{2.3}$$

However, computing and inverting the Hessian is usually infeasible for large models.

An attractive alternative is the *natural gradient* [24, 25], which replaces the Euclidean metric with the geometry of probability distributions. The key object is the *Fisher information matrix* (FIM),

$$F(\theta) = \mathbb{E}\big[\nabla_\theta \log p_\theta(x) \, \nabla_\theta \log p_\theta(x)^\top\big], \tag{2.4}$$

where p_θ denotes the model distribution. The natural gradient update is

$$\theta^{(t+1)} = \theta^{(t)} - \eta \, F(\theta^{(t)})^{-1} \nabla_\theta \hat{\mathcal{R}}(\theta^{(t)}). \tag{2.5}$$

This preconditioning makes updates invariant to reparameterizations of θ and often improves conditioning of the optimisation. In practice, computing F^{-1} exactly is intractable, but approximations (e.g. stochastic reconfiguration in variational Monte Carlo, Kronecker-factored approximations in deep learning) make the method viable [6].

2.2.3 The variational tangent space: geometric tensor, tangent kernel, and effective dimension

For a variational quantum state the Fisher metric takes a particularly concrete form, and that form is the analytical lens used throughout the results of this thesis. Write the per-sample log-derivatives, the *score*, of an ansatz Ψ_θ as

$$O_{ki} = \frac{\partial \log |\Psi_\theta(\mathbf{x}_k)|}{\partial \theta_i}, \qquad k = 1, \dots, B, \;\; i = 1, \dots, P, \tag{2.6}$$

for a batch of B configurations drawn from $|\Psi_\theta|^2$ and P parameters θ (here θ collects *all* variational parameters). Its centred Gram matrices define two dual objects. Throughout, O is taken *column-centred*, $O_{ki} \mapsto O_{ki} - \frac{1}{B}\sum_{k'} O_{k'i}$, which is what makes the Gram matrix a covariance rather than a second moment and is how it is formed in the implementation (§4.4). The *quantum geometric tensor* (QGT) is then

$$S = \tfrac{1}{B} \, O^\top O \in \mathbb{R}^{P \times P}, \tag{2.7}$$

is the Fisher–Rao metric of the state written in terms of the amplitude score. Every state studied here is real and positive up to the determinant sign, at zero magnetic field, so the amplitude score is the whole story; for a general complex wavefunction the quantum geometric tensor also carries phase sensitivity, and the identification with the classical Fisher matrix is then only of its real part. It is proportional to the Fisher matrix $F(\theta)$ of the previous subsection specialised to $p_\theta = |\Psi_\theta|^2$: because $\nabla_\theta \log |\Psi_\theta|^2 = 2 \nabla_\theta \log |\Psi_\theta|$, one has $F = 4S$, the factor of four being a convention absorbed into the learning rate.

Stochastic reconfiguration is natural-gradient descent under this metric, $\dot{\theta} = -S^{-1}\nabla_\theta E$ [26, 27]. Its Gram dual, the *neural tangent kernel* (NTK) [28],

$$K = OO^\top \in \mathbb{R}^{B\times B}, \tag{2.8}$$

shares the nonzero spectrum of S up to the overall factor B (its nonzero eigenvalues are $B\lambda_a$), and is cheaper to diagonalise when $B < P$.

The analysis is organised around two scalars read from that spectrum. The *condition number* $\kappa(S) = \lambda_{\max}/\lambda_{\min}$ measures how anisotropic the tangent metric is, and therefore how far a natural-gradient step departs from a Euclidean one. It is the quantity one would expect to decide when SR is worth its cost; Chapter 7 measures it and finds that it does not, which is how the estimability criterion of that chapter arises. The *effective dimension*

$$d_{\text{eff}}(S) = \frac{\left(\sum_a \lambda_a\right)^2}{\sum_a \lambda_a^2} \tag{2.9}$$

is the participation ratio of the spectrum: it counts how many independent tangent directions carry appreciable Fisher weight locally around the solution, regardless of the raw parameter count P. A small d_{eff} means the local tangent sensitivity is concentrated in a few collective directions; a large one means it is spread across many. Because these are properties of the tangent geometry, not of the raw parameter count, they permit architectures, optimisers and training paradigms to be compared on a common footing—provided the comparison is made under matched conventions. In the comparisons of §7.1.1 the measured d_{eff} proved stable across seeds and across the choice of probe measure; that is an empirical finding at those settings, not a general property.

The participation ratio d_{eff} is parameterisation-dependent. It is computed from the QGT eigenvalues in the ordinary Euclidean parameter coordinates, and those eigenvalues are *not* invariant under an arbitrary reparameterization: rescaling a single parameter $\theta_1 \mapsto c\,\theta_1$ leaves the physical state untouched but changes d_{eff} (for a diagonal metric it moves from 2 toward 1 as $c \to \infty$). d_{eff} is therefore *not* a coordinate-free intrinsic dimension of the wavefunction; it is a property of the (architecture, parameterisation, normalisation) triple. We use it accordingly—as a comparison between fixed architectures under matched conventions and a common probe measure—and read a smaller d_{eff} as "this architecture concentrates its Fisher weight into fewer active directions," not as an absolute manifold dimension.

2.2.4 Tangent-space diagnostics

Convergence speed is not what matters about the optimiser here. A natural-gradient step is only as good as the metric it inverts, so the estimator of S becomes part of the method rather than an implementation detail, and Chapter 7 turns that into a three-regime result. Separately, $\kappa(S)$ and d_{eff} can be read off a trained state, which makes them diagnostics as much as convergence aids, and they are used that way from Chapter 6 onward.

2.3 Smooth networks for Laplacian objectives

A feed-forward network composes affine maps with elementwise nonlinearities,

$$h^{(0)} = x, \qquad h^{(\ell)} = \sigma\big(W^{(\ell)}h^{(\ell-1)} + b^{(\ell)}\big), \qquad f_\theta(x) = W^{(L)}h^{(L-1)} + b^{(L)},$$

and without the nonlinearity σ the composition collapses to a single affine map. For most applications the choice of σ is settled by convention or by a sweep. Here the objective settles it, because the local energy $E_L = \hat{H}\Psi_\theta/\Psi_\theta$ contains a Laplacian and training therefore backpropagates through second derivatives of the network.

The activation. $\mathrm{ReLU}(x) = \max(0, x)$ has a second derivative that vanishes almost everywhere and is distributional at its kinks, so it cannot represent the pointwise curvature a Laplacian-based local energy asks for; it is excluded at once. Less obviously, being C^∞ does not suffice either. The relevant property is the reach of the higher derivatives: an activation whose second and third derivatives decay quickly away from the origin contributes little curvature over most of its input range, so the Laplacian of a deep composition stays bounded and its variance finite. Tanh and the logistic sigmoid are perfectly smooth, yet their higher derivatives are sharply peaked and change sign, and curvature accumulates unevenly through

depth. We use the SiLU$(x) = x\,\sigma(x)$ family instead. Figure 2.1 shows GELU and Mish, which are the members of that family whose derivative shapes we compared and which SiLU closely resembles; their second and third derivatives are broader and decay more slowly than those of tanh or the logistic sigmoid. This is a design heuristic rather than a theorem: the figure establishes the shapes of the derivatives, not a bound on the Laplacian variance of a deep composition, and we did not run an activation ablation.

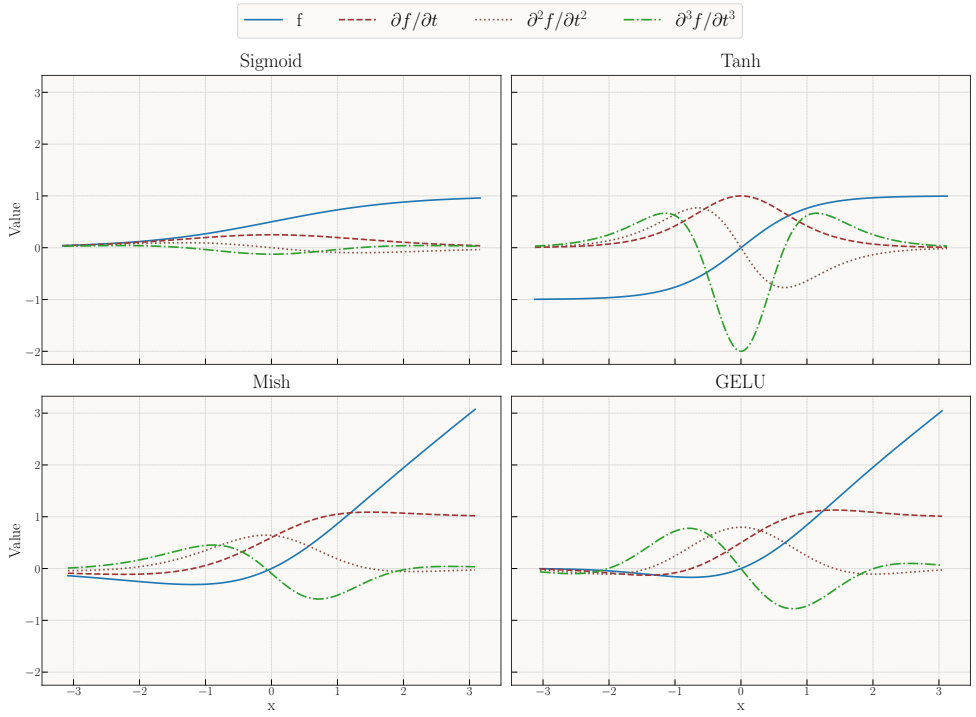


Figure 2.1: Common activation functions and their first three derivatives. All of GELU, Mish, tanh and the logistic sigmoid are C^∞, so the distinction is not smoothness but the *reach* of the higher derivatives: for GELU and Mish the second and third derivatives are broad and decay slowly to zero, while for tanh and sigmoid they are sharply peaked near the origin and change sign, so curvature accumulates unevenly when such units are composed through depth. ReLU is shown for contrast; its second derivative is a distribution and it cannot be used in a Laplacian-based objective at all.

Initialisation. The standard schemes (Xavier/Glorot, scaling weight variance as $1/(n_{\text{in}} + n_{\text{out}})$, and Kaiming/He, scaling as $2/n_{\text{in}}$ for rectified units) are designed to keep forward activations and backward gradients from shrinking or exploding through depth [29, 30]. That is necessary but not sufficient when the loss contains second derivatives, because a scheme that stabilises $\partial f/\partial x$ says nothing directly about $\partial^2 f/\partial x^2$. We use Xavier-style initialisation with the smooth activations above, and—more consequentially than any initialisation choice—zero-initialise the final layer of the backflow, so that the ansatz begins exactly at the Slater nodes and the learned nodal deformation grows from nothing rather than from noise (§3.5.3).

None of this makes the Laplacian well conditioned; it only stops the network from making it worse. The conditioning itself comes from the coordinate scaling, the soft-core pair features and the analytic cusp of Chapter 3, and is analysed in Appendices A and B.

2.4 Architectures for many-body wavefunctions

The feed-forward network of the previous section maps a fixed-size input vector to an output. A many-body wavefunction is not a generic function of such a vector: identical particles impose a permutation symmetry, and correlation is intrinsically *relational*—what one particle should do depends on where the others are. A plain multilayer perceptron acting on the concatenated coordinates $(\mathbf{x}_1, \ldots, \mathbf{x}_N)$ respects neither property; it would have to learn the symmetry from data, and its parameter count would grow with

N. This section introduces the two architecture classes that build the required structure in, permutation-invariant pooling and message passing, and situates the specific message-passing network used in this thesis as one instance of the latter.

2.4.1 Permutation symmetry and equivariance

For identical particles the modulus $|\Psi|$ is invariant under exchange (the sign is carried by the antisymmetric Slater reference of Chapter 3). The two learnable objects of the ansatz therefore have distinct symmetry requirements. A scalar *correlator* f must be permutation-*invariant*,

$$f(P_\sigma R) = f(R),$$

while a per-particle *backflow* displacement field $\mathbf{g} = (\mathbf{g}_1, \ldots, \mathbf{g}_N)$ must be permutation-*equivariant*,

$$\mathbf{g}(P_\sigma R) = P_\sigma\, \mathbf{g}(R),$$

for any permutation σ (here P_σ relabels the particles). Enforcing these by construction is both more data-efficient than learning them and yields a parameter count independent of N, because the same shared maps are applied to every particle and pair.

2.4.2 Permutation-invariant pooling: DeepSets

The simplest way to build an invariant function is to embed each particle independently and pool the embeddings symmetrically. This is the DeepSets construction [31]: any suitably regular permutation-invariant function admits the form

$$f(R) = \rho\Big(\textstyle\sum_i \phi(\mathbf{x}_i)\Big),$$

with a per-particle embedding ϕ, a symmetric pool $\sum_i$, and a readout ρ. A two-body extension encodes pairs, $\phi(\mathbf{x}_i, \mathbf{x}_j)$, and pools over them, capturing pairwise correlation. What characterises this class is that each particle, or pair, is embedded *independently* and only then pooled: no information is exchanged between particles before the pool. We refer to such networks as *separable*, and they serve as the non-message-passing baseline for the correlator.

2.4.3 Message passing and graph networks

To let a particle's representation depend on the others' in an iterated, higher-order way, one places the particles on a graph—nodes are particles, edges are pairs—and passes learned messages along the edges [32, 33]. In a generic round ℓ, each edge gathers information from its endpoint nodes and each node aggregates its incoming edge messages,

$$\mathbf{m}_{ij}^{(\ell)} = \phi_e\big(\mathbf{h}_i^{(\ell)}, \mathbf{h}_j^{(\ell)}, \mathbf{e}_{ij}^{(\ell)}\big), \qquad \mathbf{h}_i^{(\ell+1)} = \phi_v\Big(\mathbf{h}_i^{(\ell)},\ \mathrm{Agg}_{j\neq i}\, \mathbf{m}_{ij}^{(\ell)}\Big),$$

with shared maps ϕ_e, ϕ_v and a symmetric aggregation Agg, so that permutation equivariance holds by construction; a final readout produces an invariant scalar (correlator) or an equivariant field (backflow). Both DeepSets and message passing use pairwise information, but only message passing lets that information *propagate*: after several rounds a particle's representation reflects not just its neighbours but their neighbours' responses. This iterated propagation is what we call the *relational channel*, and asking what it buys, in the correlator and in the backflow, is one of the central questions of this thesis. Message passing has been used in exactly this spirit for neural quantum states of extended fermion systems [9, 10].

2.4.4 The CTNN: one message-passing instance

The message-passing correlator and backflow studied here are realised by a particular network we refer to as the *CTNN*: a copresheaf-style message-passing network that maintains explicit *node and edge* features and transports information between them through learned maps, rather than recomputing and discarding one-shot messages. The correlator iterates this exchange over a down/up "V-cycle"; the backflow applies a single such copresheaf round. Its concrete update equations, hidden dimensions, and parameter counts are specified in the Methods chapter (§3.5.2, Eqs. (3.13)–(3.15)). Two things follow. First, the CTNN is *one* concrete realisation of the message-passing class defined above, and the comparison run in this thesis is between two specific inductive biases—the CTNN against a DeepSets-type separable baseline—not

between the classes they belong to. Conclusions are stated for message passing *as instantiated by the CTNN, relative to the separable baseline studied here*; whether they generalise to other message-passing neural quantum states is a plausible extrapolation and not something these experiments test. Wherever we contrast a "message-passing" network with a "separable" one, the separable member is a DeepSets-type network and the message-passing member is the CTNN. Second, we write "copresheaf-*style*" deliberately: the name reflects the node/edge transport structure of the updates, not a claim to a formal sheaf-theoretic construction.

2.4.5 Relevance to the ansatz

Both learnable objects of the ansatz—the Jastrow correlator and the coordinate backflow—can be built in either form: as a separable DeepSets-type network, or as the message-passing CTNN. Holding everything else fixed and varying only this choice isolates the effect of the relational channel, which Chapter 7 analyses through the geometry of the tangent kernel introduced in §2.2.3.

2.5 Approximation theory and its limits

The usual justification for using a neural network as a variational ansatz is that it can represent anything. That argument needs disposing of early, since this thesis is about what happens after it stops being relevant.

The universal approximation theorems say that a single hidden layer with a non-polynomial activation is dense in $C(K)$ on compacta: for any continuous f and any $\varepsilon > 0$ there exist a width and a parameter vector achieving $\sup_{x\in K} |f(x) - f_\theta(x)| < \varepsilon$ [34–36]. They establish one thing, best put negatively: there is no *fundamental* representability obstruction. They say nothing about the particular 20k–236k networks used here, nor about whether gradient-based training will find such a representation, at what sample cost, or how the answer depends on the parameterisation. The existence proofs are non-constructive, the widths they guarantee are astronomical, and the same function can be represented by two networks whose optimisation behaviour is entirely different—which is, in miniature, the comparison Chapter 7 makes.

One further result does more than assert possibility. In high dimension a generic continuous target requires a network size growing exponentially in d, so approximation alone cannot explain why these methods work at all in Nd dimensions. The escapes are all statements about *structure*, not capacity: targets of finite Barron norm admit dimension-independent rates [37]; compositional targets admit efficient representation by deep networks when the composition matches the architecture; and imposing a symmetry removes the redundant copies the network would otherwise have to learn, reducing the burden on the representation rather than the dimension of the domain. The last of these is the one we exploit directly, and the middle one is why the message-passing architectures of §2.4 deserve attention: their usefulness comes from a compositional form matching the relational structure of the physics, not from added capacity.

The rest of the thesis therefore assumes representability and measures the other things: trainability, conditioning, and the geometry of the parameterisation.

2.6 Residual objectives and their difficulties

In scientific applications the governing equations already encode most of what is known about a system, and a model fitted only to data throws that information away. *Physics-informed neural networks* (PINNs) [38] put it back by penalising the residual of the equation itself: for a differential operator $\mathcal{N}$ and a network u_θ, the loss is a sampled sum of $|\mathcal{N}[u_\theta](\xi_j)|^2$ over collocation points $\{\xi_j\}$, together with boundary and data terms. Training a network against a differential operator predates the name by two decades [39]; what [38] added was automatic differentiation, which evaluates ∇u_θ and Δu_θ exactly and cheaply enough that the residual can drive the optimisation directly.

Four difficulties follow from differentiating through the operator, and each recurs by name in the results of this thesis [40].

Approximation in Sobolev norms. Universal approximation controls function error in L^2 or L^∞, but a strong residual depends on *derivatives*. Making it small requires approximation in $W^{\ell,p}$ with ℓ the order

of the equation, which forces smooth activations for $\ell \geq 2$ and architecture choices that control higher derivatives. This is why the activation here is chosen for its second derivative, not its values.

Residual-to-error stability. A small residual implies a small solution error only where the forward problem is stable. This is why a small residual is never taken here as evidence of a good state without an independent energy evaluation.

Sampling. Collocation minimises a sampled surrogate of a continuum loss, and the empirical residual can be small while the population residual is not. In this setting the question becomes whether an importance-weighted estimator has a finite variance at all, which is what bounds the Markov-chain-free programme of Chapter 7.

Conditioning. Residual losses couple terms of disparate scale and produce ill-conditioned gradient dynamics; in the linearised view, convergence is governed by the spectrum of a problem-dependent kernel. That spectrum is the subject of the entire tangent-kernel analysis.

The last of these determined the design of the collocation programme. A strong-form residual that differentiates through the Laplacian meets a structural obstruction as soon as it is asked to train a nodal surface, because the learning signal lives where E_L diverges. The weak form keeps the local energy as a detached reward and never differentiates it, and so avoids the obstruction; Appendix C shows why, and it is the reason the collocation programme takes the form it does.

2.7 Constraints carried into the method

The two traditions of this chapter and the last have been describing the same objects in different words. The quantum geometric tensor of variational Monte Carlo is the Fisher metric of information geometry; stochastic reconfiguration is natural-gradient descent under it; the neural tangent kernel is that metric's Gram dual; and the sampling measure a physicist chooses for a Metropolis chain is what a statistician calls a proposal distribution. Neither community needs the other's vocabulary in order to do its own work, which is why the identifications are easy to miss. They matter here because the architecture determines the spectrum of that metric, the optimiser inverts it, and the sampling measure determines how well it can be estimated.

Four properties of this problem bear on that metric.

The objective contains a second-order operator. The local energy carries a Laplacian, so training differentiates through second derivatives of the network. This is what dictates smooth activations, controlled initialisation, and a determinant that is never asked to do anything violent (§2.3).

The interaction is singular, and the state carries nodes. The Coulomb term diverges at coalescence, and the trial function vanishes on a surface whose location is itself being learned. Both are places where $E_L = \hat{H}\Psi/\Psi$ can blow up, and the cusp conditions of §1.3.4 say that the first divergence can be cancelled analytically if the short-range form is built in, not learned.

The measure is part of the method. The objective is an expectation under $|\Psi_\theta|^2$, so the sampling distribution enters the gradient, the metric, and the estimate of both. Whether that metric can be estimated at all turns out to decide whether preconditioning helps or hurts (§2.2.3).

Correlation is relational. What one electron should do depends on where the others are, and the two architecture classes of §2.4 differ precisely in how far that dependence is allowed to propagate.

The first two are met by construction in Chapter 3, which hard-wires antisymmetry and the cusp and conditions the geometry so that the learned part can remain small and smooth. The third is treated in Chapter 4, where the sampling measure is a design variable rather than a fixed background. The fourth is not settled here: both architectures are constructed, and the comparison between them is reported in Chapter 7.

Part III

Methods

No great thing is created suddenly, any more than a bunch of grapes or a fig.

EPICTETUS
Discourses

Chapter 3

The trial wavefunction

The trial state is constructed to satisfy fermionic antisymmetry and the electron–electron cusp analytically, and to keep the learned terms smooth under the Laplacian. Antisymmetry requires a determinant. The Coulomb divergence admits an analytic cancellation, so it is imposed, not learned. Second derivatives enter the loss, so the features fed to the network are bounded and smooth and the coordinates are scaled to keep those derivatives $\mathcal{O}(1)$ across three decades of confinement. The form of the learned part is not fixed by these constraints; both candidate architectures are constructed here and compared in Chapter 7. Unless stated otherwise we work in atomic units.

3.1 Preliminaries

Hamiltonian and units. We study N spin-$\frac{1}{2}$ fermions in two spatial dimensions confined by a harmonic trap and interacting via Coulomb repulsion,

$$H = \sum_{i=1}^{N} \Big(-\tfrac{1}{2}\nabla_{\mathbf{r}_i}^2 + \tfrac{1}{2}\omega^2 r_i^2 \Big) + \sum_{1\le i<j\le N} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}, \qquad (\hbar = m_e = e = 1). \tag{3.1}$$

Spin does not appear in H; eigenstates factorize into a spatial function with good L_z and a spin function with good (S, S_z).

Coordinates and scaling. We collect positions as $R = (\mathbf{r}_1, \dots, \mathbf{r}_N) \in \mathbb{R}^{2N}$ and use trap scaling $\tilde{\mathbf{r}}_i = \sqrt{\omega}\,\mathbf{r}_i$, $\tilde{R} = \sqrt{\omega}\,R$. This aligns typical length scales to $\mathcal{O}(1)$, improves conditioning of second derivatives, and makes the noninteracting ground state ω-independent. When useful, we also use pair distances $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$ and their scaled versions $\tilde{r}_{ij} = |\tilde{\mathbf{r}}_i - \tilde{\mathbf{r}}_j|$.

Sectors and focus. We primarily report *closed-shell* cases (unique spin singlet at the noninteracting level). Open shells introduce a degenerate manifold and additional choices for the reference; we keep this to brief remarks only where needed.

Spin–orbitals and basis. One-particle spatial orbitals are the Cartesian two-dimensional harmonic-oscillator eigenstates $\{\phi_{n_x n_y}(\mathbf{r})\}$ with $E_{n_x n_y} = \omega(n_x + n_y + 1)$, $n_x, n_y \in \mathbb{N}_0$ (the basis used in the implementation; for the closed shells studied it spans the same filled shells as any complete basis of those shells). Spin–orbitals are $\varepsilon_\alpha(x) = \phi_{n_x n_y}(\mathbf{r})\chi_\sigma(s)$, $\sigma \in \{\uparrow, \downarrow\}$.

3.2 Overview of the ansatz

Our trial state combines exact antisymmetry, an analytic short-range cusp, and a compact neural residual, while optionally evaluating the Slater determinant on backflowed coordinates:

$$\Psi_{\theta,\beta}(R) = \mathrm{SD}\big(\tilde{R}'\big)\ \exp\Big(\sum_{i<j} u_{\sigma_i\sigma_j}(r_{ij}) + W_\theta(\tilde{R})\Big), \tag{3.2}$$

$$\tilde{R}' = \tilde{R} + \Delta_\beta(\tilde{R}). \tag{3.3}$$

Here SD enforces antisymmetry; u hard-wires the Kato cusp; W_θ (the NQS correlator) learns smooth medium/long-range structure; and Δ_β (backflow) adjusts the nodes by smoothly warping the coordinates used inside the determinants; its per-particle displacements $\Delta_{\beta,i}$ are written $\Delta\mathbf{x}_i$ in the results chapters. We deliberately feed the NQS with $\tilde{R}$ (not $\tilde{R}'$) to avoid self-reinforcing feedback and keep local-energy variance low.

Design principles. (i) *Separate guaranteed physics from learned details:* antisymmetry and the cusp are hard-wired; the network learns the rest. (ii) *Condition the geometry:* work in trap units and use soft-core pair features so that first/second derivatives are bounded. (iii) *Keep the learned object small and smooth:* once the cusp is explicit, a compact NQS suffices across (N, ω).

3.3 Slater reference

For closed shells we use a restricted product of spin-block determinants of the lowest $N/2$ harmonic oscillator orbitals (trap units):

$$\mathrm{SD}(\tilde{R}) = \det\big[\phi_a(\tilde{\mathbf{r}}_i)\big]_{i\in\uparrow,\ a\le N/2}\ \det\big[\phi_a(\tilde{\mathbf{r}}_j)\big]_{j\in\downarrow,\ a\le N/2}. \tag{3.4}$$

The correlator reweights amplitudes *within* nodal pockets; without backflow, the nodes are those of (3.4).

3.4 Neural correlator W_θ

As with the backflow, the correlator comes in two forms that are compared directly in §7.1: a *separable* network that encodes particle- and pair-features independently and pools them, and a copresheaf *message-passing* (CTNN) network. We describe the separable network first, since it is the baseline and supplies the safe features and analytic cusp shared by both, and then the message-passing one.

3.4.1 Separable correlator (DeepSets baseline)

The separable NQS aggregates *particlewise* and *pairwise* embeddings with permutation-invariant pooling and appends two global statistics, all in $\tilde{R}$. A core safety requirement is that pair channels satisfy $ds_k/d\tilde{r} \to 0$ as $\tilde{r} \to 0$ so that W_θ never injects spurious $1/\tilde{r}$ terms into $\nabla \log \Psi$ or $\Delta \log \Psi$.

Particle branch (DeepSets).

This part of the network is typically referred to as Deep Sets [31]. The permutation invariance of quantum many-body mechanics is naturally captured by this architecture. A neural network ϕ creates a latent embedding for each particle individually. All embeddings are pooled together to form a global representation, which is used later in the readout. Formally, we have

$$\mathbf{h}_i^\phi = \phi(\tilde{\mathbf{x}}_i) \in \mathbb{R}^{d_L}, \qquad \overline{\mathbf{h}}^\phi = N^{-1}\sum_i \mathbf{h}_i^\phi \in \mathbb{R}^{d_L}.$$

Pair branch (soft-core, even channels).

Although the particle branch could theoretically produce pair correlations, in practice it struggles to represent sharp near-field structure. We therefore add a dedicated pair branch that builds features from interparticle distances. This network processes all pairs $(i, j), i \neq j$ individually and pools the resulting latent embeddings. The problem is that raw distances $\tilde{r}_{ij}$ have unbounded derivatives at coalescence, which destabilizes Laplacian evaluations. We therefore build *safe* pair features with bounded first and second derivatives as $\tilde{r} \to 0$. With $\tilde{r}^{\text{soft}} = \sqrt{\tilde{r}^2 + \varepsilon^2}$ ($\varepsilon \in [0.15, 0.30]$ in trap units), we build six "safe" features

$$s_1 = \log\big(1 + (\tilde{r}^{\text{soft}}/\varepsilon)^2\big), \quad s_2 = \tfrac{\tilde{r}^2}{\tilde{r}^2+\varepsilon^2}, \quad s_3 = (\tilde{r}^{\text{soft}}/\varepsilon)^2 \exp\big[-(\tilde{r}^{\text{soft}}/\varepsilon)^2\big],$$
$$\text{rbf}_g = \exp(-g\, s_1), \quad g \in \{0.25, 1, 4\}, \qquad \mathbf{u}_{ij} = [s_1, s_2, s_3, \text{rbf}_{0.25}, \text{rbf}_1, \text{rbf}_4] \in \mathbb{R}^6. \tag{3.5}$$

An MLP ψ maps $\mathbf{u}_{ij} \mapsto \mathbf{h}^{\psi}_{ij} \in \mathbb{R}^{d_L}$. A short-range gate

$$\chi(\tilde{r}) = \frac{\tilde{r}^2}{\tilde{r}^2 + r_g^2}, \quad r_g \approx 0.3, \quad \chi(0) = \chi'(0) = 0,$$

suppresses learned responses *at* coalescence; downstream we use $\chi\, \mathbf{h}^{\psi}_{ij}$.

3.4.2 Analytic short-range cusp (shared)

The analytic cusp below is shared by both correlator variants (and is independent of the backflow). Because the safe features are necessary in the pair branch, the NQS cannot represent the Kato cusp condition exactly. We therefore add an analytic cusp term to the exponent in (3.2), written in *physical* coordinates so that its slope at contact is manifestly correct:

$$u_{\sigma\sigma'}(r) = \gamma_{\sigma\sigma'}\, r\, e^{-r/a_{\text{ho}}}, \qquad a_{\text{ho}} = 1/\sqrt{\omega}, \qquad \gamma_{\uparrow\downarrow} = \tfrac{1}{d-1}, \quad \gamma_{\uparrow\uparrow} = \tfrac{1}{d+1}. \tag{3.6}$$

Because the linear prefactor is the physical separation r, the slope at contact is $\partial_r u|_0 = \gamma_{\sigma\sigma'}$, exactly the Kato condition in general d; for $d = 2$ this gives $\gamma_{\uparrow\downarrow} = 1$ and $\gamma_{\uparrow\uparrow} = \frac{1}{3}$, the standard two-dimensional quantum-dot values, derived—including the parallel-spin case, where antisymmetry already forces a linear node—in Appendix B.2. Only the exponential taper carries the trap length a_{ho}, suppressing long-range interference so the NQS need not unlearn cusp structure at medium and long range; in trap units ($\tilde{r} = \sqrt{\omega}\, r$) the same term reads $u = (\gamma_{\sigma\sigma'}/\sqrt{\omega})\, \tilde{r}\, e^{-\tilde{r}}$.

3.4.3 Separable readout

With contributions from the particle and pair branches, we form a global representation for the final readout. In addition to the pooled embeddings $\overline{\mathbf{h}}^{\phi}$ and $\overline{\mathbf{h}}^{\psi}$, we also include two global statistics that help the network gauge overall system size and density. We average pairs by default or apply degree-normalised radial attention

$$w_{ij} = \exp[-(\tilde{r}_{ij}/r_c)^p], \quad \hat{w}_{ij} = w_{ij} \Big/ \sum_{k \neq i} w_{ik}, \quad (r_c, p) = (0.4, 6),$$

then form $g_i = \sum_{j \neq i} \hat{w}_{ij}\, \mathbf{h}^{\psi}_{ij}$ and $\overline{\mathbf{h}}^{\psi} = N^{-1} \sum_i g_i$. Two system-level scalars in trap units,

$$\mathbf{g}(\tilde{R}) = \Big[\tfrac{1}{Nd} \sum_i \|\tilde{\mathbf{x}}_i\|^2,\ \tfrac{2}{N(N-1)} \sum_{i<j} s_1(\tilde{r}_{ij}) \Big] \in \mathbb{R}^2,$$

are concatenated with $\overline{\mathbf{h}}^{\phi}$ and $\overline{\mathbf{h}}^{\psi}$ and mapped to a scalar by a tiny MLP ρ:

$$W_\theta(\tilde{R}) = \rho\Big(\overline{\mathbf{h}}^{\phi} \parallel \overline{\mathbf{h}}^{\psi} \parallel \mathbf{g}(\tilde{R})\Big). \tag{3.7}$$

3.4.4 Message-passing correlator (CTNN)

The message-passing correlator keeps the same node/edge embeddings, safe pair features, and analytic cusp, but replaces the independent pooling of the separable network with the copresheaf transport of

§2.4.4: at each round the endpoint nodes are transported into an edge update and the updated edges are transported back and aggregated onto the nodes,

$$\mathbf{h}_{ij} \mathrel{+}= F_e\big(\mathbf{h}_{ij} \,\|\, \rho_{v\to e}\mathbf{h}_i \,\|\, \rho_{v\to e}\mathbf{h}_j\big), \qquad \mathbf{h}_i \mathrel{+}= F_v\Big(\mathbf{h}_i \,\|\, \operatorname*{Agg}_{j\neq i} w_{ij}^{\mathrm{spin}}\, \rho_{e\to v}\mathbf{h}_{ij}\Big), \tag{3.8}$$

with residual updates and per-round transport maps. Unlike the single-round backflow, the correlator stacks these rounds in a multiscale "V-cycle": $n_\downarrow$ down-pass rounds, a linear coarsening to a bottleneck width, then $n_\uparrow$ up-pass rounds with skip connections back to the down-pass features (defaults $n_\downarrow = n_\uparrow = 2$). A permutation-invariant readout over the final node and edge statistics returns the scalar $W_\theta(\tilde{R})$, exactly as in (3.7). Holding the Slater reference, cusp, and training fixed and swapping only the separable readout for this V-cycle isolates the relational channel analysed in §7.1.

3.5 Backflow

Backflow deforms the coordinates *used only inside the Slater reference* so that the nodal surface can adapt to interparticle correlations:

$$\tilde{R}' \;=\; \tilde{R} + \Delta_\beta(\tilde{R}), \qquad \Psi_{\theta,\beta}(R) \;=\; \mathrm{SD}(\tilde{R}')\, \exp\Big(\sum_{i<j} u_{\sigma_i\sigma_j}(r_{ij}) + W_\theta(\tilde{R})\Big). \tag{3.9}$$

The correlator W_θ and cusp u are evaluated on $\tilde{R}$ to avoid feedback loops in the local energy; only the determinant "sees" $\tilde{R}'$. This choice cleanly separates (i) amplitude reweighting within nodal pockets (handled by $u + W_\theta$) from (ii) nodal *geometry* (handled by backflow).

Design goals. (i) *Stability near coalescence:* messages are built from mollified radii and short-range gates so that $\partial\Delta_\beta/\partial\tilde{r}$ stays bounded; (ii) *Permutation symmetry:* permutation equivariance holds by construction (shared maps and symmetric aggregation); (iii) *Close to identity at initialisation:* a zero-initialised last layer and a positive learnable scale keep $\Delta_\beta \approx 0$ at the start. Rotational symmetry, by contrast, is only *approximate*: the displacement head reads out a d-vector from node features that include the raw coordinates, so the map is not rotation-equivariant by construction, and it is encouraged instead by the geometry-based edge features and by rotationally-augmented sampling. A global translation is not a symmetry of the trap in any case; the zero-mean projection (3.18) is a regulariser that removes a spurious centre-of-mass drift, not an imposed physical symmetry.

We study *two* backflow architectures. Both map the configuration to a per-particle displacement field and share the output bounding, spin masking, and centre-of-mass projection below; they differ only in how a particle's displacement is made to depend on the others. The conventional network is the baseline of the architecture comparison in Chapter 7; the copresheaf CTNN is the production ansatz.

3.5.1 Conventional backflow (single-round messages)

The baseline uses a single round of pairwise messages. For each ordered pair (i, j) with $i \neq j$,

$$\mathbf{m}_{ij} = \phi_\beta\big(\tilde{\mathbf{x}}_i,\ \tilde{\mathbf{x}}_j,\ \mathbf{r}_{ij},\ \tilde{r}_{ij}^{\mathrm{soft}},\ (\tilde{r}_{ij}^{\mathrm{soft}})^2\big) \cdot w_{ij}^{\mathrm{spin}}, \qquad \tilde{r}_{ij}^{\mathrm{soft}} = \sqrt{\tilde{r}_{ij}^2 + \varepsilon_{\mathrm{bf}}^2}, \tag{3.10}$$

with ϕ_β a small MLP and $w_{ij}^{\mathrm{spin}} \in \{0, 1\}$ a spin mask (same-spin only or all pairs). Messages are aggregated to a node state and a node network returns the displacement,

$$\mathbf{m}_i = \operatorname*{Agg}_{j\neq i} \mathbf{m}_{ij}, \qquad \delta\tilde{\mathbf{x}}_i = \psi_\beta\big(\tilde{\mathbf{x}}_i \,\|\, \mathbf{m}_i\big), \qquad \mathrm{Agg} \in \{\mathrm{sum}, \mathrm{mean}, \mathrm{max}\}. \tag{3.11}$$

There is no persistent edge state: each message is recomputed from the raw geometry and discarded after aggregation.

3.5.2 CTNN backflow (copresheaf)

The message-passing architecture instead maintains explicit *node and edge* features and transports information between them through learned linear maps—the copresheaf-style step of §2.4.4. Node features are embedded from the position and a spin scalar, edge features from the relative geometry,

$$\mathbf{h}_i^{(0)} = W_v\,[\tilde{\mathbf{x}}_i,\, s_i], \qquad \mathbf{h}_{ij}^{(0)} = \mathrm{MLP}_e\big(\mathbf{r}_{ij},\, |\mathbf{r}_{ij}|,\, |\mathbf{r}_{ij}|^2\big). \tag{3.12}$$

A node-to-edge transport $\rho_{v\to e}$ feeds the endpoint nodes into an edge update; an edge-to-node transport $\rho_{e\to v}$ returns the updated edges to the nodes, which are refined by a residual node update; a linear head reads out the displacement:

$$\mathbf{h}'_{ij} = \mathrm{MLP}'_e\big(\mathbf{h}_{ij}^{(0)} \,\|\, \rho_{v\to e}\mathbf{h}_i^{(0)} \,\|\, \rho_{v\to e}\mathbf{h}_j^{(0)}\big), \tag{3.13}$$

$$\mathbf{m}_i = \operatorname*{Agg}_{j\neq i} w_{ij}^{\mathrm{spin}}\, \rho_{e\to v}\big(\mathbf{h}'_{ij}\big), \tag{3.14}$$

$$\mathbf{h}_i = \mathbf{h}_i^{(0)} + \mathrm{MLP}_v\big(\mathbf{h}_i^{(0)} \,\|\, \mathbf{m}_i\big), \qquad \delta\tilde{\mathbf{x}}_i = W_\Delta\, \mathbf{h}_i. \tag{3.15}$$

The learned transport maps $\rho_{v\to e}, \rho_{e\to v}$ are the channel whose ablation collapses the backflow at the Wigner crystal (Chapter 7). The backflow uses a single copresheaf round; the message-passing *correlator* of the architecture comparison (`CTNNJastrowVCycle`) iterates the same style of update over a down/up "V-cycle".

3.5.3 Constraints, symmetries and scaling

Both variants take inputs in trap units ($\tilde{\mathbf{x}} = \sqrt{\omega}\,\mathbf{x}$), bound the displacement, and start at the Slater nodes,

$$\Delta_\beta(\tilde{R}) \;=\; \tanh\big(\delta\tilde{X}\big)\cdot s_\beta, \qquad s_\beta = \mathrm{softplus}(s_\beta^{\mathrm{raw}}) > 0, \tag{3.16}$$

with the last linear layer zero-initialised so that $\Delta_\beta \approx 0$ initially and s_β initialised in $[0.02, 0.10]$. This parameterisation stabilises the SR and energy tails and protects the Laplacian.

Near coalescence raw $1/\tilde{r}$ factors make $\nabla\cdot\Delta_\beta$ spiky. We mollify $\tilde{r}_{ij}$ as in (3.10) and optionally gate the message,

$$\chi_{\mathrm{bf}}(\tilde{r}_{ij}) = \frac{\tilde{r}_{ij}^2}{\tilde{r}_{ij}^2 + r_g^2}, \qquad \chi_{\mathrm{bf}}(0) = \chi'_{\mathrm{bf}}(0) = 0, \quad r_g \approx 0.3, \tag{3.17}$$

or use degree-normalised radial attention $\hat{w}_{ij} = w_{ij}/\sum_{k\neq i} w_{ik}$ with $w_{ij} = \exp[-(\tilde{r}_{ij}/r_c)^p]$. Either keeps the learned response smooth at contact and limits the backflow's contribution to the local-energy variance. The mollification scale $\varepsilon_{\mathrm{bf}} = 10^{-6}$ is fixed, not tuned, and is a numerical guard rather than a modelling choice: $\sqrt{\tilde{r}^2 + \varepsilon_{\mathrm{bf}}^2}$ differs from $\tilde{r}$ by less than $\varepsilon_{\mathrm{bf}}$ everywhere, so it changes no value the network can resolve and exists only to keep $\nabla\tilde{r}$ finite at exact coincidence. The physical short-range scale is instead the pair-feature softening $\varepsilon \in [0.15, 0.30]$ of (3.5), four to five orders of magnitude larger.

Permutation symmetry is automatic. To remove a spurious centre-of-mass drift the flow is projected to zero mean,

$$\Delta_\beta \;\leftarrow\; \Delta_\beta \;-\; \frac{1}{N}\sum_{i=1}^{N}\Delta_{\beta,i}, \tag{3.18}$$

taken over particles within each configuration, not across the batch. Messages are built from relative vectors $\mathbf{r}_{ij}$, but the map is not translation-invariant or rotation-equivariant by construction, since the displacement head reads a d-vector from absolute coordinates. The two variants apply the projection at different points: the copresheaf network applies (3.18) inside its forward pass, while the conventional network returns the raw displacement and the caller projects when $\tilde{R}'$ is formed. The shifted coordinates are identical either way, but a diagnostic evaluated on a network's *returned* field sees a centre-of-mass component in one case and not the other, which is why the two rank columns of §7.1.2 sit under ceilings differing by two.

Because only the determinant is evaluated at $\tilde{R}'$, backflow moves the *nodes* while the correlator exponent stays on $\tilde{R}$, and no Jacobian term arises. This Feynman–Cohen construction [41] targets fixed-node error directly; its modern quantum-Monte-Carlo form is due to Holzmann and Ceperley [42]. Compute and memory scale as $O(BN^2(d+H))$ for batch size B and hidden width H, which is what limits N in practice.

3.6 Architecture sizes and hyperparameters

Table 3.1 collects the representative settings. The two comparisons in Chapter 7 stand on different footings in this respect.

The *correlator* comparison is capacity-controlled. The two variants are held at comparable parameter counts, so a difference in outcome cannot be attributed to one simply being larger, and where §7.1.1 departs from matched capacity it does so in the *separable* network's favour, scaling it from 20k to 164k parameters against a fixed 80k message-passing network without closing the gap.

The *backflow* comparison is not. At equal width the copresheaf network is intrinsically the heavier object—it carries a persistent edge state where the conventional network carries only a per-particle aggregate—so at the shared production setting (message width 128, node MLP 128×3) it holds 182k parameters against 51k, a factor of 3.6. No matched backflow pair was trained. What separates architecture from capacity there is therefore not a matched control but the conventional network's own behaviour across ω, which §7.1.2 sets out where the measurement is reported. All settings below are those of the accompanying code the accompanying code.

Table 3.1: The three sets of models trained in this thesis. They exist for different purposes and are never compared with one another. Widths are hidden dimensions; "rounds" are message-passing steps ($n_\downarrow$+$n_\uparrow$ for the V-cycle). Within the architecture study the correlator variants are not capacity-matched, which is why §7.1.1 also reports the matched 89.4k separable network and the full 20k–164k ladder; the backflow variants are not matched either, and cannot be at equal width, for the reason given above. Settings follow the accompanying code (`scripts/run_depth_analysis.py`, `scripts/source_state_check.py`).

Programme	Correlator	Backflow	Total
Physics production (§6.1)	separable; width 128, 2 layers; 54k	conventional; msg 128, node 128×3; 51k	105k
	(as above)	message-passing; same widths; 182k	236k
Architecture study (§7.1)	separable; pair-MLP 64×4, out 32; 47.7k	*as production; not matched*	—
	message-passing; node/edge 32, bottleneck 16, 2+2 rounds; 79.8k		—
Collocation (§7.3)	message-passing V-cycle; 25k	none	25k
	(as above)	copresheaf; h_{bf}=128; +24k	49k

Readout for every correlator: hidden 64, 2–3 layers, SiLU. Optimiser: Adam at $\mathrm{lr} = 5 \times 10^{-4}$ (backflow) and 5×10^{-5} (Jastrow); CG-SR with Tikhonov damping λ, a trust-region cap and CG restarts. Collocation batch $n_{\mathrm{coll}} \sim 3 \times 10^3$ with oversampling $m = 8$–16; the collocation families are described further in §4.5.3.

Implementation correspondence. For replication: the separable correlator is `PINN` and the message-passing one `CTNNJastrowVCycle`; the conventional backflow is `BackflowNet`, whose ϕ_β and ψ_β are its message and node MLPs, and the copresheaf backflow is `CTNNBackflowNet`, whose embeddings, transport maps and update MLPs realise (3.13)–(3.15). Aggregation is selectable (`sum|mean|max`); the bounded output of (3.16) is `out_bound="tanh"` with $s_\beta = \mathrm{softplus}$(`bf_scale_raw`). The comparison tiers are produced by `scripts/run_depth_analysis.py` and the production widths by `scripts/source_state_check.py`. Schedule and solver options carry the names `alpha_start`, `alpha_end`, `alpha_decay_frac`, `max_param_step`, `damping`, `micro_batch` and `grad_clip`.

Chapter 4

Optimisation

4.1 Primary objective and two–stage training

Given a parameterised wavefunction Ψ_θ, the variational ground–state energy is

$$E(\theta) = \frac{\langle \Psi_\theta | H | \Psi_\theta \rangle}{\langle \Psi_\theta | \Psi_\theta \rangle} = \mathbb{E}_{\tilde{R}\sim\pi_\theta}\big[E_L(\tilde{R})\big], \qquad \pi_\theta(\tilde{R}) = \frac{|\Psi_\theta(\tilde{R})|^2}{\int |\Psi_\theta|^2}. \tag{4.1}$$

The local energy is evaluated in *physical* coordinates, consistently with the Hamiltonian (3.1),

$$E_L(R) = -\tfrac{1}{2}\sum_{i=1}^{N}\Big[\Delta_{\mathbf{r}_i} \ln \Psi_\theta + \|\nabla_{\mathbf{r}_i} \ln \Psi_\theta\|^2\Big] + \tfrac{1}{2}\,\omega^2 \sum_{i=1}^{N} r_i^2 + \sum_{i<j} \frac{1}{r_{ij}}, \tag{4.2}$$

and the zero–variance principle holds: $\mathrm{Var}_{\pi_\theta}[E_L] = 0$ iff Ψ_θ is an eigenstate. The network *features* are computed in trap units $\tilde{\mathbf{r}} = \sqrt{\omega}\,\mathbf{r}$ for conditioning, but the local energy itself is formed in physical units, so no ω factors are absorbed into the kinetic, trap, or Coulomb terms. (In trap units the same operator reads $E_L = \omega\big[-\tfrac{1}{2}\sum_i(\Delta_{\tilde{\mathbf{r}}_i} \ln \Psi_\theta + \|\nabla_{\tilde{\mathbf{r}}_i} \ln \Psi_\theta\|^2) + \tfrac{1}{2}\sum_i \tilde{r}_i^2\big] + \sum_{i<j} \sqrt{\omega}/\tilde{r}_{ij}$; the kinetic and trap terms scale by ω and the Coulomb term by $\sqrt{\omega}$.)

Our optimisation follows two stages that mirror the implementation:

1. **Residual–based pretraining (stage I).** We fit the residual of E_L to a moving scalar target E_{eff} using the loss $\mathcal{L}_{\mathrm{res}}(\theta) = \mathbb{E}\big[(E_L(\tilde{R}) - E_{\mathrm{eff}})^2\big]$. We begin with a *self–target* (stabilizer) $E_{\mathrm{eff}} = \mu$ where $\mu = \mathbb{E}[E_L]$ over the current batch. This minimises the *sample* variance of E_L under the current collocation measure—which is not the same object as $\mathrm{Var}_{|\Psi|^2}(E_L)$ unless the two measures coincide—and prevents the network from chasing a potentially inaccurate external target before it can represent low–variance pockets. We then anneal toward a trusted reference E_{DMC} via

   $$E_{\mathrm{eff}}(\alpha) \;=\; \alpha\, E_{\mathrm{DMC}} + (1-\alpha)\,\mu, \qquad \alpha \in [0,1], \tag{4.3}$$

   with a smooth cosine schedule $\alpha = \alpha(t)$ that ramps from α_{start} to α_{end} over a prescribed fraction of epochs. Setting the `objective` to `"residual"` uses $E_{\mathrm{eff}} = \mu$; `"energy"` uses $E_{\mathrm{eff}} = E_{\mathrm{DMC}}$; and `"energy_var"` uses the annealed blend above.

2. **Stochastic Reconfiguration (stage II).** After residual pretraining, we refine with SR (natural–gradient VMC). Writing $\bar{O}(\tilde{R}) = \partial_\theta \log \Psi_\theta(\tilde{R}) - \mathbb{E}_\pi[\partial_\theta \log \Psi_\theta]$ for the centred score,

   $$S \;=\; \mathbb{E}_\pi\big[\bar{O}^\top \bar{O}\big], \qquad g \;=\; 2\,\mathbb{E}_\pi\big[(E_L - \mathbb{E}_\pi[E_L])\,\bar{O}\big], \tag{4.4}$$

   we solve the damped normal equations

   $$(S + \lambda I)\,\Delta\theta \;=\; -\,g, \tag{4.5}$$

   using (preconditioned) conjugate gradients with restarts, a trust–region scale on the final step, and batchwise centering/whitening of O.

This pipeline exploits the zero–variance principle for stable shaping of W_θ (and backflow) before applying the geometry–aware SR update.

4.2 Configuration–space sampling

Guiding principle (permutation invariance). In a many–body fermionic system, *which* particle indices are close is irrelevant; what matters is *how many* close pairs and at which length scales. Consequently, we can design samplers that (i) emphasize *configurational motifs* (clusters, shells, dimers, long tails) and not specific labels, and (ii) deliberately permute particles and randomly rotate configurations to explore the space efficiently without biasing towards any ordering.

4.2.1 Stratified capped–simplex mixture

We draw collocation batches $X \in \mathbb{R}^{B\times N\times d}$ from a five–component mixture,

$$\text{(centre, tails, mixed, shells, dimers)},$$

with nonnegative weights $w \in \Delta^4$ projected to a *capped simplex* $\{\, w : \sum_k w_k = 1,\ 0 \le w_k \le 0.3 \,\}$ to enforce coverage. Each component proposes x as follows (all in physical units; the code internally scales by $a_{\text{ho}} = 1/\sqrt{\omega}$):

- **Centre:** narrow isotropic Gaussian around the trap centre (samples near the density core).
- **Tails:** wide Gaussian (probes low–density outskirts and large–r behaviour).
- **Mixed:** hybrid batches mixing narrow and moderate spreads per configuration (tests robustness to inhomogeneous spreads).
- **Shells:** points placed on K hyperspherical shells with trap–scaled radii $\{r_k\}$ and occupancy $\{q_k\}$, with small radial/tangential jitter (captures ring/shell order at weak traps).
- **Dimers:** we induce a few close pairs per configuration by sampling short interparticle offsets with random directions and log–uniform distances (enforces near–coalescence coverage without singularities).

We then apply a random in–plane rotation (for $d = 2$), and a random permutation of particle indices; both preserve the target expectations by symmetry.

Adaptive mixture and shell occupancy. Per epoch we compute simple difficulty scores and update w and q by exponentiated–gradient (EG) steps with momentum, temperature, and an exploration term; w is then projected back to the capped simplex. Mixture difficulty uses the residual $(E_L - E_{\text{eff}})^2$ per component; shell difficulty uses a proxy

$$\underbrace{V_i}_{\text{harm.+Coulomb per particle}} + \underbrace{\gamma_{g^2}\,\|\nabla_{\mathbf{x}_i}\log\Psi\|^2}_{\text{stiff gradients}} + \underbrace{\gamma_{\text{cusp}}\,r_{\min}^{-1}}_{\text{coalescence pressure}},$$

aggregated by nearest shell index. This shifts probability mass toward underfit regions *without* collapsing coverage thanks to the caps.

Hard–example injection. We maintain a small buffer of previously challenging configurations (based on residuals) and stochastically inject them, with mild multiplicative and additive jitter, into the next batch. This provides a curriculum–like pressure while avoiding overfitting to outliers.

Robustness tweaks. We trim extreme local–energy outliers by quantiles (default 3%) *before* forming losses/updates; we also micro–batch the evaluation to keep memory bounded, and apply gradient clipping to f_θ (and backflow) parameters.

4.3 Residual–based training (stage I)

For a batch $\{\tilde{R}_b\}_{b=1}^{B}$ we compute $E_L(\tilde{R}_b)$ and the batch mean $\mu = \frac{1}{B}\sum_b E_L(\tilde{R}_b)$. The residual loss is

$$\mathcal{L}_{\text{res}}(\theta;\alpha) = \frac{1}{B}\sum_{b=1}^{B}\big(E_L(\tilde{R}_b) - E_{\text{eff}}(\alpha)\big)^2, \qquad E_{\text{eff}}(\alpha) = \alpha\, E_{\text{DMC}} + (1-\alpha)\,\mu. \tag{4.6}$$

- **Self–target warmup ($\alpha = 0$).** Minimising $\mathbb{E}[(E_L - \mu)^2]$ is variance minimisation *under the sampling measure in use*; μ is treated as a detached constant, so no gradient flows through the batch mean. It teaches the network to produce smooth, low–variance amplitudes inside nodal pockets and reduces sensitivity to early sampler noise.
- **Anneal to reference ($\alpha \uparrow$).** As capacity grows, we blend in the external target, gently steering the mean energy while retaining the variance pressure. The cosine ramp avoids abrupt objective switches.
- **Reference-target residual ($\alpha = 1$).** Stage I may end with $E_{\text{eff}} = E_{\text{DMC}}$. Despite its name in the code, this is not variational energy minimisation: it is a squared residual against a fixed scalar, and it can in principle drive $\langle E_L\rangle$ *up* toward a target the ansatz could beat. The variational optimisation is stage II.

In practice we compute $\nabla_\theta \mathcal{L}_{\text{res}}$ by autodiff, use micro–batches, apply optional gradient clipping, and log per–component usage and shell radii in physical units.

Local–energy derivatives in practice. We support three Laplacian backends:

1. **Exact:** chunked second partials (reference–accurate, heavier).
2. **HVP–Hutchinson:** $\Delta \log \Psi = \mathbb{E}_v[v^\top H_{\log\Psi} v]$ with Rademacher v (fast and stable; fallback to FD if a probe is non–finite).
3. **FD–Hutchinson:** centred finite differences of directional gradients (robust in corner cases; slightly noisier).

We average a small number of probes (2–16 in ablations), and trim quantiles before forming losses.

4.4 Stochastic Reconfiguration (stage II)

Stochastic reconfiguration [26, 43] preconditions the energy gradient with the (inverse) quantum geometric tensor, and is the imaginary-time / natural-gradient step for a variational state [24, 27]. It uses the covariance form of the energy gradient

$$\partial_{\theta_k} E = 2\,\mathrm{Cov}_\pi\big(E_L, O_k\big) = 2\Big(\langle E_L O_k\rangle - \langle E_L\rangle\,\langle O_k\rangle\Big), \tag{4.7}$$

and approximates a natural–gradient step by solving $(S + \lambda I)\Delta\theta = -g$ with the quantum geometric tensor S of §2.2.3. Implementation details:

- **Centering/whitening.** We centre O per batch and whiten by the diagonal of S before CG to improve conditioning.
- **Trust region.** The final step is scaled to satisfy a maximum parameter norm.
- **Damping.** A small λ stabilises low–variance directions.

Because our backflow only alters the Slater coordinates (and W_θ stays on safe features of $\tilde{R}$), the SR statistics remain well–behaved even when nodes shift.

Optimiser variants compared. Three first-order updates appear in the results, and their comparison is itself a finding (§7.2):

- **Adam** [22]: a diagonal adaptive step, used as the robust default, with separate learning rates for the Jastrow and backflow blocks.
- **Plain SR**: the natural-gradient step above with a fixed diagonal damping λ.
- **CG-SR**: the same step solved matrix-free by preconditioned conjugate gradients, with Tikhonov damping and a trust-region cap. It is most useful where S is anisotropic but still statistically resolved. Whether that condition holds, and what follows when it does not, is §7.2.

Solving the SR step without forming S is what makes it affordable at $\mathcal{O}(10^5)$ parameters. The same obstacle is addressed from the other side by the minSR and kernel-formulation reductions, which exploit the fact that the number of samples, rather than the number of parameters, sets the true rank of the problem [44, 45]. Whether the extra cost buys anything over a diagonal preconditioner is measured in §7.2.

4.5 Sampling paradigms and the weak-form collocation objective

The alternative pipeline described in this section draws its batches from an analytic proposal rather than a Markov chain—the SR route of §4.4 samples $|\Psi_\theta|^2$ by Metropolis—so the choice of *sampling measure* must be stated explicitly: the results treat it as a variable in its own right. Two paradigms appear in this thesis. In *variational Monte Carlo* (VMC) the batch is drawn from $|\Psi_\theta|^2$ itself, usually by a Metropolis chain, so the estimators of the QGT and of the gradient share the very metric they are meant to precondition. In *collocation* the batch is drawn from a fixed analytic proposal $q(\mathbf{R})$ and reweighted to $|\Psi_\theta|^2$ by importance weights $w_i = |\Psi_\theta(\mathbf{R}_i)|^2/q(\mathbf{R}_i)$, so that the gradient training requires no Markov chain. A Metropolis probe is used only outside the gradient loop—for periodic checkpoint selection and for the final energy certification.

For the collocation objective we use a *weak form*: the local energy E_L enters only as a reward in a score-function (REINFORCE) gradient [46],

$$\nabla_\theta \langle E \rangle = 2\,\mathbb{E}_{\mathbf{R}\sim|\Psi_\theta|^2}\big[(E_L(\mathbf{R}) - b)\,\nabla_\theta \log|\Psi_\theta(\mathbf{R})|\big], \tag{4.8}$$

with a variance-reducing baseline b (a running mean of E_L), and the Laplacian inside E_L is never differentiated through. This is deliberate. Differentiating the Laplacian through a backflow-shifted determinant is the source of the nodal instability analysed in Appendix C; the weak form keeps the full local-energy signal as a reward while sidestepping that pathology. Two robustness controls stabilise the reward: local energies are clipped to a robust median/MAD location–scale band ($\sim 5\sigma$), and the most extreme rewards are quantile-trimmed before the gradient is formed.

4.5.1 Analytic proposal and importance resampling

The collocation points are drawn from an analytic mixture of isotropic Gaussians centred at the trap minimum and reweighted to $|\Psi_\theta|^2$ [16],

$$q(\mathbf{R}) = \frac{1}{K}\sum_{k=1}^{K} \mathcal{N}\big(\mathbf{R};\, 0,\, \sigma_{f,k}^2/\omega\; I_{Nd}\big), \quad \sigma_f \in \{0.8, 1.3, 2.0\}, \qquad w_i = \frac{|\Psi_\theta(\mathbf{R}_i)|^2}{q(\mathbf{R}_i)}, \tag{4.9}$$

where the $1/\omega$ scaling ties the proposal width to the oscillator length $\ell = 1/\sqrt{\omega}$ and the three components cover core, bulk, and tails. A batch of n_{coll} points is obtained by multinomial resampling on $p_i = w_i/\sum_j w_j$ from $m\,n_{\text{coll}}$ candidates ($m \approx 8$–16). Reweighting quality is tracked by the effective sample size $\text{ESS} = (\sum_i w_i)^2/\sum_i w_i^2$ and, as a heavier-tailed diagnostic, the Pareto shape parameter $\hat{k}$ of the weights [47]. The second is not a second opinion on the first. Importance weights are heavy-tailed, and $\hat{k}$ estimates the tail index of the generalised Pareto distribution fitted to the largest of them. The thresholds are statements about the moments of the *fitted* tail. For $\hat{k} < 1/2$ the weights behave as though

the variance were finite and the reweighted estimator obeys a central limit theorem. Between 1/2 and 1 the variance behaves as infinite though the mean does not, and convergence is slow enough that $\hat{k} \approx 0.7$ is the practical limit of trustworthiness. Above 1 the fitted tail has no finite mean.

That last statement needs care, because in this setting the underlying ratio does have a finite mean by construction: $w = |\Psi_\theta|^2/q$ with q a proper density gives $\mathbb{E}_q[w] = \int |\Psi_\theta|^2 < \infty$ whenever Ψ_θ is square-integrable, so asymptotically $\hat{k} < 1$. A measured $\hat{k} > 1$ is therefore a *finite-sample* diagnostic: it says the sample is deep in a pre-asymptotic regime where the realised estimator behaves as if no finite mean existed, not that the expectation fails to exist [47]. We use it in that sense throughout. An ESS can look survivable while $\hat{k}$ says the estimator has no variance to speak of, which is why we track both. Both degrade when q ceases to overlap $|\Psi_\theta|^2$, and that degradation is the failure mode which bounds the paradigm at large N and weak ω (Chapter 7, §7.3.4). Three controls manage the tails: weight tempering ($w \to w^\alpha$, $\alpha < 1$), log-weight clipping at an upper quantile, and ESS-adaptive oversampling that raises m when the achieved ESS falls below a floor. The first two are deliberate bias–variance trades: they change the distribution the reward is averaged over, so the reweighted energy readout is no longer an estimate of $\langle E_L \rangle_{|\Psi_\theta|^2}$. That is why the final certification is always a separate evaluation under $|\Psi_\theta|^2$ (§7.2). An instability-rollback reverts to the last good checkpoint and decays the learning rate if the energy jumps beyond a threshold.

4.5.2 Physics-informed proposal for the Wigner regime

The origin-centred proposal (4.9) places its mass where a Wigner molecule has almost none: at strong correlation the density collapses onto a shell at the classical ring radius r_0, and $|\Psi_\theta|^2$ has essentially no support near the origin, so the ESS collapses. We therefore replace q by a proposal built from the Wigner shell itself—a radial Gaussian centred at r_0 (set by balancing confinement against Coulomb repulsion) times an angular distribution over the ring, so that proposed configurations resemble electrons spread around the ring instead of clustered at the centre. Where the ring itself develops angular structure the uniform angular factor is replaced by a Dirichlet model of the angular gaps. Difficult confinements are reached by *cascade warm-starting*: parameters converged at one ω initialise the next, following $\omega : 1.0 \to 0.5 \to 0.1 \to 0.01 \to 0.001$. What this construction achieves, and where it stops, is §7.4.

4.5.3 Ansatz families used in the collocation programme

Three ansatz families are trained under this objective in Chapter 7, all on the same Slater–Jastrow core $\Psi(\mathbf{R}) = \det[\phi_i(\mathbf{r}_j + \Delta\mathbf{r}_j)]\, e^{J(\mathbf{R})}$ and differing only in what dresses it. They are deliberately smaller than the production networks of §3.6, because this programme is limited by gradient noise and memory, not by capacity.

Jastrow-only. The correlator alone at ∼25k parameters, in its message-passing form (`CTNNJastrowVCycle`), acting on node features (single-particle orbital values) and edge features (soft-core pair distances). It is used both standalone and as the initialiser for the backflow runs.

Backflow + Jastrow. Adds the copresheaf displacement network (`CTNNBackflowNet`), ∼49k parameters in total at h_{bf}=128. Its edge tensor scales as $\mathcal{O}(N^2 h_{\mathrm{bf}})$, which is the memory bottleneck at large N: at N=20 the hidden width must be cut to $h_{\mathrm{bf}} \in \{48, 64\}$.

Neural Pfaffian. Replaces the determinant with $\mathrm{Pf}[A_{ij}]$, a more general antisymmetric form admitting richer nodal topology [48].

4.6 Units, scaling, and batch construction

All features are computed in trap units $\tilde{\mathbf{x}} = \sqrt{\omega}\,\mathbf{x}$ so that length scales and gates (e.g. the short–range gate in the NQS and backflow) behave consistently across ω. The sampler internally widens/narrows Gaussians and shell radii using $a_{\mathrm{ho}} = 1/\sqrt{\omega}$ in a fixed, explicit way, so that the five components cover comparable *dimensionless* regimes for $\omega \in [0.01, 1]$. Proposal candidates are drawn independently, so the batch carries no serial Markov-chain autocorrelation; the retained batch is not i.i.d. in the strict sense, since multinomial resampling can repeat candidates and the hard-example buffer reinjects earlier ones. What the design buys is the absence of a mixing time, not statistical independence.

4.7 The two workflows

Both routes start from the same ansatz and end at an energy evaluated under $|\Psi_\theta|^2$. They differ in the measure that defines the loss in between.

SR–VMC (§4.3–§4.4)	**Collocation (§4.5)**
stratified mixture → residual pretraining	pretrained checkpoint
↓	↓
$\lvert\Psi_\theta\rvert^2$ sampling (Metropolis)	analytic proposal q, importance reweighting
↓	↓
SR step $(S + \lambda I)\Delta\theta = -g$	weak-form REINFORCE gradient (Adam)
↓	↓
energy under $\lvert\Psi_\theta\rvert^2$	Metropolis probe for checkpoint choice, then energy under $\lvert\Psi_\theta\rvert^2$

The right-hand route places no Markov chain in any gradient; the probe that selects its checkpoints sits outside the update loop entirely. The left-hand route, which carries the ground-state energies of §6.1, keeps the early dynamics variance-dominated and sampler-aware and then transitions to a geometry-aware natural-gradient refinement—a division that matters most at weak traps, where the shell and dimer coverage of the stratified mixture is what makes the early stage converge at all.

4.8 Statistical uncertainties

Every energy and observable we report carries a statistical error, and two distinct sources contribute. Within a single evaluation, successive MCMC samples are autocorrelated, so a naive standard error over B samples understates the sampling uncertainty; this is what blocking addresses. Across training runs, the optimisation is itself stochastic and lands on different states from different seeds; blocking says nothing about that, and it is assessed separately by repeating runs at independent seeds wherever the claim depends on it (§7.3.4, §7.1.1). We therefore estimate the error on a scalar mean $\bar{A} = \frac{1}{B}\sum_i A(\mathbf{R}_i)$ by *blocking*: successive samples are averaged into blocks of growing size and the plateau of the block standard error as the block size exceeds the autocorrelation time is taken as the uncertainty [49, 50]. For the final ground-state energy this is the dominant reported error; a parenthetical digit, e.g. 155.8738(7), denotes one standard error on the last quoted figure.

For quantities that are *nonlinear* functions of sample statistics (participation ratios, effective ranks, PCA cosines, and the coupling diagnostics of Chapter 6) a blocked standard error is not available in closed form, and we use a *block bootstrap* instead: blocks (of the plateau size above) are resampled with replacement, the statistic is recomputed on each resample, and its spread across resamples is the reported error (§5.2.3). Where a number is quoted as a seed average, we additionally report the spread across independent random seeds, which captures optimisation variability and not only sampling variability; the two are distinguished in the captions. Reference energies drawn from the literature carry the uncertainties quoted by their sources.

Chapter 5

Analysis

This chapter describes *what* we measure on fresh $|\Psi_\theta|^2$ samples, *how* we compute the diagnostics, and *why* each quantity is informative. Results and numbers are presented in Chapter 6.

5.1 Structural diagnostics and Wigner markers

Unless stated otherwise, curves are accumulated in *trap units* $\mathbf{y} = \sqrt{\omega}\,\mathbf{x}$ to compare shapes across ω, while all headline distances are reported in *Bohr* (physical units $\mathbf{x}$). Production chains are thinned; we split samples into halves for stability checks (acceptance window, JSD between halves, and reproducibility under restarts). Using trap units to compare shapes across confinements is standard in parabolic dots; see the overview in [4].

5.1.1 Pair distribution and radial statistics

We estimate an *effective* pair distribution—a shell-area-corrected pair-distance density—in trap units. For a finite trapped dot there is no homogeneous-system normalisation, because the one-body density varies strongly across the cloud, so A_{eff} below is a finite-dot convention rather than a thermodynamic volume and $g(r)$ does not tend to unity at large r. Peak positions and relative profiles, which is what the analysis uses, are insensitive to that choice; absolute amplitudes are not.

$$g(r_\alpha) \;=\; \frac{A_{\text{eff}}}{N(N-1)}\,\frac{\left\langle \sum_{i<j} \mathbf{1}\{r_{ij} \in [r_\alpha^-, r_\alpha^+)\}\right\rangle}{2\pi r_\alpha\,\Delta r_\alpha}, \qquad r_{ij} = \|\mathbf{y}_i - \mathbf{y}_j\|_2, \tag{5.1}$$

with $A_{\text{eff}} = \pi R_{0.95}^2$ and $R_{0.95}$ the 95% quantile of single-particle radii in trap units. This $g(r)$-based structural analysis follows established practice in mesoscopic 2D electron systems and quantum dots [1, 4, 11]. The pair-distance probability is therefore denoted $P_{\text{pair}}(r_{ij}) \propto r_{ij}^{d-1} g(r_{ij})$. It is used for pair-structure diagnostics, distinct from the one-body radial probability $P_1(r) \propto r\rho_1(r)$, from which we report the headline mode r_{mode}, mean $\bar{r}$, standard deviation σ_r, and FWHM (all in Bohr). A compact one-body width-to-scale marker is

$$\gamma \;=\; \sigma_r / r_{\text{mode}}, \tag{5.2}$$

which tracks the width of P_1 relative to its modal radius. It is interpreted alongside shell separation and pair/angular diagnostics, rather than as a universal monotone Wigner marker [1, 11].

5.1.2 Automatic shell detection and topologies

For each frame we sort the single-particle radii $s_1 \le \cdots \le s_N$ (trap units), compute gaps $g_k = s_{k+1} - s_k$, and flag the frame as *two-shell* if the largest gap exceeds a robust threshold:

$$\max_k g_k \;\ge\; \tau\,\text{median}(g_k), \qquad \tau \in [2.5, 3.5] \text{ (default 3.0)}. \tag{5.3}$$

The cut radius is the midpoint of the maximal gap; the inner multiplicity n_{in} defines the topology $(n_{\text{in}}, N - n_{\text{in}})$.

More than two shells. At N=20 and weak confinement a single cut is not enough, and the detector is applied recursively. Having placed one cut, we re-apply the same robust-gap test *within* each resulting group, accepting a further cut only if that group contains at least three particles and its own maximal gap again exceeds $\tau\,\mathrm{median}(g_k)$ computed over the whole frame—so the threshold is global and a shell is never split by a gap that is small on the scale of the configuration. Recursion stops when no group qualifies, which for the systems here never yields more than three shells. A frame is labelled by the resulting occupancy vector $(n_1, \ldots, n_S)$ ordered outward, so that the two-shell case above is the S=2 member of the same family; the N=20 deep-Wigner topology $(1, 7, 12)$ is an S=3 assignment with cuts after the first and eighth particle. Topology fractions and their robustness under τ are reported exactly as in the two-shell case. Shelling and discrete ring occupancies are well known in parabolic Coulomb clusters and quantum dots [2–4]; our detector provides a reproducible, data-driven assignment in that spirit. We report: (i) the fraction of two-shell frames, (ii) the histogram over n_{in} among two-shell frames, and (iii) for selected topologies (e.g. $(1, 5)$ or $(3, 9)$) the fraction among *all* frames.

Shell-resolved reconstruction of $g(r)$. On two-shell frames we decompose pairs into II, IO, and OO and accumulate histograms $g_{\mathrm{II}}, g_{\mathrm{IO}}, g_{\mathrm{OO}}$. With empirical weights $w_{\mathrm{II}}, w_{\mathrm{IO}}, w_{\mathrm{OO}}$ (combinatorial counts) we reconstruct $\tilde{g}(r) = w_{\mathrm{II}} g_{\mathrm{II}} + w_{\mathrm{IO}} g_{\mathrm{IO}} + w_{\mathrm{OO}} g_{\mathrm{OO}}$ and report $\cos\angle(g, \tilde{g})$. Values ≈ 1 indicate that the observed $g(r)$ shape is explained by shell geometry, not sampling artefacts, in line with shell-based descriptions in [2–4].

5.1.3 Orientational order and "ring stiffness"

On each detected ring (inner or outer) we take the particle angles $\{\phi_k\}_{k=1}^{n}$ in the lab frame. No rotation is removed at this stage: the modulus $|\Phi_m|$ below is already invariant under a global rotation, and the registration of Eq. (5.7) is applied only where the *phase* is needed, which is the cross-shell comparison. Bond-orientational order is

$$\Phi_m \;=\; \frac{1}{n}\sum_{k=1}^{n} e^{\,\mathrm{i}m\phi_k}, \qquad m = n, \tag{5.4}$$

so that $|\Phi_m| \in [0, 1]$ measures n-fold orientational order (e.g. m=5 for $(1, 5)$, m=9 for the outer ring of $(3, 9)$), analogous to standard bond–orientational order parameters used in 2D Coulomb/Yukawa crystals [14]. As a complementary, dimensionless *angular Lindemann* number we use

$$\lambda_\phi \;=\; \frac{\mathrm{std}(\Delta\phi)}{\mathbb{E}[\Delta\phi]}, \qquad \Delta\phi \;=\; \text{nearest-neighbour angular spacing on the ring.} \tag{5.5}$$

Smaller λ_ϕ indicates a stiffer (more Wigner-like) ring. This is the quantity abbreviated "Lind" in Chapter 6; where a shell index is attached, as in λ_{in} and λ_{out}, it is the same estimator evaluated on that shell alone.

Registered and lab-frame order. Two further scalars are used for the multi-shell cases, and both are needed because $|\Phi_{n_s}|$ is defined shell by shell while the question at N=20 is whether the *molecule* is ordered. For a frame with shells $s = 1, \ldots, S$ and occupancies n_s we define a configuration-level order parameter in two frames. In the *lab* frame,

$$C_{\mathrm{global}} \;=\; \Big|\,\Big\langle \tfrac{1}{S'}\textstyle\sum_{s:\,n_s\geq 3}\Phi_{n_s}\Big\rangle_{\mathrm{frames}}\Big|, \tag{5.6}$$

the modulus of the *frame-averaged* complex order parameter, averaged over the S' shells carrying at least three particles. Because a freely rotating molecule visits all orientations, the complex phases cancel and $C_{\mathrm{global}} \to 0$ however ordered the molecule is internally: it measures *pinning* to the lab frame, not order. In the *co-rotating* frame each configuration is first registered by an estimated physical rotation. Because a rigid rotation by α sends $\Phi_{n_s} \mapsto e^{\mathrm{i}n_s\alpha}\Phi_{n_s}$, the phase of a shell's order parameter advances at n_s times the rotation angle, and the angle itself must be recovered from the outermost shell as $\hat{\alpha} = \arg\Phi_{n_S}/n_S$ (modulo $2\pi/n_S$) before it can be applied to the others:

$$\tilde{C}_{\mathrm{global}} \;=\; \Big|\,\Big\langle \tfrac{1}{S'}\textstyle\sum_{s:\,n_s\geq 3}\Phi_{n_s}\,e^{-\mathrm{i}n_s\hat{\alpha}}\Big\rangle_{\mathrm{frames}}\Big|, \qquad \hat{\alpha} = \tfrac{\arg\Phi_{n_S}}{n_S}. \tag{5.7}$$

Dividing the outer phase by n_S is essential: removing $\arg\Phi_{n_S}$ from every shell alike would only correspond to a common spatial rotation when all occupancies coincide, and would otherwise mix shells that rotate

together into an apparent misalignment. This retains the *relative* angular arrangement of the shells while discarding the global orientation, so a rigid molecule tumbling freely gives $\tilde{C}_{\mathrm{global}}$ near its single-frame value and C_{global} near zero. The pair $(\tilde{C}_{\mathrm{global}}, C_{\mathrm{global}})$ is therefore the direct test of the rotating Wigner-molecule picture, and it is reported that way in §6.2.5. Finally, $L_{n,\mathrm{reg}}$ denotes the angular Lindemann number λ_ϕ evaluated on the shell of occupancy n after the same registration.

Two of these quantities are insensitive to the registration convention and one is not, and the distinction bounds what may be concluded from each. The shellwise modulus $|\Phi_{n_s}|$ and the nearest-neighbour spacing statistics behind λ_ϕ are invariant under any global rotation, so they carry no dependence on how $\hat{\alpha}$ is estimated. The *cross-shell* quantity $\tilde{C}_{\mathrm{global}}$ does depend on it, since its whole content is the relative alignment between shells of different occupancy. The N=20 values reported in §6.2.5 were produced by an earlier analysis pass whose code has not been retained, so we cannot confirm which convention it used; the per-shell numbers there are unaffected, but the cross-shell $\tilde{C}_{\mathrm{global}}$ should be recomputed under Eq. (5.7) before it is relied on quantitatively.

The random-angle null. None of these is interpretable without a baseline. For n particles at independent uniform angles, $n\Phi_n$ is a two-dimensional random walk of n unit steps, so $\langle|\Phi_n|^2\rangle = 1/n$ exactly, and the mean modulus follows from Kluyver's integral,

$$\big\langle|\Phi_n|\big\rangle_{\mathrm{random}} = \frac{1}{n}\int_0^\infty \frac{1 - J_0(t)^n}{t^2}\,dt, \tag{5.8}$$

which we evaluate numerically. The familiar Rayleigh form $\sqrt{\pi}/(2\sqrt{n}) \approx 0.886/\sqrt{n}$ is its large-n limit and undershoots it at the ring sizes that occur here, by 2.6% at $n = 3$, 1.3% at $n = 5$ and 0.4% at $n = 19$—the same size as the effects being tested at strong confinement, which is why the exact value is used throughout. Because shells are assigned by radial rank alone and no angular cut is applied, it is also exactly the null of our pipeline: redrawing every angle uniformly while keeping each frame's radii reproduces it within its bootstrap error in all fifteen (N, ω) cells (Appendix F.2). A five-particle ring with no angular order whatever returns $|\Phi_5| \approx 0.40$. We therefore report $|\Phi_n|$ together with its ratio to this null wherever the ordering claim depends on it (Appendix F.2). This is a Lindemann–type orientational metric tailored to finite rings, motivated by the use of Lindemann criteria to diagnose intershell rotational (angular) melting in mesoscopic Coulomb clusters and dots [3, 11].

5.1.4 Scaling check for $N = 2$

To connect with the classical strong-coupling limit we fit

$$r_{\mathrm{mode}}(\omega) \;=\; C\,\omega^{\alpha} \tag{5.9}$$

by linear regression in $\big(\log\omega, \log r_{\mathrm{mode}}\big)$ and compare the slope with the classical reference $\alpha_{\mathrm{cl}} = -2/3$ (two charges in a harmonic trap at strong coupling; see, e.g., the reviews in [4]). Approach toward α_{cl} as $\omega \downarrow$ is one of our Wigner crossover indicators, consistent with the crossover phenomenology reported in [1, 11].

5.1.5 Reporting and stability checks

For all scalars and histograms we report standard errors calculated over thinned blocks or frames. We verify: (i) acceptance in the target window, (ii) Jensen–Shannon divergence and ℓ_2 distance between halves of production (stationarity), (iii) robustness of topology fractions and $|\Phi_m|$ under $\tau \in [2.5, 3.5]$ and mild pre-smoothing of radii, and (iv) reproducibility of $g(r)$ and $P(r)$ under independent sampler restarts. All diagnostics operate directly on saved bundles $\{\mathbf{X}_{\mathrm{Bohr}}, r, g(r)\}$.

5.2 Representation analysis

We probe what the networks learn by evaluating fixed, trained models on fresh $|\Psi_\theta|^2$ samples. Unless stated otherwise, features are computed in *trap units* $\mathbf{y} = \sqrt{\omega}\,\mathbf{x}$ to match the correlator's internal scaling. All estimates use thinned MCMC frames, split into two equal blocks A and B; PCA fits on A and reports on B (and vice versa) to avoid circularity. We use `float64` throughout.

Feature taps. Let $z_\theta \in \mathbb{R}^D$ denote the concatenated correlator representation

$$z_\theta = \big[\overline{\phi} \mid \overline{\psi} \mid \mathbf{g}\big],$$

where $\overline{\phi}$ are per-particle summaries, $\overline{\psi}$ are pairwise summaries, and $\mathbf{g}$ are global/extras. The readout ρ_θ is a two-layer MLP, not a linear map, so

$$f_{\text{base}}(z) = \rho_\theta(z), \qquad f = f_{\text{base}} + f_{\text{cusp}}, \tag{5.10}$$

and the ablations below re-evaluate ρ_θ on modified inputs instead of manipulating a weight vector. Backflow returns a displacement field $\Delta x \in \mathbb{R}^{N\times d}$ that enters only the Slater determinant; the correlator and cusp are evaluated on the unshifted $\tilde{R}$ (§3.5). The coupling diagnostics below additionally probe the correlator on the shifted coordinates as a counterfactual sensitivity.

5.2.1 Representation geometry

Preprocessing and PCA. Given a sample matrix $Z \in \mathbb{R}^{K\times D}$ (rows are frames), we *mean-centre* the columns, $Z_c = Z - \mu$, and take the SVD of Z_c to obtain orthonormal loadings $U = [u_1, \dots, u_D]$; scores are $S = Z_c U$. Centring but not rescaling is deliberate: the readout ρ_θ was trained on the raw features, so every ablation below is performed in raw coordinates and the mean is restored before ρ_θ is applied. Where a per-column standardisation is used—for the branch-power decomposition of §5.2.2, which compares loadings across branches carrying different natural scales—it is stated explicitly. PCA is fitted on block A and reported on the held-out block B unless noted.

Entropy effective rank. Let $\{\lambda_i\}_{i=1}^{D}$ be the eigenvalues of $\mathrm{Cov}(Z_c)$ (on the held-out block). With $p_i = \lambda_i / \sum_j \lambda_j$, the *entropy effective rank* is

$$r_{\text{eff}}(Z) \;=\; \exp\Big(-\sum_{i=1}^{D} p_i \log p_i\Big),$$

which equals 1 for perfectly one-dimensional representations and increases with spectral spread. Two spectral counts appear in this thesis and they are not the same number. $r_{\text{eff}} = \exp(H_1)$ above is the exponential of the Shannon entropy of the spectrum; the *participation ratio* $d_{\text{eff}} = (\sum_a \lambda_a)^2 / \sum_a \lambda_a^2$ of §2.2.3 is the exponential of the Rényi-2 entropy. Since Rényi entropies decrease in their order, $d_{\text{eff}} \le r_{\text{eff}}$ always, with equality only for a flat spectrum. Appendix E reports r_{eff} for feature covariances and Chapter 7 reports d_{eff} for the tangent metric and for the displacement field, so the same object can carry two slightly different counts in the two chapters; each table names which is used.

5.2.2 Probes and ablations

Principal-component ablations of the head. To test how many latent axes the readout truly uses, we project the centred features onto the top-k PCs and restore the mean before re-evaluating the trained readout *without refitting*,

$$\Pi_k = \sum_{i=1}^{k} u_i u_i^{\top}, \qquad z^{(k)} = \mu + \Pi_k\,(z - \mu), \qquad \hat{f}^{(k)} = \rho_\theta\big(z^{(k)}\big). \tag{5.11}$$

Restoring μ matters: ρ_θ is a nonlinear map trained on uncentred inputs, so feeding it a centred or standardised vector would evaluate it off its training domain and the resulting error would measure that displacement rather than the loss of PC content. As a null, the same projection is repeated onto k *random* orthonormal directions and the two error curves are compared; a top-k curve that is not below the random-k curve would mean the leading PCs carry no privileged information for the readout. We report a *normalised* error on block B,

$$\text{rel-MAE}(k) \;=\; \frac{\mathbb{E}_B \,|\, \hat{f}^{(k)} - f_{\text{base}} \,|}{\mathrm{std}_B(f_{\text{base}})},$$

together with the learning curve in k.

PC1 block-power decomposition. To attribute the leading axis to branches, we partition u_1 conformably with $z = [\overline{\phi}\,|\,\overline{\psi}\,|\,\mathbf{g}]$ and report

$$\text{power}(\overline{\phi}) = \frac{\|u_{1,\overline{\phi}}\|_2^2}{\|u_1\|_2^2}, \quad \text{power}(\overline{\psi}) = \frac{\|u_{1,\overline{\psi}}\|_2^2}{\|u_1\|_2^2}, \quad \text{power}(\mathbf{g}) = \frac{\|u_{1,\mathbf{g}}\|_2^2}{\|u_1\|_2^2}.$$

These are computed on standardised features so that branches carrying different natural scales are comparable. One confound remains and should be kept in mind when reading them: a branch holding d_b of the D coordinates would already carry about d_b/D of the squared loading mass under an isotropic random direction, so a large share partly reflects a wide branch. The shares below are therefore read as *relative* shifts across ω at fixed architecture—where the branch widths are constant and the confound cancels—and not as absolute statements that one branch dominates.

Linear probes to coarse physics. We regress interpretable summaries per configuration on the first k PC scores $S^{(k)}$ using ridge regression (tiny penalty $\alpha = 10^{-6}$):

$$y \approx S^{(k)}\beta, \qquad y \in \big\{\bar{r},\ \mathrm{Var}(r),\ \Pr(r < r_0),\ \text{shell contrast}\big\}.$$

We report R^2 on the held-out block; r_0 is a fixed near-contact cutoff in trap units.

Near-field gradient concentration. For each frame we compute the minimum pair distance $r_{\min}$ (trap units) and the gradient of the residual head with respect to particle coordinates, $\nabla f_{\text{base}}(\mathbf{y})$. The quantity reported is an *enrichment factor*: the mean gradient power among the closest-contact frames divided by the mean over all frames,

$$\mathcal{E}(q) \;=\; \frac{\mathbb{E}\big[\,\|\nabla f_{\text{base}}\|_2^2 \;\big|\; r_{\min} \le Q_q\,\big]}{\mathbb{E}\big[\|\nabla f_{\text{base}}\|_2^2\big]}, \tag{5.12}$$

with Q_q the q-quantile of $r_{\min}$. This is a ratio of a *conditional* mean to an unconditional one and is therefore unbounded above, unlike an indicator-weighted mass fraction, which could not exceed unity. $\mathcal{E} \approx 1$ means gradient power is spread in proportion to frame count, so the analytic cusp is handling most short-range structure; $\mathcal{E} \gg 1$ means residual stiffness is concentrated near contact.

5.2.3 Backflow–correlator coupling

Backflow geometry and energy-locality shares.

Field spectrum. We vectorize the backflow displacement per frame, $\delta = \mathrm{vec}(\Delta x) \in \mathbb{R}^{Nd}$, standardize across frames, and compute the entropy effective rank $r_{\text{eff}}(\Delta x)$ of Sec. 5.2.1. Chapter 7 reports the participation ratio of the same spectrum instead, which is the smaller of the two counts; the ceiling $2N - d$ set by the zero-mean projection (3.18) is common to both.

Where backflow acts. On the same thinned frames we evaluate local energies with and without backflow, $E_{\text{loc}}^{\text{BF}}$ and $E_{\text{loc}}^{\text{noBF}}$, using the *same* coordinates. We form the signed correction $\Delta E_{\text{loc}} = E_{\text{loc}}^{\text{BF}} - E_{\text{loc}}^{\text{noBF}}$ and report near-field shares at quantile q of $r_{\min}$:

$$\text{NF-}\Delta E(q) \;=\; \frac{\mathbb{E}\big[|\Delta E_{\text{loc}}|\,\mathbf{1}\{r_{\min} \le Q_q\}\big]}{\mathbb{E}\big[|\Delta E_{\text{loc}}|\big]}.$$

(Using absolute values isolates magnitude; signed averages are also available.)

Alignment of correlator representations. We collect two standardized feature matrices $\tilde{Z}^{\text{noBF}}$ and $\tilde{Z}^{\text{BF}}$ on identical frames (each standardized by its own μ, σ), compute their PC1 loadings u_1^{noBF} and u_1^{BF}, and report the cosine similarity

$$\cos\big(u_1^{\text{noBF}}, u_1^{\text{BF}}\big) \;=\; \frac{\langle u_1^{\text{noBF}},\, u_1^{\text{BF}}\rangle}{\|u_1^{\text{noBF}}\|_2\,\|u_1^{\text{BF}}\|_2}.$$

Cosines near 1 indicate that backflow acts as a structured reweighting along the same dominant axis, rather than rotating the correlator's manifold.

Centred kernel alignment. The PC1 cosine compares one direction. To compare the two feature *geometries* as a whole we use linear centred kernel alignment [51]. With $\tilde{Z}^{\mathrm{noBF}}, \tilde{Z}^{\mathrm{BF}} \in \mathbb{R}^{K\times D}$ column-centred on the same K frames,

$$\mathrm{CKA}\big(\tilde{Z}^{\mathrm{noBF}}, \tilde{Z}^{\mathrm{BF}}\big) \;=\; \frac{\big\| (\tilde{Z}^{\mathrm{noBF}})^{\top} \tilde{Z}^{\mathrm{BF}} \big\|_F^2}{\big\|(\tilde{Z}^{\mathrm{noBF}})^{\top} \tilde{Z}^{\mathrm{noBF}}\big\|_F \, \big\|(\tilde{Z}^{\mathrm{BF}})^{\top} \tilde{Z}^{\mathrm{BF}}\big\|_F} \;\in [0,1], \tag{5.13}$$

which is invariant to orthogonal transformations and to isotropic scaling of either representation, and equals 1 only when the two Gram matrices are proportional. It is the quantity reported in Table E.2. Two cautions attach to it. CKA compares the *geometry of the feature cloud*, so a high value says the correlator's representation is unchanged by the presence of backflow—not that the backflow occupies the same subspace, and not that the two components are functionally independent. And CKA is known to saturate near 1 whenever both representations are dominated by a few shared high-variance directions, which is exactly the regime here (§5.2.1); values above 0.99 should therefore be read as "no detectable reorganisation" rather than as a quantitative distance.

Uncertainty and stability. For scalars we report standard errors over blocks. For PCA-based quantities and cosines we use a block bootstrap (resampling blocks with replacement) to obtain 68%/95% CIs. Stability checks include: (i) swapping train/held-out blocks for PCA, (ii) varying the near-field quantiles $q \in \{1, 5, 10\}\%$, (iii) re-running analyses on independent sampler restarts.

Reproducibility package. All diagnostics consume the saved bundles $\{\mathbf{X}_{\mathrm{Bohr}},\, r,\, g(r)\}$ and the model snapshot, and emit JSON/CSV tables with r_{eff}, rel-MAE(k) curves, block-power shares, R^2 probes, near-field shares, $r_{\mathrm{eff}}(\Delta x)$, and noBF/BF cosines. As in Sec. 5.1, we log JSD/ℓ_2 splits to confirm stationarity.

5.3 Tangent-kernel diagnostics

The representation diagnostics above describe the ansatz's *features*; a second family, developed with the geometry of §2.2.3, describes its *tangent space*, and it is these that carry the architecture, optimiser, and paradigm comparisons of Chapter 7. From a batch we assemble the centred score matrix $O_{ki} = \partial_{\theta_i} \log|\Psi_\theta(\mathbf{x}_k)|$ and form the quantum geometric tensor $S = O^{\top}O/B$ from the *centred* O (or, when $B < P$, the tangent kernel $K = OO^{\top}$, whose nonzero spectrum equals that of S up to the overall factor B). From the spectrum $\{\lambda_a\}$ we report the effective dimension $d_{\mathrm{eff}} = (\sum_a \lambda_a)^2 / \sum_a \lambda_a^2$ and the condition number $\kappa(S) = \lambda_{\max}/\lambda_{\min}$. Cross-architecture comparisons of d_{eff} are made on a *common probe set* (all models scored on the same configurations, not on each model's own $|\Psi|^2$) so that the comparison reflects the ansatz and not its sampling measure, and we report only participation ratios that have converged in the batch size.

A physical-operator dictionary. Counting tangent directions says how many; naming them says which. We therefore project the leading tangent modes onto a small, fixed dictionary of one- and two-body operators chosen before the fits were run, each written as a scalar function of a configuration whose parameter-space image is obtained by the same score construction as O:

$$\begin{aligned}
&\mathcal{D}_1 = \textstyle\sum_i \tilde{r}_i^{\,2} && \text{breathing / monopole, the generator of } \partial_\omega;\\
&\mathcal{D}_2 = \textstyle\sum_i \tilde{r}_i^{\,2}\cos 2\theta_i && \text{quadrupole, the lowest shape distortion;}\\
&\mathcal{D}_3 = \textstyle\sum_{i<j} e^{-\tilde{r}_{ij}^2/2\sigma_h^2} && \text{correlation hole, with } \sigma_h \text{ the fitted hole width;}\\
&\mathcal{D}_4 = \textstyle\sum_{i<j} \tilde{r}_{ij} && \text{pair-linear, the leading long-range pair response.}
\end{aligned} \tag{5.14}$$

Writing $D \in \mathbb{R}^{B\times 4}$ for the centred matrix of these four functions on the probe batch and $v_a \in \mathbb{R}^B$ for the ath tangent-kernel eigenvector, the *naming* score of a mode is the coefficient of determination of its least-squares regression on the dictionary,

$$R_a^2 \;=\; 1 - \frac{\|(I - P_D)\, v_a\|^2}{\|v_a\|^2}, \qquad P_D = D(D^{\top}D)^{-1}D^{\top}, \tag{5.15}$$

and the *dictionary-orthogonal tangent mass* is the eigenvalue-weighted share of the leading A modes lying outside that span,

$$M_\perp \;=\; \frac{\sum_{a\le A}\lambda_a\,\|(I-P_D)v_a\|^2}{\sum_{a\le A}\lambda_a}, \qquad A=10. \tag{5.16}$$

Finally, *correlation-hole capture* is R^2 of Eq. (5.15) computed against the single operator $\mathcal{D}_3$, i.e. the share of the leading mode explained by the hole channel alone.

The dictionary is a *chosen, non-exhaustive* probe, and nothing here establishes that it spans the physically meaningful directions. A large $M_\perp$ means the tangent space is poorly accounted for by these four operators; it does not mean the remaining directions are unphysical. The comparison supports a relative statement between two architectures scored against the same dictionary on the same probe batch, and that is how Chapter 7 uses it.

Reporting the condition number. $\kappa(S)=\lambda_{\max}/\lambda_{\min}$ is quoted with two caveats that its definition forces. Our batches have $B<P$, so S is rank-deficient by construction and $\lambda_{\min}$ is the smallest *sample-supported* eigenvalue, not an intrinsic one; the value of κ therefore depends on B and is a property of the empirical metric the optimiser actually sees, not of the variational manifold. That is the relevant object for the question we ask of it, whether preconditioning by S^{-1} helps, but it means κ may not be compared across batch sizes, and we hold B fixed wherever we compare it.

To ask whether two trained models are the *same* state rather than merely equal in energy, we use a symmetric, importance-sampled squared overlap,

$$\mathcal{S}^2(A,B)=\big\langle \Psi_B/\Psi_A\big\rangle_{|\Psi_A|^2}\,\big\langle \Psi_A/\Psi_B\big\rangle_{|\Psi_B|^2},$$

each factor evaluated by a log-sum-exp mean over samples of the corresponding state. It equals 1 for identical states, and on the analytic two-electron ground state the trained ansatz reproduces $\mathcal{S}^2 = 1.000000$, which anchors the tooling.

One convention must be stated, because the ratio Ψ_B/Ψ_A is signed for a fermionic state. We evaluate the ratios from $\log|\Psi|$, so the estimator is strictly a *modulus* overlap, $\langle|\Psi_B/\Psi_A|\rangle\langle|\Psi_A/\Psi_B|\rangle$. For the comparisons made here this is an accurate proxy for the signed overlap: both states share the same Slater reference, and the correlator and cusp enter as a strictly positive exponential, so the sign of Ψ is fixed by the determinant and the two states agree in sign except on the measure-suppressed set where their backflows carry a configuration across a node differently. The modulus overlap is an *upper* bound on the signed one, so $\mathcal{S}^2\to 0$ is conservative while $\mathcal{S}^2\approx 1$ rests on the structural argument above rather than on the bound. Overlap, energy and $\mathrm{Var}(E_L)$ are properties of the state and are unaffected by how correlation is divided between the correlator and the backflow; the energetic price of ablating a component is not, since two parameterisations that divide correlation differently report different ablation costs.

Part IV

Results

The argument of Part IV

The two results chapters run a loop between solver and physics.

We first show that the trained states agree closely with every external energy reference that exists for this system—an agreement rather than an independent validation, and one that stops at $\omega = 0.1$ for $N \geq 6$, below which the energies are predictions. That agreement is the precondition for using the states as instruments in the physics analysis that follows. Read for the physics they contain, they show a crossover from compact dot to finite Wigner molecule that is *staged*: the cloud expands and its shells stiffen radially well before the electrons acquire angular order within them.

We then turn from the state to the solver. Message passing changes the geometry of the variational representation even where the total energy cannot tell two networks apart—under the parametrisations used here the correlator concentrates its tangent weight on fewer and more interpretable directions, and the message-passing backflow avoids a rank collapse that overtakes a smaller conventional one. Architecture, optimiser and sampling measure then meet in a single object, the geometry of the wavefunction's log-derivatives, and the evidence suggests that natural-gradient preconditioning earns its cost only while that geometry can still be estimated from a batch.

The loop closes when a generic collocation proposal loses the Wigner state. The structural measurements of Chapter 6 supply the remedy: a proposal built from the observed shell geometry carries Markov-chain-free training to $\omega = 0.0035$, where a transfer instability in the Slater–Jastrow balance takes over.

Three differently sized sets of models appear below—the production states that carry the physics, the matched ladder of the architecture study, and the smaller collocation networks—and numbers from one are never compared with numbers from another. Table 3.1 gives all three. The four research questions of §I are answered here:

RQ4 — Physics	§6.2
RQ1 — Architecture	§7.1
RQ2 — Optimiser	§7.2
RQ3 — Markov-chain-free paradigm	§7.3–§7.4

The least initial deviation from the truth is multiplied later a thousandfold.

ARISTOTLE
On the Heavens

Chapter 6

Energies and Wigner-molecule structure

This chapter is about the *state*: whether it is accurate enough to trust, and what physics it turns out to contain. Why the solver behaves as it does is the subject of Chapter 7.

The quantum dot is an unforgiving test. The Coulomb interaction is singular wherever two electrons meet, and the kinetic energy carries second derivatives, which magnify any error in the wavefunction's curvature. The confinement ω then sweeps the physics across three decades of correlation strength: from a compact, kinetic-energy-dominated droplet at $\omega = 1$ to a dilute Wigner molecule at $\omega = 10^{-3}$, where the electrons settle several oscillator lengths apart and their mutual repulsion rather than the trap determines the structure of the state. We work at $N \in \{2, 6, 12, 20\}$ electrons and $\omega \in \{1.0, 0.5, 0.1, 0.01, 0.001\}$.

Section 6.1 benchmarks the energies against every published reference that exists: fixed-node diffusion Monte Carlo for $N \in \{6, 12, 20\}$ [52, 53] and the analytic two-electron compiled reference values for N=2 [52, 54]. Those references reach down to $\omega = 0.1$ for every N, and to $\omega = 0.01$ for N=2; below that, for $N \geq 6$, no published value exists, and the boundary is marked wherever it occurs. Section 6.2 then uses the trained wavefunction as an instrument and follows the Wigner crossover from two electrons to twenty. Its result is that the crossover is staged: the cloud expands and its shells stiffen radially well before the electrons acquire angular order inside them. How the network internally represents these states is a question about the solver, not about the physics, and is deferred to Chapter 7 and Appendix E.

6.1 Benchmark energies

Training is residual-based pretraining to a well-conditioned initialiser followed by a Stochastic Reconfiguration tail, run for 10^3–3×10^3 iterations on batches of $\sim 3 \times 10^3$ samples, that refines the last digits. Wherever this thesis says "SR tail" without qualification, SR is applied to variational Monte Carlo under $|\Psi|^2$ sampling; the same optimiser reappears in Chapter 7 on the collocation objective, where the measure differs and so does its role.

Two production ansätze carry this chapter. They share the separable correlator of §3.4.1 at ≈54k parameters and differ only in the backflow: a *conventional* single round of pairwise messages aggregated per particle (≈51k, total ≈105k), or a *message-passing* field that maintains node and edge features and transports information between them (≈182k, total ≈236k). They are named *conv. BF* and *MP BF* for that difference. The counts are independent of N, since interactions enter through permutation-equivariant features, not per-particle weights, and they are not the widths of the architecture study of §7.1 (Table 3.1).

6.1.1 Energies

Across every externally benchmarked system, the trained states agree with the available fixed-node reference energies to between 10^{-5} and 5×10^{-4} in relative terms, with no systematic deterioration as particles are added or the trap is opened (Tables 6.1 and 6.2, Fig. 6.1).

Everything after this section depends on it. It is less than a validation of the wavefunction, for reasons set out below, but it is the precondition for treating the state as an instrument in §6.2 rather than as a claim still under defence.

Where the agreement holds matters as much as its size. A method that matches at $\omega = 1$ and drifts at $\omega = 0.1$ differs from one that holds across the benchmarked range, since the physics changes character between those points while the architecture does not. Below $\omega = 0.1$ no reference of any kind exists for $N \geq 6$, and the values reported there are variational predictions rather than validated ones; that boundary is marked wherever it occurs.

Where a reference exists it is also the target that residual pretraining anneals toward (§4.1), so the agreement is not a fully independent validation. The SR tail that follows strengthens it considerably, since it minimises the variational energy itself and no longer sees the target; but the final states still descend from reference-guided training. Nor does closeness to a reference certify the state. The variational principle guarantees that the expectation does not fall below the exact E_0, not that it lies near it, and a flexible ansatz can reproduce an approximate reference without describing every property of the exact state correctly. §8.1 takes this up properly.

Table 6.1 begins with the easiest case, and its purpose is diagnostic, not competitive: two electrons are few enough that the architecture can be tested nearly in isolation. It reports two-electron energies from *residual-only* training, with and without the backflow. For N=2, backflow is *not* required to reach reference accuracy; it slightly reduces the variance and smooths convergence, and nothing more. The closed-shell two-electron singlet has no nontrivial same-spin fermionic node for backflow to repair. This matters because it establishes a baseline against which the backflow's later contribution, at larger N and weaker traps, can be read as a genuine effect rather than as an artefact of having added parameters.

Table 6.1: Ground-state energies (Hartree) for two electrons in a harmonic trap using *residual-only* training—that is, *without* the SR tail. To be read against Table 6.2, which reports the same systems after SR refinement: the comparison isolates what pretraining alone achieves, and the residual bias at $\omega = 0.01$ visible here is the one the SR tail removes. The reference column is the same as in Table 6.2; see its notes for provenance. Parentheses denote 1σ on the last digits.

ω	correlator + BF	correlator only	Reference
1.00	2.99998(3)	2.999940(5)	3.00000(1)
0.50	1.65975(2)	1.65979(3)	1.65977(1)
0.10	0.440795(3)	0.440833(1)	0.44079(1)
0.01	0.0740724(6)	0.0741679(6)	0.073839(2)

Table 6.2 compares diffusion Monte Carlo (DMC) references with our *final* conv.-BF energies after residual pretraining and an SR tail. From the ultra–shallow two-electron case (ω=0.01) to the many-electron systems, absolute deviations are small; for N=2 they lie within DMC uncertainties, and for $N \in \{6, 12\}$ the relative deviations span 0.008–0.05% (see also Fig. 6.1).

Table 6.2: Variational energies (Hartree) from residual pretraining plus an SR tail, for the conventional-backflow and message-passing-backflow ansätze. Reference values for $N \geq 6$ are Høgberget's fixed-node diffusion Monte Carlo energies [52], as tabulated in Ref. [53]; provenance and the N=2 attribution are given in the notes below. Parentheses denote the 1σ blocked statistical error on the last digit(s) for the single optimised state reported; they do not include the variation between independent training runs, which was not measured for these production states, and at N=20 in particular each entry is one training run. "Rel. dev." is $100\,(E - E_{\mathrm{ref}})/E_{\mathrm{ref}}$, labelled deviation and not error because the $N \geq 6$ references are themselves fixed-node upper bounds, so a negative entry is not necessarily wrong (§8.1). **These are production energies and the two ansätze are not capacity matched** ($\approx$ 105k against $\approx$ 236k); the controlled architecture comparison is §7.1, and this table is not an architecture ablation. Rows whose reference column reads *prediction*—all $\omega = 0.001$, and $\omega = 0.01$ for $N \geq 6$—have no published value to compare with. They are variational predictions, not validated energies, and in the deep-Wigner rows they are the lowest states found at fixed spin projection (§6.2).

N	ω	Reference	conv. BF	rel. dev.	MP BF	rel. dev.
2	1.00	3.00000	2.99999(1)	−0.0003	3.00000(2)	+0.0000
	0.50	1.65977(1)	1.65976(2)	−0.0006	1.65976(1)	−0.0006
	0.10	0.44079(1)	0.44079(1)	+0.0000	0.440796(4)	+0.0014
	0.01	0.073839(2)	0.073838(1)	−0.0014	0.0738382(5)	−0.0011
	0.001	*prediction*	0.0137948(8)	—	**0.013778(1)**	—
6	1.00	20.15932(8)^{H}	20.1610(6)	+0.0083	**20.1585(3)**	−0.0041
	0.50	11.78484(6)^{H}	11.7895(4)	+0.0395	**11.7847(2)**	−0.0012
	0.10	3.55385(5)^{H}	3.5549(1)	+0.0295	**3.55388(5)**	+0.0008
	0.01	*prediction*	0.69125(2)	—	**0.69036(1)**	—
	0.001	*prediction*	0.140913(1)	—	**0.140832(1)**	—
12	1.00	65.7001(1)^{H}	65.717(1)	+0.0257	**65.69556(5)**	−0.0069
	0.50	39.1596(1)^{H}	39.1786(8)	+0.0485	**39.1604(3)**	+0.0020
	0.10	12.26984(8)^{H}	12.2731(2)	+0.0266	**12.2743(2)**	+0.0363
	0.01	*prediction*	2.48620(5)	—	**2.47363(4)**	—
	0.001	*prediction*	0.515823(3)	—	**0.515365(4)**	—
20	1.00	155.8822(1)^{H}	—	—	**155.8738(7)**	−0.0054
	0.50	93.8752(1)^{H}	—	—	**93.8789(5)**	+0.0039
	0.10	29.9779(1)^{H}	—	—	29.9888(2)	+0.0364
	0.01	*prediction*	—	—	**6.14645(5)**	—
	0.001	*prediction*	—	—	**1.293033(6)**	—

Reference-energy provenance: values marked ^{H} are fixed-node diffusion Monte Carlo energies computed by Høgberget [52] and tabulated in Ref. [53]. The N=2 values come from the reference column of the same compilation, which draws on both Høgberget's DMC and Taut's analytic solutions [54]; since Taut's construction is quasi-exactly solvable only at particular frequencies, we do not assign individual N=2 rows between the two, beyond noting that 3.00000 at $\omega = 1$ is the exact analytic value. The foundational DMC study of these dots is Pederiva *et al.* [12] (spanning $N \leq 13$; see also its erratum [55]), cited here for the method, not for these particular numbers. We took the values from the compilation in the master's thesis of Haas [56], which reproduces the same benchmarks; the attribution above is to their origin. Rows marked *prediction* have no published value.

Figure 6.1: Relative deviation of the conv.-BF energies from the external references (Table 6.2) at every (N, ω) where one exists: fixed-node DMC [52] for N=6 and 12, and the compiled DMC and analytic [54] values for N=2. Deviations are at the 10^{-3} % level for N=2, comparable to the reference's own precision, and 0.008–0.05 % for N=6 and 12, with no trend in ω.

The same benchmark, without a Markov chain. Everything above was trained with a Markov chain: residual pretraining, then an SR tail that draws its samples from $|\Psi|^2$ by Metropolis. A second route removes the chain from the gradient entirely, replacing it with an analytic proposal and importance weights. It reaches comparable energies over part of this range. Its performance, its failure boundary and the mechanism behind both are the subject of §7.3, and the energies are tabulated there (Table 7.5) and not here, because they are not interpretable until that mechanism is in hand.

Where the tables stop. One subtlety matters here, because the MP-BF ansatz sits *below* the quoted DMC reference in four cells, and at N=12 and N=20, $\omega = 1$, by more than ten combined standard errors (for example 65.6956 vs. 65.7001 Ha at N=12). This does not violate the variational principle, which bounds the energy from below only by the *exact* ground state, $E_{\mathrm{VMC}} \geq E_0$; the DMC benchmarks are *fixed-node* calculations and carry their own nodal bias. A better nodal surface in the network would produce exactly this pattern. It is not the only thing that would. A difference in the convention for the Hamiltonian or its units, a time-step or population extrapolation in the reference, an underestimated statistical error on either side, or an error somewhere in the chain of compilations by which the reference reached us (see the notes to Table 6.2) would do the same, and nothing here distinguishes them. The clean test is a fixed-node DMC calculation that uses the network's own nodes as its trial surface, which we have not run. The sub-reference values are therefore recorded as deviations and not interpreted.

Wherever an external check exists, the pattern is the same. For N=2 the ansatz reaches reference accuracy at every ω for which a reference exists, and for $N \in \{6, 12\}$ the relative deviation stays within 0.008–0.05% across the benchmarked range ($\omega \geq 0.1$; Fig. 6.1). Below $\omega = 0.1$ there is no DMC reference for $N \geq 6$, and we make no claim of absolute accuracy there; what we can say is that the energies join smoothly onto the benchmarked regime and, in the deep-Wigner limit, follow a power law whose fitted exponent, 0.689 for N=6 (§7.4), sits close to the classical Wigner value 2/3. The efficiency is as notable as the accuracy: the SR tail converges in 10^3–3×10^3 iterations of $\sim 3{\times}10^3$ samples each, a modest optimisation budget for a fully many-body wavefunction. We attribute this stability to a small set of conditioning choices— trap-unit scaling, soft-core pair features with $ds/dr \to 0$ at coalescence, and an

6.2 Wigner–molecule crossover

Up to this point the wavefunction has been the thing under examination. Here it becomes the instrument.

The result of this section is that the crossover is staged. As the trap weakens the cloud first expands and its shells stiffen *radially*—the width-to-radius ratio falls, discrete occupancies emerge, the energy partition approaches classical force balance—while the electrons remain very nearly disordered in angle within those shells. Angular structure then develops in two steps of its own: a small, gradually growing correlation that is measurable well before the Wigner regime, and a large bond-orientational ordering confined to the final decade of confinement. Against the random-angle baseline the shell bond order runs at 0.99–1.01 of its null value from $\omega = 1$ down to $\omega = 0.1$, reaches 1.08 at $\omega = 0.01$ for N=6 (and stays within 1.02 at N=12 and N=20), and only at $\omega = 10^{-3}$ jumps to 1.16–1.63 on every ring at every particle number. Radial localisation and angular crystallisation are not one transition, and the diagnostics below are arranged to separate them.

Which states these are. Every state in this section is the lowest we find with $N_\uparrow = N_\downarrow$ held fixed, starting from the closed-shell singlet of the non-interacting problem. That is not a formality here. Fixing the spin projection does not fix the total spin, since an $S = 1$ state also has an $S_z = 0$ component, and we never compare sectors of different S. In parabolic dots the interaction is known to drive spin transitions as it strengthens—for instance $S = 0 \to S = 1$ at $N = 6$ in the strongly correlated regime [1]—so a change of ground-state spin is plausible precisely at the weakest traps, where the structural conclusions below are strongest. The deep-Wigner states are therefore the lowest states found within a fixed spin-projection sector, not established global ground states, and every statement below is about them. There is a reason to expect the *charge* geometry to be less sensitive than the energy ordering—deep in the Wigner regime exchange is weak and states of different S become nearly degenerate over the same classical arrangement—but that is an expectation we have not tested, not a result.

The subsections take that in order. §6.2.2 shows the cloud expanding and acquiring visible radial structure; §6.2.3 shows interactions taking over and the trap–interaction partition approaching its classical form; §6.2.4 shows the shells stiffening radially; and §6.2.5 shows angular order finally appearing, tested against a null in which the angles are random. §6.2.6 collects what the diagnostics agree on and what they do not establish.

What we are looking for cannot be seen in the observable one would naturally plot. In a rotationally symmetric trap, the electrons of a Wigner molecule localise in *relative* coordinates (concentric rings, polygon-like angular correlations) while the whole pattern is free to rotate. Average over that rotation, as any lab-frame density does, and the rings smear into something close to uniform. The order is there; the density hides it. Every diagnostic below exists to look past that averaging.

We combine one-body densities, label-invariant shell topology, rotation-registered angular order, shell radii and gaps, pair statistics, and Lindemann-type disorder, across $N = 2, 6, 12, 20$ and $\omega \in \{1, 0.5, 0.1, 0.01, 0.001\}$ [1, 4]. No one of them is decisive on its own. They fail in different ways, which is why they are reported together.

6.2.1 Observables and diagnostics

For each (N, ω) we draw $\mathbf{X} \sim |\Psi|^2$ and report (i) the pair distribution $g(r)$, (ii) the single-particle radial density $P(r) \propto r\,\rho(r)$, and summary statistics: r_{mode}, $\langle r \rangle$, σ_r, FWHM, quantiles q_{10}, q_{50}, q_{90}, and a localisation ratio $\gamma = \sigma_r / r_{\mathrm{mode}}$ (smaller $\gamma \Rightarrow$ stiffer localisation). For multi-electron systems we additionally monitor shell-resolved bond-orientational order $|\Phi_m|$ and a dimensionless Lindemann-type ratio for positional fluctuations on each ring [14].

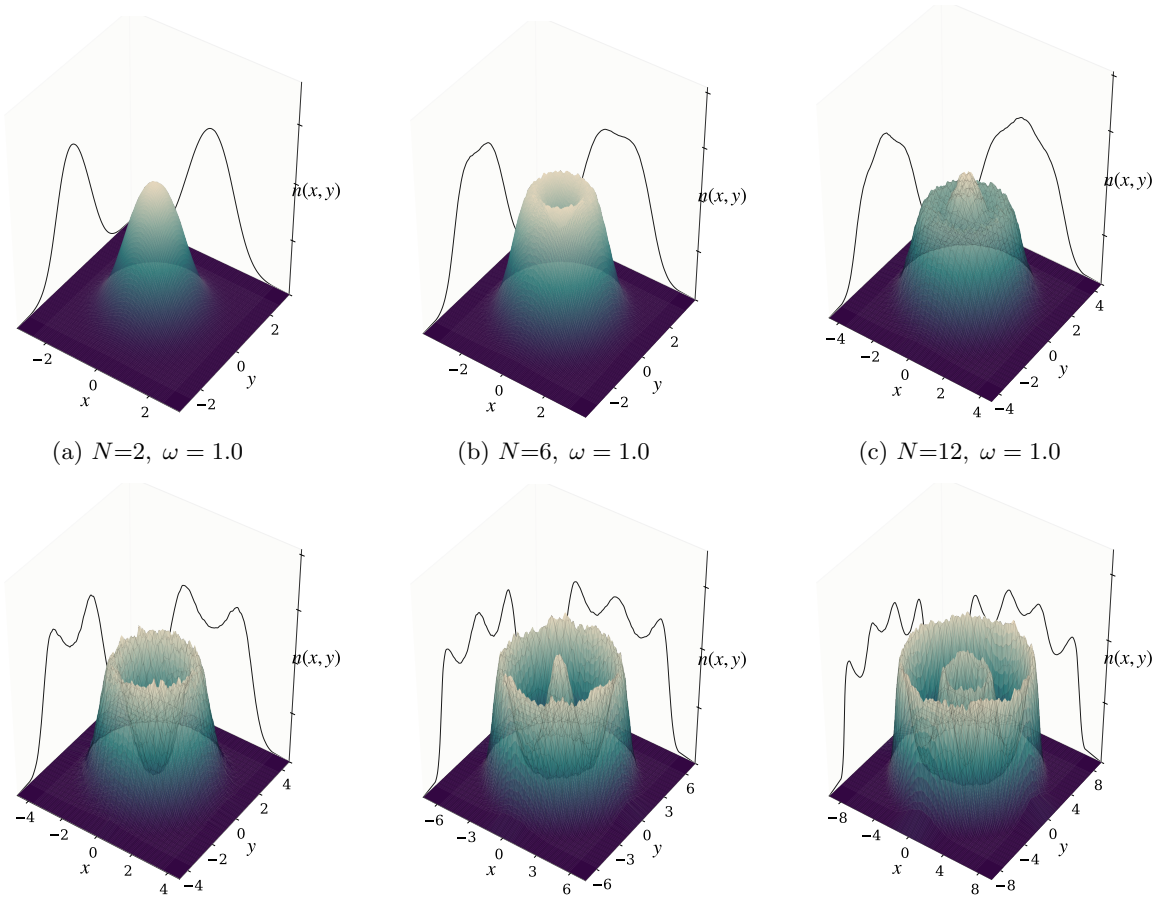
Density parameter r_s. Following Egger *et al.* [1], we estimate the Wigner–Seitz parameter from the first maximum r^* of the spin-summed pair correlation, $r_s \equiv r^*/a_B^*$. The Fermi–liquid $\to$ Wigner–molecule crossover occurs near $r_s \simeq 4$ in parabolic dots; our weak-trap cases lie well beyond this threshold [1, 11].

6.2.2 One-body densities: visual crossover

Figures 6.2 and 6.3 compare one-body densities at strong confinement ($\omega = 1$) and weak confinement ($\omega = 10^{-3}$) for $N = 2, 6, 12, 20$. At $\omega = 1$ the density remains compact and the dominant change with N

is progressive shell filling under the harmonic trap. At $\omega = 10^{-3}$ it broadens by two orders of magnitude and develops pronounced *radial* structure at every N.

What that structure can and cannot establish follows from the symmetry argument above. The lab-frame density is an average over all orientations of the molecule, so it retains radial information and destroys angular information by construction. These figures are therefore evidence of *radial shell localisation* and are silent on whether the electrons are angularly ordered within those shells—a ring of six electrons at fixed 60° spacing and six electrons sliding freely around the same ring produce the same picture. The angular question is taken up in §6.2.5, where it needs a co-rotating frame and a null model to answer.



(a) N=2, $\omega = 1.0$ (b) N=6, $\omega = 1.0$ (c) N=12, $\omega = 1.0$

(d) N=2, $\omega = 0.001$ (e) N=6, $\omega = 0.001$ (f) N=12, $\omega = 0.001$

Figure 6.2: One-body densities for $N = 2, 6, 12$ at $\omega = 1.0$ (top) and $\omega = 0.001$ (bottom). The vertical axis is the one-body density $n(x, y)$ over the two Cartesian components of a single particle's position, in trap units; the curves on the back walls are its marginals. As confinement weakens the densities broaden and develop pronounced *radial* shell structure. Being lab-frame averages over all orientations of the molecule, they cannot show angular order: that requires the registered diagnostics of §6.2.5.

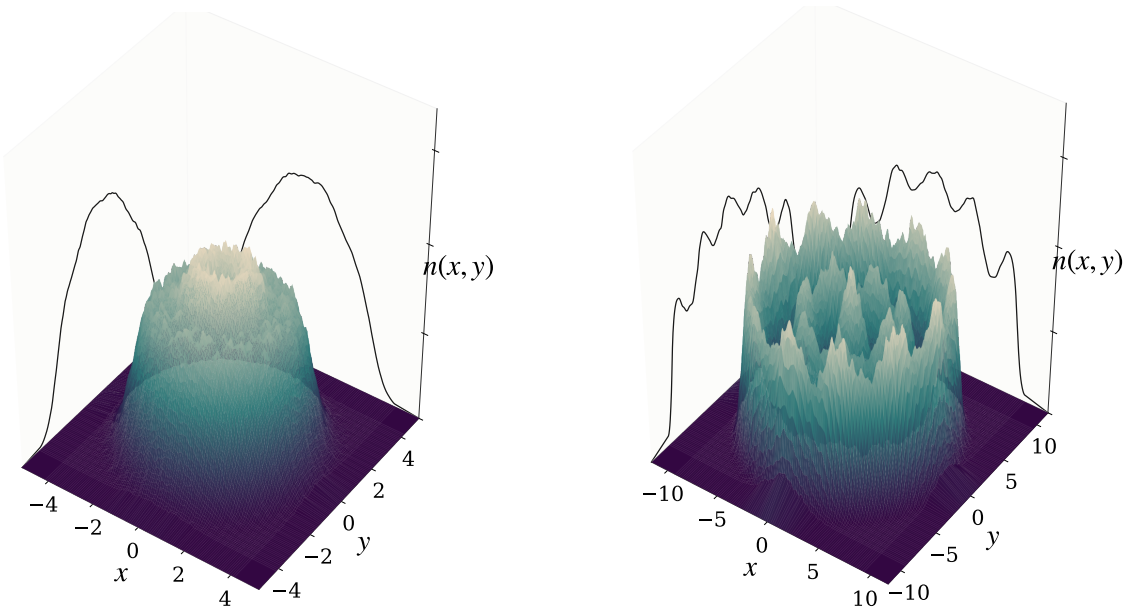
6.2.3 Energy-based crossover diagnostics

The energy partition carries two messages, and only two.

Interactions take over. The interaction-to-kinetic ratio $\Gamma = \langle V_{\text{int}} \rangle / \langle T \rangle$ rises across $\omega \in [1, 10^{-3}]$ from 0.9 to 8.6 at N=2, from 2.4 to 24.5 at N=6 and from 3.6 to 41.1 at N=12 (Fig. 6.4a). The dot stops being a kinetic-energy-dominated droplet and becomes a Coulomb-dominated one.

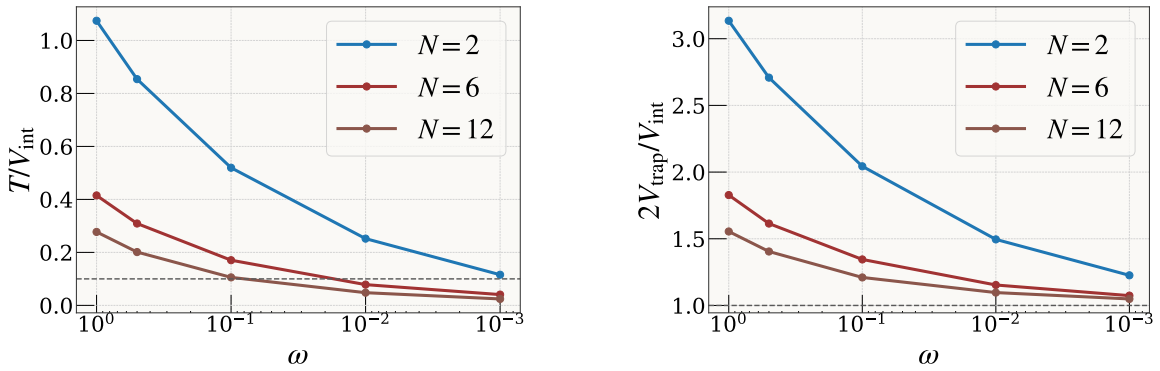
Classical force balance emerges. At $\omega = 10^{-3}$ the trap–interaction partition $2\langle V_{\text{trap}} \rangle / \langle V_{\text{int}} \rangle$ reaches 1.23, 1.07 and 1.05 at $N = 2, 6, 12$, so the twelve-electron dot satisfies the classical condition $V_{\text{int}} \approx 2V_{\text{trap}}$ to within 5% (Fig. 6.4b). The approach is monotone and improves with N, as it should: the classical limit is reached faster when there are more particles to arrange.

Both are consistent with real-space expansion. The rms radius extracted from $V_{\text{trap}} = \frac{1}{2}\omega^2 \sum_i \langle r_i^2 \rangle$



(a) N=20, $\omega = 1.0$ (b) N=20, $\omega = 0.001$

Figure 6.3: One-body densities for N=20 at strong and weak confinement; the vertical axis is $n(x, y)$ as in Fig. 6.2. The weak-trap case shows strong radial broadening and clear multi-shell organisation. As above, this is radial evidence only.



(a) T/V_{int} vs. ω for N=2, 6, 12. The horizontal line marks T/V_{int}=0.1.

(b) $2V_{\text{trap}}/V_{\text{int}}$ vs. ω for N=2, 6, 12. The horizontal line marks the classical virial ratio $2V_{\text{trap}} = V_{\text{int}}$.

Figure 6.4: Energy-based Wigner diagnostics as functions of trap frequency. As ω decreases the dots become interaction dominated ($T/V_{\text{int}} \ll 1$) and the trap–interaction partition approaches the classical virial form, most clearly for N=12.

grows from 1.1–2.0 a_B^* at $\omega = 1$ to 70–170 a_B^* at $\omega = 10^{-3}$, faster than the non-interacting $\omega^{-1/2}$ law, which is interaction-driven swelling. The fitted exponents for the individual energy components and for the radius, which support these statements without adding to them, are collected in Appendix F.1.

6.2.4 Radial localisation and pair correlations

Over the weakest three confinements, the typical one-body radius follows $r_{\mathrm{mode}} \sim \omega^{\alpha}$ with $\alpha \simeq -0.626$ at N=2—close to the classical two-body value $-2/3$, which nothing in the ansatz knows about. At N=2 the width-to-scale ratio $\gamma = \sigma_r / r_{\mathrm{mode}}$ falls as well, from 0.49 to 0.29, so the pair stiffens rather than merely dilating. At N=6 the same global ratio barely moves (Table 6.3), and radial stiffening there has to be read from quantities that resolve the shell, not from this one.

Table 6.3: Radial localisation across the confinement range. r_{mode} is the mode of the single-particle radial density in Bohr; $\gamma = \sigma_r / r_{\mathrm{mode}}$ is the width-to-scale ratio. Full diagnostics—mean, width, FWHM, quantiles, ring-angle spread and pair Lindemann ratio—are in Appendix F. The N=6 values at $\omega \leq 0.01$ come from an earlier run than the shell analysis and Fig. 6.6 (Appendix G).

	N=2		N=6	
ω	r_{mode}	γ	r_{mode}	γ
1.00	1.44	0.491	2.09	0.442
0.50	2.30	0.437	3.07	0.445
0.10	3.60	0.480	8.17	0.429
0.01	14.19	0.396	33.15	0.420
0.001	64.29	0.287	113.55	0.421

The exponent is not the sharpest form of this check. Because r_{mode} is the mode of the *single-particle* radial density, it mixes the central electron of a $(1,5)$ topology with the ring and is not the ring radius. The shell-resolved outer radius is, and comparing it with the classical Wigner ring $1.334\,\omega^{-2/3}$ tests a prefactor as well as a power. That ratio falls monotonically—1.247, 1.188, 1.098, 1.032, 1.017 from $\omega = 1$ to 10^{-3}—so at the weakest trap the quantum ring sits 1.7% outside the classical one, and the fitted exponent on that quantity is -0.637 over the full grid and -0.650 over the weakest three confinements (Appendix F.4). Nothing in the ansatz, the objective or the sampler knows the number 1.334.

The two sizes stiffen differently, and the difference is physical. At $\omega = 10^{-3}$ the N=2 pair sits at separation $r_{12} \approx 122$ with Lindemann ratio $\gamma_{r_{12}} \approx 0.20$ and a relative angle sharply peaked at π: a dilute, anti-aligned dimer. At N=6 the global γ plateaus near 0.42, and the evidence for stiffening comes from the shell-resolved and pair quantities instead: the outer ring's own relative radial width falls from 0.32 at $\omega = 1$ to 0.15 at 10^{-3} (Table F.3), and the ring's angular fluctuations and pair Lindemann ratio fall monotonically. A single-particle width is the weakest of the radial diagnostics: it mixes the central electron with the ring, and it is the one quantity here that does not resolve the shell structure it is being asked about.

6.2.5 Shell topology, angular order, and the rotating Wigner molecule

The densities plotted above look like smooth rings, yet the electrons are localised. Both are true, and the reason is symmetry. Because the Hamiltonian is rotationally invariant, so is its ground state, and any lab-frame average necessarily smears an angularly ordered configuration into a featureless ring. The order is real; it simply lives in relative coordinates, and looking for it in the lab frame is looking in the one place it is guaranteed not to be. Everything below follows from taking that seriously.

Concretely, a Wigner molecule's internal angular order can only be observed after registering frames to a co-rotating reference. For a shell s with n_s particles we compute a rotation-registered bond-orientational parameter $|\Phi_{n_s}|$ and a corresponding Lindemann-type ratio quantifying angular stiffness. Lab-frame densities remain ring-like (the "rotating Wigner molecule" scenario), while the co-rotating frame reveals crystal-like polygon order [1, 4].

N=6: single ring with sector mixing. At $\omega = 0.01$ the system fluctuates between a $(1,5)$ sector (fraction ≈ 0.69) and a $(0,6)$ single-ring sector (fraction ≈ 0.31), with bond order $|\Phi_5| = 0.453 \pm 0.214$ and

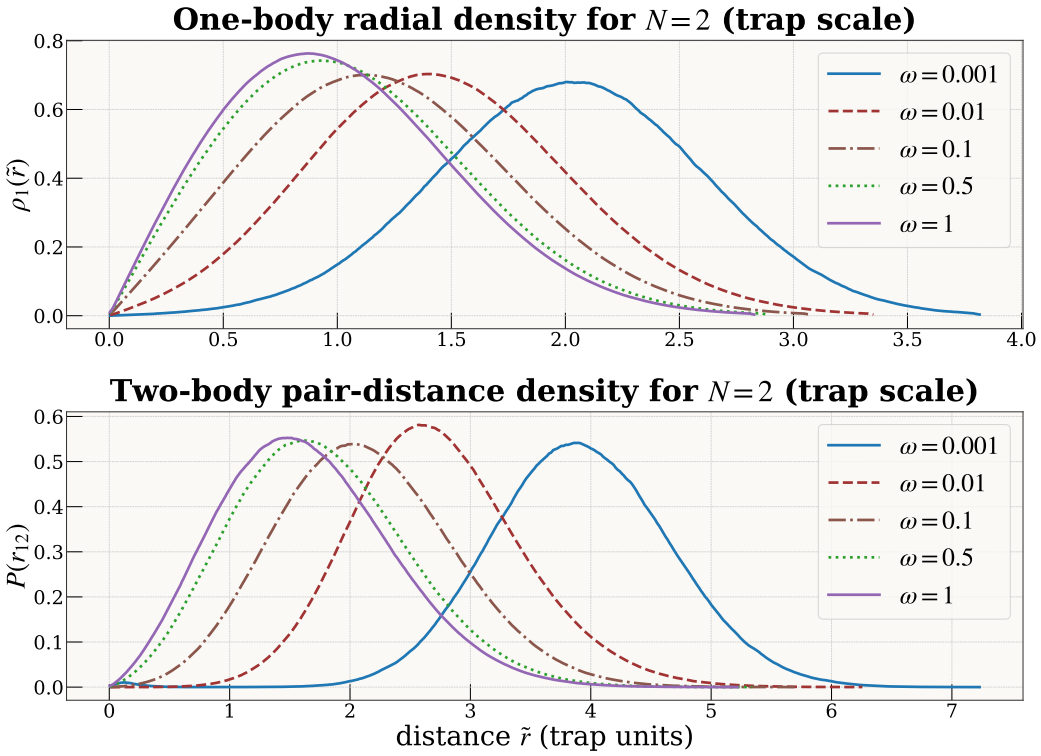


Figure 6.5: One-body radial density $\rho_1(\tilde{r})$ (top) and pair-distance density $P(r_{12})$ (bottom) for N=2 across ω, in trap units $\tilde{r} = \sqrt{\omega}\, r$.

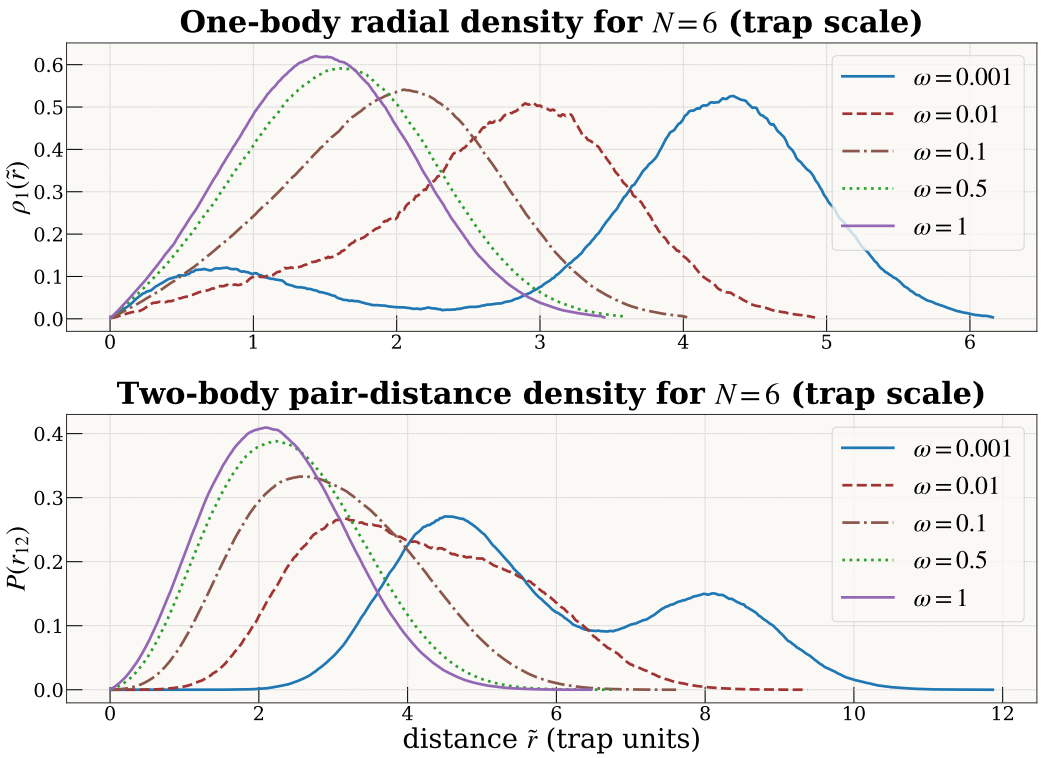


Figure 6.6: As Fig. 6.5, for N=6. At $\omega = 10^{-3}$ the one-body density separates into the central electron and the ring, and the pair distribution splits into centre–ring and nearest-neighbour distances (first peak) and cross-ring distances (second).

$|\Phi_6| = 0.466 \pm 0.206$ respectively. Two conventions apply here and below. The quoted $\pm$ is the frame-to-frame standard deviation of $|\Phi_m|$, not a standard error on its mean, which with 3×10^4 frames is smaller by two orders of magnitude. And these are *sector-conditioned* values, computed on the frames assigned to that sector, which is why they differ slightly from the modal-topology values tabulated in Appendix F.2. Both must be read against the random-angle null, the exact mean of $|\Phi_n|$ for independent uniform angles (Eq. (F.1)), which is 0.402 for n=5 and 0.366 for n=6: the ratios here are 1.13 and 1.27, so the measured bond order at $\omega = 0.01$ exceeds the random-angle value by a modest margin rather than sitting on it. At $\omega = 10^{-3}$ the ring stiffens further: $(1, 5)$ fraction ≈ 0.80, $(0, 6) \approx 0.20$, with $|\Phi_5| = 0.597 \pm 0.224$ (Lind $\approx$0.225) and $|\Phi_6| = 0.663 \pm 0.183$ (Lind $\approx$0.205)—now far above the random-angle null, by factors of 1.49 and 1.81. This is the confinement at which n-fold angular order becomes *large* rather than merely resolvable (Appendix F.2). The distinction matters because with 10^4–10^6 frames a 2% excess over the null is already resolved at tens of standard errors; detectability is therefore not what separates the regimes, magnitude is. Classically $(1, 5)$ is the ground-state topology and $(0, 6)$ a proximate competitor; the observed quantum mixture mirrors this hierarchy [2, 3]. In both weak-trap cases the global $g(r)$ is reproduced to numerical precision by the appropriate inner–inner / inner–outer / outer–outer (II/IO/OO) mixtures.

N=12: two shells and commensurability. At $\omega = 0.01$ two-shell formation is already frequent (fraction 0.761 at radial-gap threshold τ=3.0), distributed across sectors $(1, 11) = 32.6\%$, $(2, 10) = 25.7\%$, $(3, 9) = 13.7\%$. At $\omega = 10^{-3}$ shelling becomes ubiquitous ($f_{\mathrm{2shell}} \approx 0.99$–$0.92$ for $\tau = 2.0$–3.0) and the *commensurate* $(3, 9)$ split emerges as modal (43.3% of quality-cut frames), with strong bond order on both rings: $|\Phi_3| = 0.687 \pm 0.229$ ($\mathrm{Lind}_{\mathrm{in}} = 0.223$), $|\Phi_9| = 0.526 \pm 0.194$ ($\mathrm{Lind}_{\mathrm{out}} = 0.224$). Whether the shell picture actually explains the pair structure needs a test that could fail, and the obvious one cannot. Recombining the inner–inner, inner–outer and outer–outer histograms with the combinatorial weights of the ideal $(3, 9)$ topology reproduces the global $g(r)$ essentially exactly—but the sector histograms and the total are accumulated from the same frames, so that recombination is close to exact by construction. It establishes that the sector bookkeeping is self-consistent, nothing more, and it is recorded in Appendix F.3 for that reason.

The test that can fail is generative and held out (Table 6.4). Shell radii and widths are fitted on half the frames; each shell is modelled as a regular n_s-gon carrying a per-frame global rotation and a fitted angular jitter; and the model is scored on the other half. It reproduces the pair distribution to cosine similarity 0.987–0.999 across the whole grid, against 0.936–0.996 for an angle-randomised null built from the same fitted radial distributions. The difference between them, $\Delta\cos$, is what *angular* order buys over shell radii alone, and it grows monotonically as the trap weakens at every particle number—at N=12 from 0.006 at $\omega = 1$ to 0.026 at $\omega = 10^{-3}$, at N=6 from 0.012 to 0.058, at N=20 from 0.003 to 0.010. Shell radii alone account for most of the pair structure at strong confinement; angular structure accounts for a growing remainder as the molecule forms. How firmly is a separate question, because the smallest values are a few thousandths, and it was tested directly (Appendix F.3). Every $\Delta\cos$ carries a block-bootstrap interval that excludes zero—at N=20, $\omega = 1$ it is $[0.0027, 0.0030]$—and the growth with $1/\omega$ is resolved at every particle number. It survives changes of histogram binning and of the train/test split essentially unchanged. What does move it is the angular-jitter model, by up to a third, and the shell threshold where that changes the partition; the size of $\Delta\cos$ is model-dependent, its sign and its trend are not.

That growth has a different shape from the bond order. $\Delta\cos$ rises by a factor of 1.1–2.1 per decade at every particle number, monotonically and without a threshold, whereas the bond order stays flat at its null and then jumps. The two are not in conflict: $\Delta\cos$ responds to *any* angular correlation that improves the pair-distance density, while $|\Phi_n|$ responds specifically to n-fold order on a registered ring. Weak angular correlations build gradually from strong confinement onward, and crystalline n-fold order emerges from them only in the final decade—three stages, not a single threshold, with the middle one measured by a quantity whose sign and trend are robust but whose size depends on the shell model.

N=20: three-shell molecule and a rapid radial reorganisation. At $\omega = 10^{-3}$ the dominant shell topology is $(1, 7, 12)$—a clear three-shell Wigner molecule with widely separated rings at radii $r_0 \approx 38$, $r_1 \approx 132$, $r_2 \approx 247$ (gaps ≈ 94 and ≈ 115). The lab-frame pinning metric remains small ($C_{\mathrm{global}} \approx 0.06$), as expected for a freely rotating molecule. The cross-shell quantity $\tilde{C}_{\mathrm{global}}$ is omitted: it depends on a registration convention whose original analysis code was not retained (§5.1.3). The per-shell moduli and angular-spacing diagnostics below are rotation-invariant and carry no such dependence. Lindemann disorder on the two occupied outer shells is low: $L_{7,\mathrm{reg}} \approx 0.35$ and $L_{12,\mathrm{reg}} \approx 0.27$.

Already at $\omega = 10^{-2}$ the radial gap detector returns a $(1, 19)$ partition rather than a three-shell one:

Table 6.4: The structural test that can fail. Shell radii and widths are fitted on half the frames; each detected shell is then modelled as a regular n_s-gon carrying a per-frame global rotation and a fitted angular jitter; and the resulting pair-distance density is scored on the *held-out* half. $\cos_{\text{poly}}$ is the polygon model's agreement with the empirical density and $\cos_{\text{uni}}$ that of an angle-randomised null with the same fitted radial distributions, so $\Delta\cos$ isolates what polygonal order contributes over shell radii alone. The interval is a 95% block bootstrap over frames that refits the model and draws a fresh model sample in every replicate; the tabulated point uses the original single model sample, whose Monte Carlo noise is part of what the interval measures, so a point can sit near an edge. Sensitivity to binning, split, shell threshold and jitter model is in Appendix F.3, with the full table and residuals.

N	ω	occ.	$\cos_{\text{poly}}$	$\cos_{\text{uni}}$	$\Delta\cos$	95% interval
6	1.0	1, 5	0.9988	0.9866	**0.0122**	[0.0122, 0.0136]
6	0.5	1, 5	0.9986	0.9836	**0.0150**	[0.0147, 0.0160]
6	0.1	1, 5	0.9974	0.9741	**0.0234**	[0.0219, 0.0236]
6	0.01	1, 5	0.9922	0.9536	**0.0386**	[0.0371, 0.0395]
6	0.001	1, 5	0.9942	0.9358	**0.0585**	[0.0556, 0.0592]
12	1.0	1, 11	0.9993	0.9937	**0.0056**	[0.0054, 0.0059]
12	0.5	1, 11	0.9990	0.9927	**0.0063**	[0.0060, 0.0066]
12	0.1	1, 11	0.9980	0.9896	**0.0085**	[0.0083, 0.0088]
12	0.01	1, 11	0.9929	0.9805	**0.0124**	[0.0124, 0.0132]
12	0.001	3, 9	0.9868	0.9604	**0.0264**	[0.0258, 0.0271]
20	1.0	1, 19	0.9992	0.9964	**0.0028**	[0.0027, 0.0030]
20	0.5	1, 19	0.9990	0.9958	**0.0032**	[0.0030, 0.0033]
20	0.1	1, 19	0.9982	0.9942	**0.0040**	[0.0038, 0.0040]
20	0.01	1, 19	0.9947	0.9900	**0.0047**	[0.0046, 0.0049]
20	0.001	1, 7, 12	0.9915	0.9820	**0.0095**	[0.0091, 0.0095]

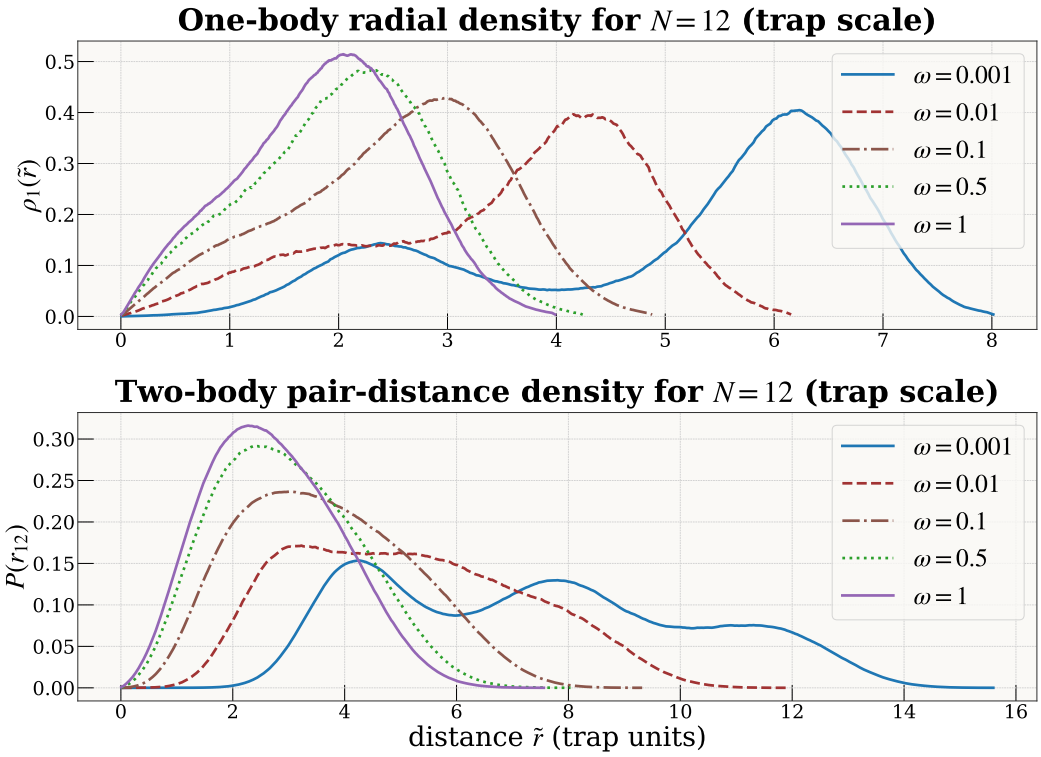


Figure 6.7: As Fig. 6.5, for N=12.

the shell radii contract by an order of magnitude ($r_0 \approx 10$, $r_1 \approx 45$), the outer-shell angular disorder rises to ≈ 0.57. The wording matters here. At $\omega = 10^{-3}$ the $(1, 7, 12)$ label describes a Wigner *topology*, because it is accompanied by separated shells, registered angular order and low Lindemann disorder. At $\omega = 10^{-2}$ and above, with angular order at its random-angle null and disorder high, $(1, 19)$ is what the *radial classifier returns* on a state that is not angularly ordered at all—a decomposition of the radial density, not a crystalline shell structure. We use the occupancy notation in both cases because the detector does, but only the weakest-trap case earns the word topology. At $\omega = 1$ the classifier returns the same $(1, 19)$ partition, with compact radii ($r_0 \approx 0.55$, $r_1 \approx 2.3$) and high angular disorder ($L_{\text{out}} \approx 0.69$). What the data show, then, is a rapid radial reorganisation somewhere between $\omega = 10^{-3}$ and 10^{-2}, not a collapse of one crystalline topology into another: the state at 10^{-2} has no demonstrated angular order to lose. How rapid, and where in that decade it happens, the grid cannot say. It samples one point per decade, and the radial classifier reports a partition at each point rather than tracking a structure between them.

6.2.6 The crossover across diagnostics

Six diagnostics, applied at four particle numbers and five confinements, agree. As ω falls from 1 to 10^{-3} the densities expand and develop rings, and the shell topology resolves into the discrete finite-N patterns that classical Coulomb clusters predict—$(1, 5)$ at N=6, $(3, 9)$ at N=12, $(1, 7, 12)$ at N=20 [2, 3]. The shells separate and stiffen. The Lindemann disorder falls monotonically throughout, while rotation-registered bond order stays within 1% of its random-angle null down to $\omega = 0.1$ and reaches 1.16–1.63 times it only at $\omega = 10^{-3}$ (Appendix F.2). Radial stiffening precedes n-fold angular ordering instead of accompanying it. Lab-frame order stays near zero throughout, as expected for a freely rotating molecule. The energy partitioning approaches the classical virial form, with Γ reaching $\sim$40 for N=12 at the weakest trap. None of these was built into the ansatz. The commensurate $(3, 9)$ sector at N=12 carries real weight in $(1, 11)$ and $(2, 10)$ besides, which is the quantum rotational mixing the rotating-Wigner-molecule picture predicts [4, 11]. At N=20 the three-shell structure at $\omega = 10^{-3}$ gives way, by $\omega = 10^{-2}$, to a radial partition of a state without angular order; the grid is too coarse to place that reorganisation more finely than within the decade. The $(1, 7, 12)$ configuration itself is not new. It is the known classical shelling, reported in quantum calculations of comparable dots, and no novelty is claimed for it. What we have not found reported elsewhere for $N \geq 12$ parabolic dots is the combination of *sector-conditioned* bond-order and Lindemann statistics, read against their random-angle null, with a held-out polygon reconstruction of $g(r)$ scored against an angle-randomised baseline.

The diagnostics do not carry equal weight, and they do not all support the same claims. Table 6.5 sets out which one carries which statement and how firmly. The strongest evidence is at the deepest Wigner points; the qualitative staging is better established than where, within the coarse ω grid, each stage sets in.

Two qualifications bound all of it. The states are the lowest we find within a fixed spin-projection sector, and we have not scanned total spin, so these are not proven global-spin ground states. And the combinatorial reconstruction of $g(r)$ establishes decomposability rather than sampling correctness; the held-out polygon fit of Appendix F.3 is the version that could have failed, and sampling correctness itself rests on the independent restart and chain diagnostics.

The two angular diagnostics stage the crossover slightly differently, and the difference is informative. Radial localisation (expansion, shell separation, falling shell-resolved width, classical force balance) develops progressively from $\omega = 1$ down. Bond-orientational order is flat at its null through $\omega = 0.1$, departs at $\omega = 0.01$ clearly only at N=6, and becomes large at 10^{-3} everywhere. The held-out polygon model, sensitive to any angular correlation rather than to n-fold order specifically, instead grows monotonically throughout, roughly doubling per decade. Read at face value, weak angular correlations build gradually across the whole range while *crystalline* n-fold order appears only in the last decade: a three-stage picture rather than a two-stage one. The two-stage separation of radial and n-fold angular order is the firmer statement; the gradual middle stage is resolved in sign and trend, but its size depends on the shell model used to measure it.

These structural facts then enter the method. The shell radius approaching its classical $\omega^{-2/3}$ scaling—within 1.7% of the classical prefactor at $\omega = 10^{-3}$, Appendix F.4—and the angular order sharpening in the last decade are exactly the quantities a sampler must respect to reach the Wigner regime without a Markov chain. An origin-centred proposal aims where the wavefunction no longer has any mass; a proposal built from the measured shell radius and angular order tracks it, and carries Markov-chain-free training far into the correlated regime (§7.4). What the trained model reveals about the physics is what

Table 6.5: Which diagnostic supports which claim about the crossover. "Firm" means a large effect in a state-level or shell-resolved quantity; "weak" marks evidence that is suggestive but small or carried by a quantity known to mix structures.

Claim	Carried by	Strength
The cloud expands faster than the non-interacting $\omega^{-1/2}$	rms radius from $\langle V_{\mathrm{trap}}\rangle$; r_{mode} exponent	firm, all N
Interactions take over	$\Gamma = \langle V_{\mathrm{int}}\rangle/\langle T\rangle$ rises tenfold	firm, $N \leq 12$
Classical force balance emerges	$2\langle V_{\mathrm{trap}}\rangle/\langle V_{\mathrm{int}}\rangle \to 1.23,\ 1.07,\ 1.05$	firm at N=12
Shells stiffen radially	shell-resolved widths (N=6 ring: $0.32 \to 0.15$) and pair Lindemann ratios; global γ only at N=2	firm; global γ weak at N=6
The ring approaches its classical radius	outer radius over $1.334\,\omega^{-2/3}$: $1.247 \to 1.017$	firm, N=6
Discrete Wigner topologies form	radial classifier with registered order: $(1,5)$, $(3,9)$, $(1,7,12)$	firm at 10^{-3}; above it, radial partitions only
Weak angular correlation builds gradually	held-out polygon model against angle-randomised null, $\Delta\cos$ (block bootstrap)	firm in sign and trend; size depends on the jitter model
n-fold angular order appears late	$\lvert\Phi_n\rvert$ against its exact null, which is also the pipeline's null	firm at 10^{-3}; absent above 0.1 (resolved departures $\lesssim$1%)
Where each stage sets in	one grid point per decade	not established

later lets us train it more cheaply.

That is the state seen from outside. What the network had to hold internally in order to produce it is a question about the solver, and Chapter 7 takes it up.

Chapter 7

Architecture, optimisation and sampling

The previous chapter showed that the ansatz works: it reaches reference quality wherever a reference exists, and the states it produces carry the staged Wigner-molecule structure the diagnostics were built to detect. Why the solver can represent those states, and what governs whether they can be trained at all, are separate and harder questions.

Three observations from that chapter remain unexplained. Two networks of the same size, trained the same way and landing on the same energy, need not be the same object. A natural-gradient step that is decisive in one training regime does nothing in another. And an importance-sampled objective reproduces a Markov chain's answer in one corner of the phase diagram and departs from it in the next.

All three can be examined through the per-sample, column-centred log-derivative matrix

$$O_{ki} \;=\; \frac{\partial \log |\Psi_{\boldsymbol{\theta}}(\boldsymbol{x}_k)|}{\partial \theta_i}, \qquad k = 1, \ldots, B, \;\; i = 1, \ldots, P, \tag{7.1}$$

and on its two Gram matrices, the quantum geometric tensor $S = O^{\top} O / B$ and the neural tangent kernel $K = O O^{\top}$ [27, 28]. That gives the chapter its structure:

> *Architecture* determines the shape of the tangent space.
> The *optimiser* acts on it through its metric.
> The *sampling measure* determines how well that metric can be estimated.

The chapter takes them in that order. The instrument throughout is the *effective dimension* of the tangent space—the participation ratio of the QGT spectrum $\{\lambda_a\}$,

$$d_{\mathrm{eff}}(S) \;=\; \frac{\left(\sum_a \lambda_a\right)^2}{\sum_a \lambda_a^2}, \tag{7.2}$$

which counts how many directions actually carry Fisher weight, whatever the parameter count says. Beside it sit the local-energy variance $\mathrm{Var}(E_L)$, zero only for an exact eigenstate; the effective sample size; and the condition number $\kappa(S)$, which is reported in §7.2 and turns out to be large everywhere and therefore diagnostic of nothing by itself.

Architectures are compared on a common probe set—the same configurations for every model, never each model's own $|\Psi|^2$—and below $\omega = 0.1$ every relative error is measured against our own converged variational energies, so it reports internal consistency rather than accuracy. The tooling is anchored on the analytic two-electron ground state, where the ansatz returns $E = 3.000127$ Ha (+0.004%) at squared overlap 1.000000.

7.1 Message passing and the variational geometry

What does a message-passing network learn that a separable one cannot? Both learnable objects of the ansatz can be built in either form, so the question is asked twice, changing one object at a time. In the first comparison only the *correlator* is swapped, a separable network against the message-passing one, while the backflow, the Slater reference, the analytic cusp, the training protocol and the capacity tier all stay as they are. In the second only the *backflow* is swapped, single-round pairwise messages against the copresheaf network, with the correlator now among the things held fixed. Neither comparison varies the two together.

7.1.1 The correlator: tangent-space compression

At $N = 6$, $\omega = 1$ the two correlators are indistinguishable in energy. Trained to DMC quality across a 20k–164k parameter ladder, every model lands within $\pm 0.1\%$ of the reference, and the best raw energy belongs to a 164k separable network—twice the message-passing parameter count. Both have saturated the variational floor.

Their tangent geometry is not indistinguishable. On a common probe set the message-passing correlator reaches a reference-quality state in $d_{\text{eff}} \approx 1.2$–$1.7$ over three seeds, where every separable network needs 2.9–3.6; the two sets do not overlap, and their closest approach is still a factor of 1.8. Capacity does not close the gap. Scaling the separable network eightfold, from 20k to 164k parameters, moves its effective dimension only from 3.6 to 2.9, and the 89k network that matches the message-passing one in size sits at 3.6. Nor does the probe measure: the gap moves by at most 0.26 whether scored on either

Two things fix the size of that effect. At the crystal, where a seeded replication exists, the three-seed sets do not overlap—$(1.95, 1.63, 1.96) \times 10^{-5}$ against $(3.40, 3.23, 3.41) \times 10^{-5}$—and their means differ by 1.8, not the 7.4 obtained by pairing single best runs, which is why the seeded ratio is the one quoted. And capacity works the other way: the production rows give the message-passing network a 1.7× size advantage, but repeating $\omega = 1$ against the capacity-matched 89.4k separable network widens the ratio to 3.4 instead of closing it. Intermediate confinements ($\omega = 0.28$, 0.05, 0.03) reproduce the ordering at ratios 1.2–1.8. It is not resolved at the smallest depth-analysis tier (12k against 9k), where the two sit within 8%, so this is a statement about trained models at production width rather than about the architectures at any size.

Table 7.1: Local-energy variance at N=6, matched training protocol, both architectures at reference-quality energy ($|\mathrm{err}| \leq 0.13\%$). Lower is better; $\mathrm{Var}(E_L) \to 0$ only for an exact eigenstate. The basis column gives the evidential status of each row: the $\omega = 0.01$ row is a three-seed mean, the rest are single runs of the production cascade. The unseeded rows establish the ordering, which never reverses, and not the precise factor (§7.1.1). The last row repeats $\omega = 1$ against the capacity-matched 89.4k separable network.

ω	CTNN	DeepSet	ratio	basis
1.0	1.09×10^{-2}	2.51×10^{-2}	2.3	single run, 79.8k vs 47.7k
0.5	4.62×10^{-3}	1.01×10^{-2}	2.2	single run, 79.8k vs 47.7k
0.1	5.63×10^{-4}	1.14×10^{-3}	2.0	single run, 79.8k vs 47.7k
0.01	1.85×10^{-5}	3.35×10^{-5}	1.8	3-seed mean, 79.8k vs 47.7k
1.0	2.75×10^{-2}	9.38×10^{-2}	3.4	matched, 79.8k vs 89.4k

Naming the leading tangent directions, at every size. The leading tangent directions are not abstract. At the exact $N = 2$ anchor an operator decomposition identifies them, to $R^2 = 1.00$, as the physical collective modes of H—a breathing mode (∂_ω), a correlation-hole mode, and the first relative excitation. This is not special to two bodies. Projecting the leading tangent mode of trained correlators onto a dictionary of physical operators (breathing/monopole, quadrupole, correlation-hole, pair-linear), seed-averaged over well-trained checkpoints (energy within 1.5% of reference), the CTNN reaches $R^2 \gtrsim 0.93$ at $N = 6$ and $R^2 \approx 1.00$ at $N = 12$ across the whole confinement range $\omega \in [0.01, 1]$, and both architectures reproduce the $N = 2$ naming at strong confinement (Fig. 7.2). In the message-passing correlator, the direction carrying most of the tangent weight is one of the handful of coordinates the Hamiltonian actually couples.

The dominant mode is not the tangent space. Figure 7.2 scores the *dominant* eigenmode against the dictionary; Fig. 7.3 asks where the eigenvalue-weighted mass of the leading modes *as a whole* lies. A network can place its largest mode squarely inside the dictionary and spend most of the rest outside it, which is what the separable network does—so the leading-mode score is generous to it. At $\omega = 1$ it is nominally the more physical of the two ($R^2 \approx 0.996$ against 0.991), with energies identical to four figures. Write the eigenvalue-weighted fraction of th0

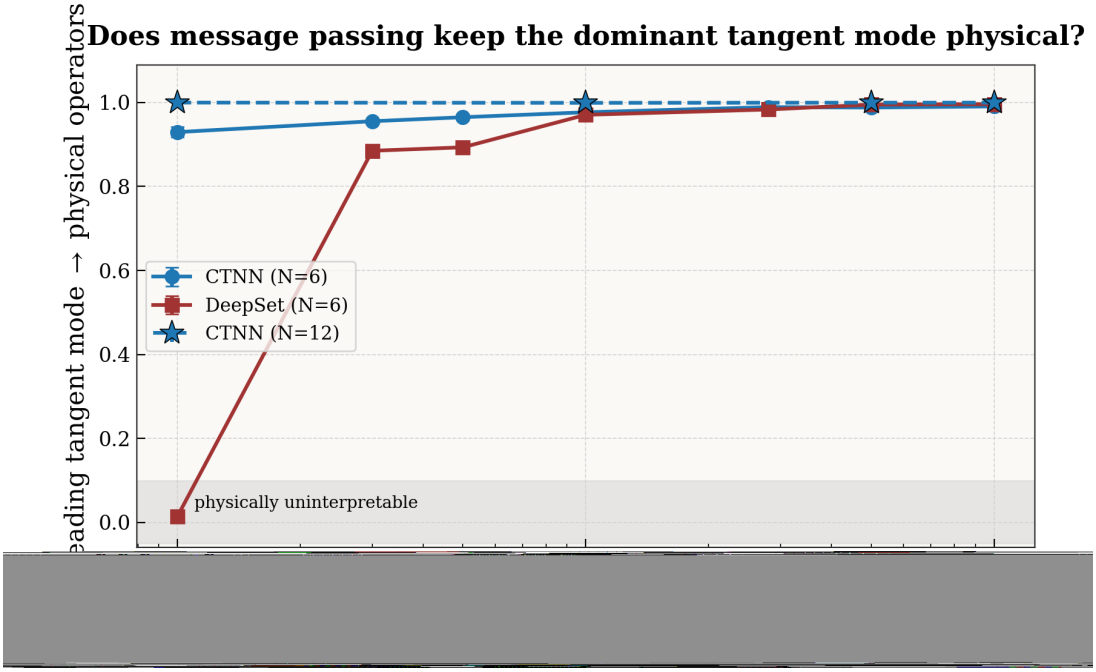


Figure 7.2: The leading tangent mode is a nameable physical mode. R^2 of the correlator's leading tangent-kernel eigenmode onto the physical-operator dictionary, seed-averaged over well-trained checkpoints. The CTNN stays physical at every ω and both sizes; the DeepSet matches it down to $\omega \approx 0.1$ and then loses the leading mode entirely in the deep-Wigner regime. Three seeds at $\omega = 0.01$.

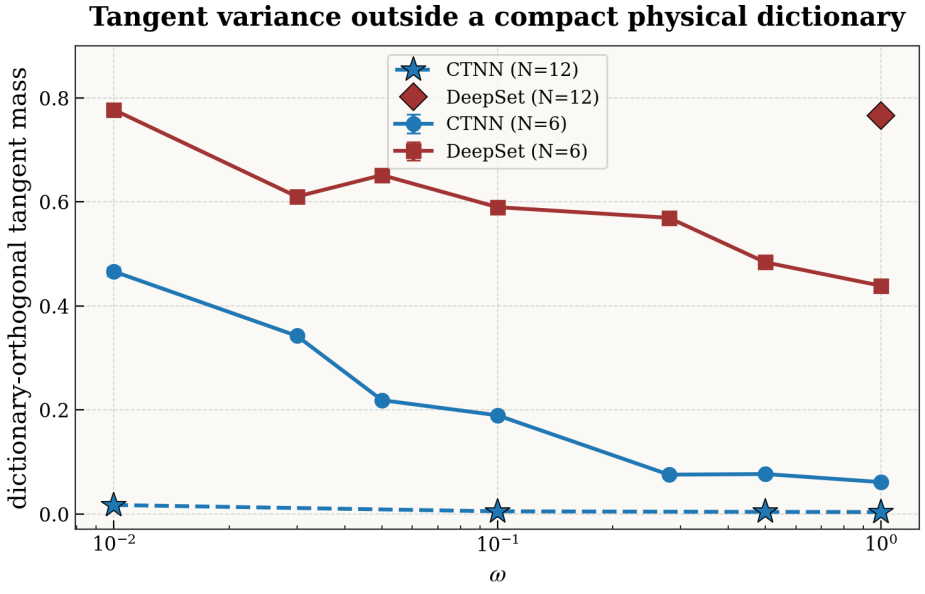


Figure 7.3: A limitation visible where the energy cannot see it. Dictionary-orthogonal tangent mass—the eigenvalue-weighted tangent variance outside the span of a compact physical-operator dictionary—against ω. At $\omega = 1$, where the two correlators agree in energy to 0.001%, the DeepSet places 44% of its tangent variance outside the dictionary and the CTNN 6%; the excess persists at every ω.

7.1.2 The backflow: rank collapse and its absence

The contrast is sharper in the backflow, and also less controlled. Here we fix the correlator, the data, the protocol and the optimiser—the matrix-free CG solve of §4.4, using the Woodbury identity to apply $(S+\lambda I)^{-1}$ through the $B \times B$ kernel rather than the $P \times P$ metric, which is what makes it affordable at $B \ll P$—and vary only the backflow architecture (conventional single-round messages vs. the copresheaf network), training both to convergence.

Unlike the correlator comparison, this one is *not* capacity-matched. At the shared production width the copresheaf network carries 182k parameters against the conventional network's 51k, because a persistent edge state is intrinsically heavier than a per-particle aggregate (§3.6); no matched pair was trained. A raw energy gap between the two is therefore not by itself evidence about architecture. What does separate the two explanations is set out after the rank measurement, and it is internal to the conventional network rather than a comparison between networks. At $\omega = 1$ both reach reference quality, but they diverge as the trap weakens (Table 7.2): the CTNN backflow stays within 0.01–0.12% across all N, ω, while the conventional backflow degrades to +0.58% at $N = 6, \omega = 0.01$ and $\sim 6\times$ the CTNN error at $N = 20, \omega = 0.01$ (+0.69% vs. +0.12%).

Table 7.2: Relative energy error (%) of the two backflow architectures, seed-averaged.

	$N = 6$		$N = 12$		$N = 20$	
ω	CTNN	conv	CTNN	conv	CTNN	conv
1.0	−0.03	+0.06	−0.01	+0.04	+0.06	+0.07
0.1	−0.05	+0.16	+0.01	+0.16	+0.02	+0.18
0.01	−0.04	+0.58	−0.01	+0.54	+0.12	+0.69

How component-level numbers are read. The measurement that follows is of a different kind from the energy, and the difference governs how much it can carry. Energy, $\mathrm{Var}(E_L)$ and the overlap between two trained states are *state-level*: properties of the wavefunction, unaffected by how the parameters are arranged. Quantities defined on a single component (the backflow's rank, its displacement magnitude, a per-component d_{eff}) are not, because the correlator and the backflow are substitutes and the division of correlation between them is a nearly flat, path-dependent direction of the loss. They record the route the training took as much as the state it reached.

That is not a hypothetical worry. An artefactual rank collapse appeared earlier in this project under joint from-scratch training, and it needs separating from the one reported below, since they are the same observable. The artefact affected *both* architectures, moved with the seed, and carried no energy penalty. The collapse reported here appears under the staged protocol used throughout the chapter, affects only the conventional backflow, is seed-robust, and comes with half a percent of energy. It is the energy that makes it a finding rather than a diagnostic curiosity.

The energy gap has a sharp structural signature. Let the *backflow rank* be the participation ratio of the displacement field $\mathbf{\Delta x}$ (how many independent directions the backflow moves electrons along). The field lives in $2N$ dimensions—12, 24 and 40 at $N = 6, 12, 20$—and the zero-mean projection of Eq. (3.18) removes two of them.

The two columns of Table 7.3 sit under different ceilings, because the diagnostic is evaluated on each network's returned field and the two apply the zero-mean projection at different points (§3.5.3): $2N - 2$ for the copresheaf network, $2N$ for the conventional one. Compare the collapse, not the level. Toward Wigner the *conventional* raw field undergoes a spectral collapse to rank 1, while the CTNN backflow stays near the ceiling (Table 7.3). The endpoint is the same at every size—rank 1, from ceilings of 12, 24 and 40 at $N = 6$, 12 and 20—so what grows with N is the height of the fall, not the sharpness of it. At $N = 6$ and $N = 12$ both seeds return exactly 1.00; the $N = 20$ entry comes from a separate re-run of that cell and is single-sourced (see the table note). The tangent dimension tells the same story from the other side: toward Wigner d_{eff} grows for CTNN ($\sim 4 \to 10$) and collapses for the conventional backflow ($\sim 6 \to 1$). The collapse to *exactly* rank 1 is seed-robust where two seeds exist ($N = 6$ and $N = 12$). Rank is not the discriminator above the crystal: at $\omega \geq 0.1$ the conventional field is nominally the higher-rank of the two (Table 7.3), though that comparison is confounded by the differing ceilings noted above, and the two architectures are then also within 0.2% of each other in energy. It is the collapse, not the level, that carries the finding.

Capacity is the obvious alternative reading. The conventional network is a third the size of the copresheaf one, so it may simply be running out of parameters at weak confinement. Its own ω-dependence argues against that. Capacity does not vary with ω, yet the same 51k network on the same protocol holds a displacement rank of 10.4–10.8 at N=6, 21.7–22.3 at N=12 and 33.7–38.7 at N=20 for every $\omega \geq 0.1$—within a unit or two of its own ceiling—and then falls to exactly 1.00 below it. A network too small throughout the grid would not behave that way. What this does not exclude is a regime-dependent limitation, since the target also becomes harder at weak confinement. Without a widened conventional backflow that confound stands, and that control—a conventional backflow at the copresheaf network's capacity—is the most important one this chapter lacks. Until it is run, the rank collapse is an observation about these two networks, not a demonstrated consequence of architecture.

This argument does not remove the capacity confound in Table 7.2. That gap is a between-network comparison at unequal size, and nothing in our data separates architecture from the extra 131k parameters. It is reported as what the two production ansätze achieve—these being the models that produced the energies of Chapter 6—and not as an architecture ablation.

Table 7.3: Backflow displacement rank (participation ratio), seed-averaged, evaluated on the fier736(data)-22 1h6]TJ/F31

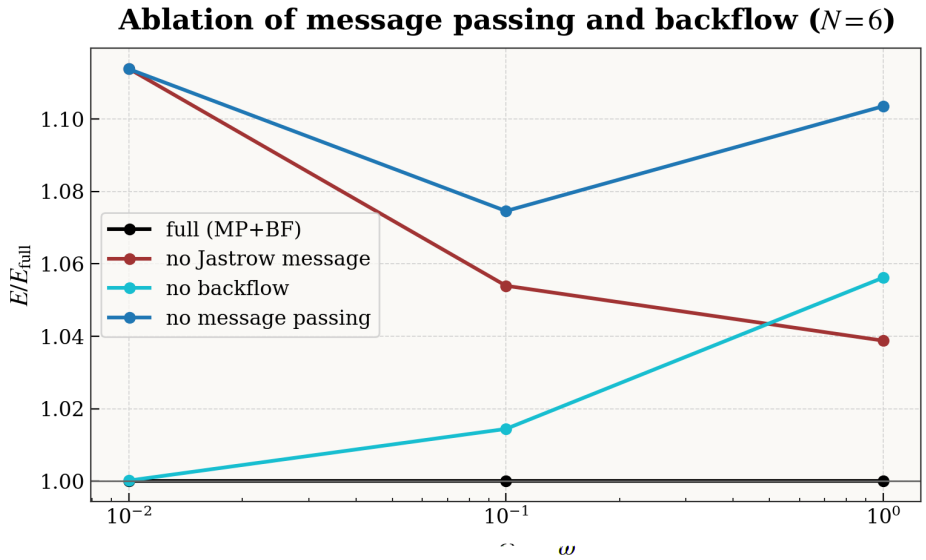


Figure 7.4: The *fixed-measure instantaneous ablation cost* of each component (trained networks, N=6). Configurations are drawn once from the full model's $|\Psi|^2$ and every arm is re-evaluated on them, so $\Delta E_{\text{probe}}/E_{\text{full}}$ is the change in the kinetic term when a component is switched off, not the energy of a retrained network. Cyan: backflow removed; dark red: Jastrow messages removed; blue: all message passing removed. The curves cross near $\omega \approx 0.4$. The division of labour shown is conditional on the training protocol (§7.1.2).

does maintain one, and moves it back and forth between nodes and edges through learned transport maps $(\rho_{v\to e}, \rho_{e\to v})$; the message-passing correlator iterates this exchange over a V-cycle, while the backflow applies a single such copresheaf round. Where a compact representation exists this *compresses* the correlation onto fewer tangent directions (Jastrow: smaller d_{eff}); where the physics demands a genuinely collective, full-rank response the single-round baseline, at the width tested, instead collapses to a single collective displacement mode (backflow: rank 1), which the copresheaf transport avoids. Both effects are visible only in the tangent-kernel geometry, not in the total energy, until weak confinement makes the geometric deficit an energetic one.

Two components failing at similar confinements is a resemblance; what makes it evidence is that moving one of them moves the other. To test this we held the backflow architecture fixed, the conventional network in both arms, swapped only the correlator, retrained each arm to convergence, and tracked both quantities down the ω-sweep.

The backflow's failure boundary moves with the correlator. With a separable correlator the two give way together near $\omega \approx 0.05$: correlation-hole capture falls from 0.68 to 0.07 and the displacement rank from $\sim$10 to 1 across the same step. With a message-passing correlator both hold half a decade further, to $\omega \approx 0.01$ (Fig. 7.5).

Each arm is a separately trained network, not one network reconfigured, so "held fixed" means the same architecture and recipe, never the same weights. Within that, intervening on one component relocates the other's failure, which is what a shared cause predicts and a coincidence does not. The correlator's grip on the physical correlation modes and the backflow's ability to sustain a full-rank displacement look like one capability seen from two sides: when the amplitude model can no longer resolve the correlation hole, the nodal deformation can no longer stay high-rank, and a better correlator postpones both.

7.1.4 What the trained representation looks like

One descriptive result belongs beside these controlled comparisons. Examining a single high-quality trained solution shows the correlator occupying a feature space of one to three effective dimensions across the entire grid and never expanding: at $\omega = 10^{-3}$, where the real-space density has developed rings and discrete occupancies, the latent rank is no larger than at $\omega = 1$. What changes across the crossover

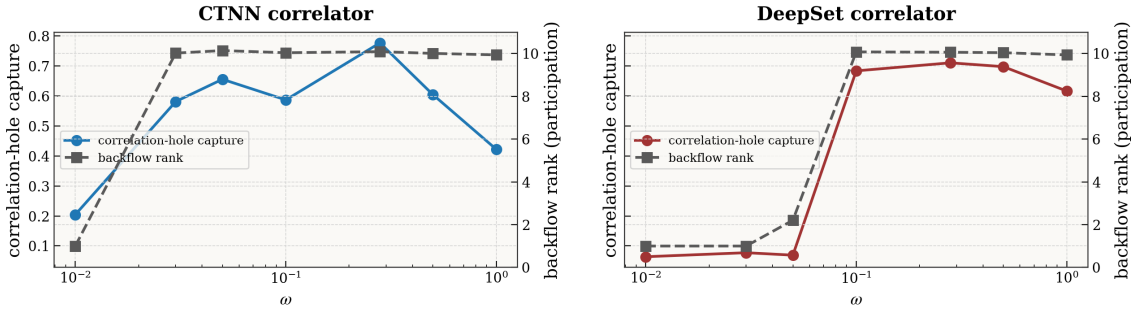


Figure 7.5: Correlator physicality and backflow rank collapse together ($N = 6$; the backflow architecture is held fixed, only the correlator changes). For each correlator, the correlation-hole capture (left axis) and the backflow displacement rank (right axis) give way at the same ω: $\omega \approx 0.05$ for the DeepSet, $\omega \approx 0.01$ for the CTNN. A correlator that keeps its physical structure lets the backflow keep its rank.

is which branch the leading axis draws on, not how many axes there are. The displacement field is the opposite kind of object, spanning nearly all the spatial directions available to it and leaving the correlator's feature geometry almost unperturbed.

These are component-level measurements on one run. They are gauge-like—the correlator and backflow are partial substitutes, so the division of labour between them records the training route as much as the state—and no architectural conclusion above depends on them. They are recorded because a low effective covariance rank is not what one would predict for a state whose real-space structure is this elaborate, and because the reorganisation happens near $\omega \approx 0.1$, the same confinement at which the energy partition crosses over (§6.2.3) and the ablation costs of Fig. 7.4 exchange order. Whether that coincidence reflects one mechanism or three we cannot say from a single run. Appendix E gives the full analysis.

Message passing therefore changes the variational geometry before it changes the energy. In the correlator, under the parametrisations used here, it shows as Fisher weight concentrated in fewer directions at strong confinement and better accounted for by a compact operator dictionary throughout; and, independently of any parametrisation, as a local-energy variance lower at every confinement by factors of 1.2 to 3.4. In the backflow the conventional network, at a third of the parameter count, collapses to rank one and loses half a percent where the message-passing field does not; without a capacity-matched conventional backflow that contrast does not separate architecture from size. The correlator swap, which does hold the backflow architecture fixed, is what links the two. What none of it establishes is that the two architectures reach different *states*: they agree in energy to four figures, and we did not measure their overlap here.

7.2 Conditioning and the sampling measure

Two design choices remain, and the tangent geometry settles both at once. One asks what a natural-gradient step buys over a diagonal one. The other asks what changes when the loss is defined by an analytic proposal instead of by $|\Psi|^2$. They are the same question twice over: stochastic reconfiguration inverts the metric S, and the sampling measure is what builds S in the first place.

7.2.1 Natural-gradient preconditioning

Stochastic reconfiguration preconditions the gradient by the inverse QGT, $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta\, S^{-1} \nabla_{\boldsymbol{\theta}} E$; geometrically it *whitens* the tangent kernel. At $N = 2$ this is exact: the SR step aligns with the imaginary-time direction to numerical precision. The question is where whitening changes the outcome.

Fixing the ansatz and varying only the optimiser from a common stage-3 base: **for $|\Psi|^2$-VMC, SR and Adam are indistinguishable**—exact at $N = 2$ (var $\sim 10^{-9}$), and at $N = 6$ the energy gap $E_{\mathrm{Adam}} - E_{\mathrm{SR}}$ is $\lesssim 0.002$ across ω and both seeds; in the matched pair at $\omega = 1$ the two agree to

four decimals in relative error. **For weak-form collocation from the same common base, SR is decisive**: there $E_{\mathrm{Adam}} - E_{\mathrm{SR}}$ is large and positive (e.g. +2.9 at $N = 6, \omega = 1$), and plain Adam does not converge the objective.

That statement is about a controlled comparison, and it needs reconciling with §7.3.1, where Adam is the production default and the entire 30-run reliability campaign uses it successfully. Both are true, and the difference between them is the rest of the pipeline. Stripped to a bare objective and a fixed proposal, the collocation gradient is badly enough behaved that a diagonal step stalls and preconditioning rescues it. The production recipe never runs in that condition: adaptive proposal refitting, 16× oversampling, reward normalisation, instability rollback and cascade warm-starting between confinements are all aimed at the gradient's conditioning at its source, and once they are in place a diagonal step suffices—though which of them does the work was not isolated (§7.3.4). On this reading the two findings are complementary—SR substitutes for sampling control, and sampling control substitutes for SR—with one asymmetry that decides the production choice: below $\omega \simeq 0.01$ the sampling that builds S collapses, so the substitution stops working in one direction only (§7.3.3). Adam is the default because it degrades gracefully where SR does not.

A large condition number alone does not explain the difference. Table 7.4 reports $\kappa(S) = \lambda_{\max}/\lambda_{\min}$ measured on the trained states under $|\Psi|^2$ sampling, and it is enormous, 10^7 to 10^{10}, for *both* architectures and at every capacity in the ladder, including precisely the models on which Adam and SR agree to four decimals. Whatever makes SR unnecessary under $|\Psi|^2$ sampling, it is not that the metric is well conditioned.

Table 7.4: Condition number of the empirical quantum geometric tensor at N=6, $\omega = 1$, measured on each trained state's own $|\Psi|^2$ samples at fixed batch $B = 768$. Because $B < P$, $\lambda_{\min}$ is the smallest sample-supported eigenvalue and κ is a property of the metric the optimiser actually sees (§5.3). For the same reason $\lambda_{\min}$, and with it κ, is sensitive to the batch size, the regularisation and the numerical threshold below which eigenvalues are discarded, so the values are comparable to one another at fixed settings and indicate orders of magnitude, not more. Adam and SR reach the same energy on every row of this table.

architecture	params	$\kappa(S)$	d_{eff}
CTNN (seed 1)	79.8k	6.9×10^7	1.46
CTNN (seed 2)	79.8k	5.2×10^7	1.39
DeepSet (small)	20.2k	3.2×10^9	3.56
DeepSet (medium)	47.7k	3.4×10^7	4.35
DeepSet (matched)	89.4k	3.2×10^{10}	3.61
DeepSet (large)	164k	1.4×10^7	4.76

Our working hypothesis is that SR matters when the tangent metric is ill-conditioned and still estimable, and the second half of that sentence does the work. Anisotropy is present under $|\Psi|^2$ sampling, where we measured it. We did not measure κ under the collocation measure at matched batch size, so the table supports only the negative half of the argument: conditioning does not distinguish the regime where SR is unnecessary. What separates the two regimes, on the evidence we do have, is whether a batch can resolve the metric. Under $|\Psi|^2$ the samples are drawn from the very measure that defines S, so the directions carrying Fisher weight are also the directions the batch populates, and Adam's diagonal step already moves along them. Under a fixed covering proposal the estimator of S is built from reweighted points that increasingly miss the target, and preconditioning supplies structure the diagonal cannot. We have not measured κ under the collocation measure at matched batch size, so this is an argument from the estimator, not a second condition-number comparison; what the table establishes is the negative half—that raw conditioning does not distinguish the two regimes.

"More ill-conditioned ⇒ more useful SR" cannot hold without bound, since inverting S requires S itself to be *reliably estimated*. SR helps when the tangent metric is anisotropic and its empirical estimate is trustworthy. Once the sampling that builds S collapses, as the importance weights do deep in the Wigner regime, the empirical S becomes noisy and rank-deficient, and preconditioning by its inverse amplifies the noise instead of the signal. This is exactly the behaviour seen at ultra-weak confinement, where CG-SR on the collocation objective becomes unstable (§7.3.3). SR pays off in the window where the metric is both ill-conditioned and estimable, and fails once estimation itself breaks down.

The VMC equivalence of Adam and SR is established here at $N = 2$ and $N = 6$, and we have not verified that it survives to $N = 12$ and $N = 20$. The mechanism we propose—that $|\Psi|^2$ sampling makes the metric *estimable* on the batch, whatever its conditioning, and so lets Adam's diagonal preconditioning

State agreement degrades with proposal overlap. We test whether the two paradigms reach the same *state* (not merely the same energy) via the symmetric overlap $\mathcal{S}^2 = \langle \Psi_B/\Psi_A \rangle_{|\Psi_A|^2} \langle \Psi_A/\Psi_B \rangle_{|\Psi_B|^2}$. At $N = 2$ they are identical ($\mathcal{S}^2 > 0.997$ for every ω, both optimisers, both seeds): the paradigm is efficiency-only, collocation paying $\sim 5\times$ the variance for the same wavefunction. At $N = 6$ they *diverge*: under Adam $\mathcal{S}^2$ falls from 0.85 at $\omega = 1$ to 0.77 at $\omega = 0.1$ (both seeds), tracking the ESS collapse, and at $\omega = 0.01$ the estimator returns values near zero but is no longer resolved—at an ESS of about five points it is a ratio of two degenerate importance-weighted averages, and we record it as *unresolved* rather than as a measured orthogonality. SR partially rescues the $\omega = 0.1$ value ($\mathcal{S}^2 \approx 0.95$) but does not close the gap. Weak-form collocation thus has a **domain of validity**—equivalent to VMC when the importance sampling is healthy, degrading as correlation strengthens. The two halves of this section meet here: the paradigm that needs SR is the one that breaks down at scale. An adaptive proposal that tracks $|\Psi|^2$ is the natural remedy, pursued in §7.4.

Why the collocation energy readout is biased. For an exact eigenstate the local energy $E_L = \hat{H}\Psi/\Psi$ is constant over configuration space, so $\mathrm{Var}(E_L) \to 0$ as the ansatz approaches the ground state—the zero-variance principle, and the reason $\mathrm{Var}(E_L)$ is a sharp quality measure. That cancellation is a property of the *full* local energy, in which the kinetic (Laplacian) and potential terms combine to a constant; it is not a property of the Laplacian term alone. The collocation objective does *not* discard the Laplacian: it is a score-function (REINFORCE) estimator that *forms* the full E_L as a detached reward and differentiates only $\log|\Psi|$ (§7.3.1). What it gives up is the *measure*, and three things then spoil the readout, in increasing order of severity. The estimator is self-normalised, since $|\Psi_\theta|^2$ is unnormalised, so it carries an $\mathcal{O}(1/n)$ finite-sample bias that vanishes with sample size. The tail controls deliberately change the target: tempering $w \to w^\alpha$ and clipping the log-weights trade bias for variance on purpose, so the estimated quantity is no longer $\langle E_L \rangle_{|\Psi|^2}$. And, dominant in the regimes that matter, proposal–target mismatch inflates the variance beyond any affordable sample size (§7.3.4). Importance sampling is not biased merely for being importance sampling; it is the last two that make the readout unusable as a stopping rule. The reliable criteria are $\mathrm{Var}(E_L)$ and the state overlap, both on $|\Psi|^2$ samples, and for the same reason no energy reported in this thesis is an importance-sampling estimate: every one is a Metropolis evaluation of $|\Psi|^2$.

The value of a natural-gradient step is therefore not predicted by the conditioning of the metric, which is large under $|\Psi|^2$ sampling on exactly the models where SR buys nothing. What predicts it, on our reading, is whether the batch can resolve the metric it inverts. The negative half of that is directly measured; the positive half—that collocation occupies an anisotropic but still estimable middle regime—is inferred, since κ was never measured under the collocation measure at matched batch size. It is a hypothesis supported by the pattern of failures, not an established criterion. A decisive test would measure the stability of the leading eigenvalues across independent batches, a bootstrap uncertainty on the resolvable part of the spectrum, its sensitivity to batch size and regularisation, the same spectrum under both $|\Psi|^2$ and the collocation measure, and the agreement between the estimated natural-gradient direction and an independently evaluated update.

7.3 Training without a Markov chain

The conditioning story has a practical edge. If the sampling measure is what makes the metric estimable, then choosing that measure is a design decision, not a fact of nature—and the obvious thing to try is a measure with no Markov chain in it at all. "Markov-chain-free" refers to the gradient throughout: no configuration entering a parameter update comes from a chain, but a Metropolis probe still selects checkpoints and certifies the final energy, and the cascades start from reference-guided pretrained states (§8.2 states the claim exactly).

Two walls. What follows is easier to hold if the destination is given first. The Markov-chain-free route runs into two limits of entirely different kinds, and almost every result in §7.3–§7.4 is located against one or the other.

The first wall is statistical. A fixed analytic proposal loses overlap with $|\Psi_\theta|^2$ as the trap opens; the importance weights grow heavy-tailed; the gradient estimator comes to be dominated by a handful of draws; the empirical metric becomes unreliable and preconditioning by it does harm. Everything on that

chain is a sampling problem, and a better proposal moves the wall—from $\omega \approx 0.01$ with an origin-centred mixture to $\omega = 0.0035$ with one built out of the measured Wigner geometry.

The second wall is not. Below $\omega = 0.0035$ a converged state fails to survive transfer to a weaker trap even when the sampler is initialised directly on the correct shell. The Wigner peak exists only through a near-cancellation between an exponentially suppressed determinant and a Jastrow that amplifies it back, and a 23% change in ω tips that balance. No change of proposal repairs it, because the proposal is not what fails.

§7.3 establishes the first wall and the mechanism behind it; §7.4 moves it and then runs into the second.

A Markov chain is what makes neural variational Monte Carlo sequential. Chains must equilibrate, successive samples are correlated, and mixing times grow with particle number and with correlation strength. They grow fastest, in other words, exactly where the physics gets interesting.

Draw the configurations instead from a fixed analytic proposal and reweight them, and the whole gradient step becomes embarrassingly parallel. The question is what that costs.

Two ways of posing the problem behave very differently, and the difference decides everything downstream. The *strong form* asks the wavefunction to satisfy the local Schrödinger equation pointwise, minimising the variance of E_L and differentiating through the Laplacian to do it. The *weak form* uses a score-function (REINFORCE) gradient in which the Laplacian enters only as a reward and is never differentiated. Appendix C shows that the strong form meets a structural obstruction the moment it is asked to train a backflow: the learning signal lives at the nodal surface, where E_L diverges for any imperfect node, and no combination of sampling, loss or preconditioning breaks an accuracy floor near 0.3%—a floor set by the Jastrow alone, with the backflow contributing almost nothing. That is a property of the objective, not of the idea. Everything below uses the weak form.

On the same Slater–Jastrow–backflow ansatz, and with no Markov chain in any gradient, the weak form comes within about a tenth of a percent of DMC at strong confinement and within three tenths of a percent of our own converged energies well into the correlated regime for $N \in \{6, 12\}$ (Table 7.5). The energies are collected here, where the method that produced them is explained, rather than beside the SR–VMC energies of §6.1: without the failure mechanism of §7.3.4 in hand, the pattern of where they hold and where they give way is not interpretable.

7.3.1 The weak-form objective

The collocation training used here is exactly the Markov-chain-free pipeline specified in the Methods: a weak-form (REINFORCE [46]) objective in which the local energy enters as a detached reward (§4.5, Eq. (4.8)), an analytic Gaussian-mixture proposal reweighted to $|\Psi_\theta|^2$ with ESS / Pareto-$\hat{k}$ monitoring [47] and tail controls (§4.5.1), and—in the Wigner regime—the physics-informed ring proposal with cascade warm-starting (§4.5.2). Two methodological findings shape what follows. First, among the optimisers (§4.4), **Adam** was the robust default across all regimes, whereas **CG-SR** converged faster at $\omega \geq 0.1$ but diverged for $\omega \leq 0.01$, where the empirical metric becomes statistically unresolved. Second, **cascade warm-starting**, initialising each confinement from the converged checkpoint of a stronger one, was decisive for reaching the deep-Wigner regime, exploiting the transfer of correlation structure across ω.

Three ansatz families were tried within this framework, all built on the same Slater–Jastrow core and differing only in what dresses it; their definitions and sizes are in §3.6. The **backflow+Jastrow** family produced the best results at N=6 and N=12 and carries every number below. The **Jastrow-only** family was the fastest and most robust to train, served as the initialiser for the backflow runs, and becomes the better choice at N=20 for the memory reason of §7.3.5. The **neural Pfaffian** [48], which replaces the determinant with a more general antisymmetric form and in principle offers richer nodal topology, reached only +0.43–0.50% at N=6, ω=1 against +0.07–0.10% for backflow+Jastrow: its extra expressivity was not exploitable under this sampling regime, because the larger parameter space amplifies importance-sampling gradient noise faster than the richer nodes repay.

Why the weak form is stable where the strong form is not. The catch-22 analysis (Appendix C) identified a fundamental obstacle for direct PDE-residual training of the many-body Schrödinger equation: the backflow learning signal is concentrated near nodes and coalescence points where the local energy E_L diverges, but these are precisely the configurations where second-derivative pathways are ill-conditioned. Backpropagating through the Laplacian in such regions produces pathological gradients that destabilise training.

The REINFORCE formulation resolves this by construction. The Laplacian enters only through the *reward* (the local energy in Eq. (4.2)), and gradients act exclusively on the score function $\nabla_\theta \log |\Psi|$. The

Table 7.5: Variational energies from Markov-chain-free collocation training, all rows using the backflow+Jastrow ansatz. No Markov chain enters any gradient; every energy is a Metropolis evaluation of $|\Psi|^2$. For N=6 and N=12 it is an independent re-evaluation of the saved checkpoint, made for this table with four separately seeded chains of 102 400 samples each and a blocking analysis; parentheses give the larger of the blocked and the between-chain standard error. For N=6 the headline number is the *reliability of the protocol*, not its luckiest run: the campaign of §7.3.4 fixes a 1200-epoch budget and the hyperparameters and varies only the seed, and both the three-seed mean ± standard deviation and the best of the three are given. The N=2 rows are single collocation runs, zero-variance extrapolated. The N=12 rows are marked †: they are single continuation runs whose final training stage used the corrected proposal density (Appendix G); no campaign was run at that size, so they are not protocol-controlled—read them as what the method reached, not as what it reaches reliably. The reference column is the external value where one exists and, where none does (N=6 at $\omega \leq 0.01$ and N=2 at $\omega = 10^{-3}$; see Table 6.2), our own converged SR–VMC energy, in which case the deviation measures internal consistency rather than absolute accuracy. Deviations are $100\,(E - E_{\mathrm{ref}})/E_{\mathrm{ref}}$.

				relative deviation (%)	
N	ω	Reference	Best energy	best	mean ± s.d.
2	1.0	3.00000	3.00035	+0.012	—
	0.5	1.65977(1)	1.66014	+0.022	—
	0.1	0.44079(1)	0.44077	−0.005	—
	0.01	0.073839(2)	0.073840	+0.001	—
	0.001	0.013778	0.013799	+0.152	—
6	1.0	20.15932(8)	20.1731(18)	+0.068	+0.082 ± 0.014
	0.5	11.78484(6)	11.7952(12)	+0.088	+0.098 ± 0.009
	0.1	3.55385(5)	3.56393(24)	+0.284	+0.293 ± 0.009
	0.01	0.69036	0.691895(80)	+0.222	+0.244 ± 0.020
	0.001	0.140832	0.141175(4)	+0.244	+0.282 ± 0.059
12†	1.0	65.7001(1)	65.7207(17)	+0.031	—
	0.5	39.1596(1)	39.1727(12)	+0.034	—
	0.1	12.26984(8)	12.28361(32)	+0.112	—

stability is subtler than the score simply being bounded. For a *parameter-dependent* (backflow) node the score itself can diverge— $\nabla_\theta \log|\Psi| \supset (\nabla_\theta D)/D$ grows like $1/d_\theta$ as the learned determinant $D \to 0$ near its node. What matters is that this divergence is *integrable* under the $|\Psi|^2$ sampling measure, which weights it by $|\Psi|^2 \sim d_\theta^2$ (and the near-node points are further suppressed by the importance screening). REINFORCE avoids the *stronger*, non-integrable divergence of the strong-form residual, in which the Laplacian term $\nabla^2 D/D$ enters the gradient directly (Appendix C). Keeping the Laplacian in the *reward* but out of the *gradient flow* is what lets the collocation pipeline train backflow and Jastrow parameters stably in regimes where direct residual minimisation fails.

The superiority of pure REINFORCE (β=0) over the hybrid objective (β>0) has a related explanation. The direct weak-form term $\tilde{E} = \frac{1}{2}\|\nabla \log \Psi\|^2 + V$ is cheaper to evaluate (no Laplacian needed). Averaged under the current $|\Psi|^2$ measure, it equals the variational energy by integration by parts. The implemented direct path, however, differentiates $\tilde{E}(R)$ at retained samples without the score contribution from the changing sampling measure; $V(R)$ then has no explicit parameter derivative and the update behaves as a kinetic-only pathwise gradient. At nonzero β, the resulting gradient conflict between the two objectives manifests as oscillatory behaviour in the energy, with the REINFORCE term pulling toward the variational minimum and the direct term pulling toward a kinetic-energy minimum that is generally distinct.

7.3.2 Collocation performance

N=6: within a few tenths of a percent across all ω. For six electrons, collocation training comes close to the reference wherever one exists, but not to within it. At $\omega \geq 0.5$ every run lands 0.07–0.10% above DMC, with evaluation errors of 0.005–0.013%, so the gap is resolved and several times the deviation of the SR–VMC route (Table 6.2). An earlier reading of these runs put the best of them at +0.013%, statistically consistent with DMC; that number came from a single 30 000-sample evaluation of the best of three seeds, and the independent re-evaluation shows it to have been the lucky tail of a noisier estimate. At $\omega = 0.1$ the robust recipe sits at +0.29%. At the most extreme confinements, $\omega \in \{0.01, 0.001\}$, where the electrons are separated by $\sim$30–120 a_0^* (a few oscillator lengths) and no DMC reference is available, the energies stay within 0.22–0.35% of our converged variational energies; this is agreement between two independent training paradigms on the same ansatz, not a certificate of absolute accuracy.

These results are achieved without any Markov chain in the training gradients. MCMC enters only outside the gradient loop: a small Metropolis probe periodically selects the best checkpoint, and the final heavy evaluation certifies the energy. The training itself uses 4096 importance-resampled collocation points per epoch with an oversampling factor of 8, requiring $\sim$32k proposal evaluations per gradient step. That is ten times the $\sim 3\times10^3$ samples per step used by the SR route, but each is a single independent amplitude evaluation rather than a step of a correlated chain that must first equilibrate. The collocation batch is also embarrassingly parallel where the chain is not, though we did not benchmark the two pipelines against a common wall-clock budget.

Staged continuation is what makes those budgets pay, and an exploratory run at ω=1.0 shows the effect cleanly: an initial BF+Jastrow joint REINFORCE stage (900 epochs, lr=5×10^{-4}), a low-LR continuation (750 epochs, lr=3×10^{-4}) and a hard-focus stage (500 epochs, lr=2×10^{-4}), each inheriting the previous checkpoint, improve monotonically through +0.08%, +0.05% and +0.010%. That run was hand-tuned and is not among the protocol-controlled numbers of Table 7.5; we report it because it is the observation that put cascade warm-starting into the campaign's robust recipe (Table 7.6).

N=12: a few hundredths of a percent at strong confinement. For twelve electrons, the BF+Jastrow collocation results track DMC closely at $\omega \geq 0.5$, at +0.031% and +0.034% for ω=1.0 and 0.5 (Table 7.5), resolved above the reference at evaluation errors of 0.003%. No reliability campaign was run at this size, so these are single continuation runs and not protocol-controlled in the way the N=6 rows are. At ω=0.1 the error increases to +0.11%, reflecting the growing difficulty of importance sampling in 24 dimensions at weak confinement. The ESS at ω=0.1 is typically 5–15 per epoch (out of 4096 kept points), indicating that the Gaussian mixture proposal still maintains usable overlap with $|\Psi|^2$.

Attempts to extend to $\omega \leq 0.01$ at N=12 were unsuccessful: both direct starts and cascade warm-starts from ω=0.1 failed to converge, with the ESS collapsing to 1 and the rollback mechanism rejecting all gradient updates. This marks the boundary of the current collocation approach for N=12.

N=20 exposes the memory–sampling trade-off. Twenty electrons in two dimensions push the collocation pipeline against a hardware wall, not a conceptual one. The backflow's edge-feature tensor scales

as $\mathcal{O}(N^2 h_{\mathrm{bf}})$, which at N=20 forces the hidden dimension, the collocation budget or the oversampling factor down far enough to degrade the gradient—so far that a Jastrow-only ansatz, freed of that cost, trains better than the backflow ansatz it should in principle beat. Since the limitation is memory and gradient noise, the N=20 results and the reversal that exposes them are reported in Appendix D.2.

7.3.3 Ablations and failed variants

Most of the method space we explored did not work. The failures are reported here beside the successes, because a working configuration tells you only that some set of choices is sufficient, whereas a failure chased far enough to name its mechanism tells you which choice was load-bearing—and that transfers to systems other than this one. Each attempt below was, at the time, an intuitively reasonable idea. Each turned out to violate an assumption the method had never been forced to state.

Table 7.6: Collocation variants tried, and what each one settled. Errors are relative to the reference of Table 7.5. The three marked † are taken up in prose below; the rest are recorded for completeness.

Attempt	Outcome	Mechanism
Pure REINFORCE (β=0) over the hybrid objective	adopted	The direct path omits the sampling-measure score contribution; at $\beta > 0$ it contributes a kinetic-only pathwise gradient that conflicts with REINFORCE, and the two gradients oscillate against each other.
Cascade warm-start from higher ω	essential	Cold starts never converged at ω=0.01 for $N \geq 6$. Cascading from ω=0.1 cut the initial error from ∼30% to ∼5% and made +0.15% reachable in ∼1000 epochs.
ESS-adaptive oversampling with instability rollback	essential	Raises the proposal budget when overlap degrades and reverts on energy spikes. Without both, ω=0.001 diverged within a few hundred epochs.
CG-SR below $\omega \simeq 0.01$ †	abandoned	+4–36% against +0.15–0.24% from Adam. Not ill-conditioning but unestimability; see below.
Langevin proposal refinement †	abandoned	Worse in every configuration tested (+153% against +5.7% at N=20, ω=0.1). The weights are computed against a proposal the samples have stopped following; see below.
Finite-difference residual loss	abandoned	+0.22% at N=6, ω=1 against +0.010% from REINFORCE. Sensitive to the step size and to noise in the Laplacian estimate.
Neural Pfaffian ansatz [48]	abandoned	+0.43% at N=6, ω=1, forty times the BF+Jastrow error. The larger parameter space amplifies importance-sampling gradient noise faster than the richer nodal topology repays.
Backflow at N=20 †	abandoned	+18% against +1.32% for Jastrow-only. A memory limit rather than a conceptual one; see §7.3.5.

Why natural gradients stop helping. CG-SR gave the fastest convergence and the best final energies at $\omega \geq 0.1$, and then failed outright below it. The obvious explanation is ill-conditioning: at weak confinement the Fisher matrix $F_{ij} = \mathbb{E}[\partial_i \log|\Psi|^2 \, \partial_j \log|\Psi|^2]$ has eigenvalues spanning many orders of magnitude, because parameters controlling short-range correlation and parameters controlling the orbital envelope act on scales separated by $1/\sqrt{\omega}$.

Ill-conditioning alone does not account for the failure, because ill-conditioning is what makes natural gradients useful in the first place; a well-conditioned metric is one Adam handles unaided. What disqualifies them is the failure to *estimate* the metric. F is built from importance-resampled points, and when the ESS is low a handful of high-weight samples dominate it, leaving it noisy and rank-deficient. Tikhonov damping stabilises the inversion at the cost of washing out the curvature information that motivated it.

Three regimes follow. Where the metric is well conditioned, SR is sound and unnecessary. Where it is anisotropic and still estimable, SR is at its most valuable—which is the collocation case of §7.2. And where importance-sampling collapse makes the empirical metric unreliable, inverting it amplifies the estimation noise and SR does active harm. The failures above are the third regime, not the second, and the practical implication is about the estimator rather than the curvature: any second-order method

applied to a weakly confined many-body system must certify that the metric it inverts is statistically resolved at the sample sizes in use.

Why pushing the samples uphill makes things worse. Langevin refinement was motivated by an intuition that is hard to argue with. If the Gaussian proposal overlaps $|\Psi|^2$ poorly, evolve the samples up the score $\nabla_{\mathbf{R}} \log |\Psi|^2$ for K steps before reweighting, and the overlap should improve. It failed in every configuration tested, and the reason is standard in the MCMC literature and easy to miss in an importance-sampling setting.

The weights $w_i = |\Psi(\mathbf{R}_i)|^2/q(\mathbf{R}_i)$ are valid only if the $\mathbf{R}_i$ came from q. After K steps from an out-of-equilibrium start the samples follow some intermediate $\tilde{q}$ that depends on K and ε, while the weights still divide by q. The resulting bias is not noise. It pushes the wavefunction toward the regions the Langevin dynamics favours, which moves $\tilde{q}$ further the same way, and the loop closes on itself—which is what the +153% in the table is. Ten to twenty steps in Nd=40 dimensions do not equilibrate. Fixing it means running to equilibrium, adding a Metropolis–Hastings step, or computing annealed weights along the non-equilibrium path. The first two are MCMC under another name; the third costs enough to forfeit the advantage that made the route attractive.

What the failures have in common. Read together these are not separate disappointments. Four of them (CG-SR, Langevin, the Pfaffian's parameter count, and the backflow at N=20) fail through one channel: a quantity that must be *estimated* from a finite sample degrades faster than the quantity it was meant to improve. Natural gradients need an estimate of the metric. Langevin needs samples that have actually equilibrated. A larger parameter space needs a correspondingly better-resolved gradient. None of these methods is wrong in principle, and each would be sound given samples we cannot afford. What fails is the estimate, and it fails precisely in the regime where the correction was supposed to help. That thread is what §7.3.4 follows deliberately, from the proposal mismatch at one end to the evaluation bottleneck at the other.

7.3.4 Reliability analysis: the gradient-quality chain

The results in Table 7.5 report the *best* energies obtained by collocation training, selected from multiple runs on different seeds. A natural question is whether these results are *typical* or whether they reflect rare fortunate outcomes from a high-variance process. This subsection reports a systematic 30-run reliability campaign designed to answer that question, and exposes a mechanistic chain that accounts for both the successes and failures of the Markov-chain-free collocation approach.

Experimental design

All runs use the BF+Jastrow ansatz (∼49k parameters) initialised from the same pretrained checkpoint. The grid is N=6 across $\omega \in \{1.0, 0.5, 0.1, 0.01, 0.001\}$ with three random seeds per cell (42, 137, 314), giving $5 \times 2 \times 3 = 30$ runs. Two training recipes are compared head-to-head:

- **Config A (baseline):** Adam optimiser with lr=5×10^{-4}, lr_{jas}=5×10^{-5}, 4096 collocation points, 8× oversampling, local-energy clipping at 5σ, no reward normalisation, no rollback, fixed Gaussian mixture proposal (Eq. (4.9)).
- **Config B (robust):** Identical to Config A plus four stabilisation mechanisms: 16× oversampling (doubling the proposal budget), reward normalisation (standardising the REINFORCE reward to zero mean and unit variance before the gradient step), instability rollback (reverting to the previous checkpoint when the energy jumps exceed 3σ, with learning-rate decay of 0.95), and an adaptive Gaussian mixture proposal (8 GMM components refitted every 30 epochs to track the evolving $|\Psi|^2$).

All runs train for 1200 epochs. A 20 000-sample Metropolis probe of $|\Psi|^2$ every 50 epochs selects the checkpoint, and the selected checkpoint was evaluated at the end of training by a fresh 30 000-sample Metropolis run. Every number below is instead an independent re-evaluation of that saved checkpoint, made for this revision with four separately seeded chains of 102 400 samples each and blocking; it reproduces the original evaluations within their errors on average, but not the best-of-three values selected on them (§7.3.4).

Main results

Table 7.7 gives all thirty runs, three seeds per cell.

Table 7.7: Reliability campaign: relative energy deviations for N=6 across five trap frequencies, scored against the reference column of Table 7.5—external at $\omega \geq 0.1$, and our own converged SR–VMC energy at $\omega \leq 0.01$, where the deviation then measures internal consistency rather than absolute accuracy. Each cell is one $(\omega, \text{recipe}, \text{seed})$ run, evaluated by independent blocked Metropolis sampling of the saved checkpoint (errors 0.003–0.02%, 0.05–0.09% for the baseline at $\omega = 10^{-3}$). Bold marks the best run at each ω. The last two columns give the mean ± standard deviation and the worst of the three seeds. The standard deviation is reported in place of a coefficient of variation because the latter is uninformative when the mean deviation approaches zero.

ω	Recipe	Seed 42	Seed 137	Seed 314	mean ± s.d.	worst
1.0	Baseline	+0.067	**+0.066**	+0.078	0.070 ± 0.006	0.078
	Robust	+0.068	+0.081	+0.096	0.082 ± 0.014	0.096
0.5	Baseline	+0.107	+0.112	+0.104	0.108 ± 0.004	0.112
	Robust	+0.103	+0.103	**+0.088**	0.098 ± 0.009	0.103
0.1	Baseline	+0.479	+0.597	+0.486	0.520 ± 0.066	0.597
	Robust	**+0.284**	+0.302	+0.293	0.293 ± 0.009	0.302
0.01	Baseline	+0.368	+0.455	+0.661	0.494 ± 0.150	0.661
	Robust	+0.246	+0.262	**+0.222**	0.244 ± 0.020	0.262
0.001	Baseline	+4.012	+6.906	+4.628	5.182 ± 1.525	6.906
	Robust	+0.252	+0.350	**+0.244**	0.282 ± 0.059	0.350

The stabilisation mechanisms buy nothing at strong confinement, where the two recipes land within a few hundredths of a percent of each other and the best runs are indistinguishable, and nearly everything at weak confinement. At $\omega = 0.1$ and 0.01 the robust recipe halves the mean error and shrinks its seed spread about sevenfold; at $\omega = 10^{-3}$ the mean error falls by a factor of eighteen, from +5.2% to +0.28%, and the *worst* robust seed beats the *best* baseline seed by an order of magnitude. What follows is a mechanism for why the difference appears where it does.

The gradient-quality chain

The reliability campaign exposes a mechanistic chain that accounts for the quality of collocation-trained wavefunctions:

$$\begin{aligned} &\text{proposal mismatch} \longrightarrow \hat{k} \text{ inflation} \longrightarrow \text{gradient noise} \\ &\qquad\qquad\qquad\quad \longrightarrow \text{optimisation stochasticity} \longrightarrow \text{run-to-run variance} \end{aligned} \tag{7.3}$$

Each link is checked against the training logs below. One caveat governs how strongly to read the arrows: the robust recipe changes four things at once (oversampling, reward normalisation, rollback and adaptive proposal refitting), so the campaign shows that the package works, not which element causes which improvement. The chain is a mechanism consistent with every measurement we have, not a set of individually isolated causal links. We did not ablate the four stabilisers separately; a factorial or stepwise ablation is what would be needed to establish the chain link by link.

Step 1: proposal–target mismatch. The REINFORCE estimator of Eq. (4.8) requires samples from $|\Psi_\theta|^2$, but collocation draws from the analytic mixture $q(\mathbf{R})$ of Eq. (4.9) and reweights. Everything depends on how well q overlaps the target. At $\omega = 1$ the electrons occupy a compact cloud of radius $\ell = \omega^{-1/2}$ and the three-component mixture covers it. At $\omega = 10^{-3}$ the cloud extends to $\sim 100\, a_0^*$ against $\ell \approx 31.6\, a_0^*$, the pair correlation develops sharp shell structure (§6.2), and the mass concentrates in a thin correlated region an isotropic Gaussian covers badly.

The mismatch shows directly in the importance weights. At ω=1.0, weights are relatively uniform: the effective sample size is typically ESS $\sim$ 5000–6000 out of the $\sim$65k *raw proposal draws* (16× oversampling of 4096 points)—the second of the two normalisations noted in §7.2—and the Pareto shape parameter $\hat{k}$ of the weight tail is 0.4–0.5, indicating a well-behaved thin-tailed distribution. At ω=0.001, the ESS

collapses to 1–6 out of the same 65k draws, and $\hat{k}$ rises above 1.0, indicating a catastrophically heavy fitted tail and a sample far from the regime in which the reweighted estimator is reliable.

Step 2: $\hat{k}$ inflation and gradient quality. The tail index $\hat{k}$ of the importance weights turns that mismatch into a number (§4.5.1): below 0.7 the reweighted estimator is trustworthy, above 1.0 the fitted tail is heavy enough that a handful of samples dictates the update. Two qualifications keep the reading honest. $\hat{k}$ is fitted to the *weights*, not to the full gradient integrand $w\,(E_L - b)\,\nabla_\theta \log|\Psi_\theta|$, whose tail index we do not measure; and because w is a ratio of proper densities its mean is finite in the limit, so $\hat{k} > 1$ diagnoses a sample far from that limit rather than a gradient that fails to exist. What the campaign establishes is severe proposal–target mismatch and a gradient estimator dominated by a few draws. That is enough to explain everything downstream, and it is all we claim.

Table 7.8 summarises the average $\hat{k}$ across the campaign. The correspondence between $\hat{k}$ and final accuracy is striking where $\hat{k}$ exceeds about 0.7, and absent below it.

Table 7.8: Average Pareto $\hat{k}$ of the importance weight distribution during training, by regime and recipe. Lower is better. The fitted tail has finite variance only for $\hat{k} < 1/2$ and a finite mean only for $\hat{k} < 1$; $\hat{k} < 0.7$ is the practical reliability threshold, and $\hat{k} > 1$ indicates a sample so far from the asymptotic regime that the estimator behaves as though no finite mean existed (§4.5.1).

	$\hat{k}$ (avg)		Mean error (%)	
ω	Baseline	Robust	Baseline	Robust
1.0	0.75	0.50	0.070	0.082
0.5	0.83	0.54	0.108	0.098
0.1	1.48	0.70	0.520	0.293
0.01	2.63	1.39	0.494	0.244
0.001	3.13	1.86	5.182	0.282

The robust recipe reduces $\hat{k}$ by a factor of 1.5 to 2.1, with the larger reductions at the confinements where the baseline is worst. At ω=1.0 this takes $\hat{k}$ from 0.75 (borderline) to 0.50 (reliable), with no measurable effect on the energy. At ω=0.1 it takes $\hat{k}$ from 1.48 (broken) to 0.70 (borderline), corresponding to the transition from mediocre (+0.52%) to marginal (+0.29%) accuracy. At ω=0.001 even the robust recipe has $\hat{k}$=1.86, but the adaptive proposal and rollback mechanisms prevent the catastrophic divergence seen in the baseline (+5.2%).

Step 3: from gradient noise to optimisation stochasticity. When $\hat{k}$ is large, most of the gradient signal comes from a few samples with extreme importance weights. Which samples are extreme depends on the random proposal draw at each epoch—a function of the random seed. Two seeds that happen to draw different extreme samples will follow different optimisation trajectories, even though they share the same architecture and hyperparameters.

This is not the usual seed dependence of stochastic gradient descent, where the law of large numbers ensures that gradient estimates are unbiased and the trajectory variance decreases with batch size. When $\hat{k} > 1$ the weight distribution is so heavy-tailed that the realised estimator behaves as though its variance were unbounded: increasing the batch size reduces the spread sublinearly, and extreme draws continue to dominate the update at every batch size we could afford. The result is optimisation trajectories that diverge qualitatively across seeds, explaining the large run-to-run variance observed at low ω.

Step 4: where the reported energy can mislead. The final energy is not an importance-sampling estimate. It is a Metropolis evaluation of $|\Psi_\theta|^2$ at the checkpoint the probe selected, and every value in Tables 7.5 and 7.7 is an independent re-evaluation of that checkpoint with four separately seeded chains and blocking. Where the reported number can mislead is one step earlier, in the selection. The checkpoint is the best of about two dozen probe readings of 20 000 samples each, and the minimum of noisy readings is biased low. Across all thirty runs the probe that chose a checkpoint read it as better than the independent evaluation finds, by 0.07% on average and in 26 of 30 cases; the original single end-of-training evaluation, by contrast, is unbiased on average (+0.001%) but noisy enough that the best of three seeds selected on it came out optimistic. Table 7.9 shows representative runs.

The first two rows are the instructive ones. A probe reading of +0.001% looked like an essentially exact wavefunction; the state sits at +0.08%, and it was the probe that was lucky. The second row was,

Table 7.9: Selection bias of the checkpoint probe. "Probe" is the Metropolis probe reading that selected the checkpoint during training; "original" the single 30 000-sample evaluation made at the end of training; "independent" the blocked re-evaluation used in the tables (error in parentheses, last digits). The gap is independent minus probe. All values are deviations in percent.

ω	Recipe	Seed	probe	original	independent	gap
1.0	Baseline	314	+0.001	+0.060	+0.078(12)	+0.077
1.0	Robust	137	+0.002	+0.013	+0.081(6)	+0.079
0.5	Robust	42	+0.037	+0.002	+0.103(12)	+0.066
0.1	Robust	137	+0.136	+0.263	+0.302(8)	+0.166
0.01	Baseline	42	+0.320	+0.500	+0.368(17)	+0.048
0.001	Baseline	314	+4.114	+4.733	+4.628(52)	+0.514

until this re-evaluation, the headline strong-confinement number of this chapter. Selection on a noisy estimate inflates any "best run", which is why the tables report independent re-evaluations and why the reliability campaign is summarised by seed means rather than by its best seeds.

Summary of the chain. The chain in Eq. (7.3) provides a unified account of the full landscape of collocation results. At high ω, the proposal matches the target well ($\hat{k} \sim 0.5$), gradients are reliable, optimisation is stable, and all seeds converge to within about a tenth of a percent of DMC. At low ω, the proposal-to-target mismatch grows, $\hat{k}$ inflates, gradients become noisy, and optimisation becomes seed-dependent. The robust recipe acts on multiple links of this chain simultaneously: higher oversampling improves the raw ESS, reward normalisation stabilises the gradient scale, rollback prevents catastrophic divergence, and the adaptive proposal reduces the fundamental mismatch. Each is aimed at a different link in the chain; which of them is necessary was not tested, and the campaign shows only that together they suffice.

7.3.5 The memory wall at N=20

The reversal of the backflow advantage at N=20 highlights an important architectural trade-off. The message-passing backflow (`CTNNBackflowNet`, §3.5.2) stores edge features of size $\mathcal{O}(N^2 \times h_{\mathrm{bf}})$, which at N=20 with h_{bf}=128 requires ∼10× the memory of the Jastrow-only network. This forces a choice: reduce h_{bf} (losing expressivity), reduce the collocation budget (losing gradient quality), or reduce the oversampling factor (losing ESS). All three options degrade the result.

The Jastrow-only network, with ∼25k parameters and $\mathcal{O}(N^2)$ edge cost (no hidden-dimension multiplier), permits full-size collocation budgets (n_{coll}=4096, oversample=8–16) even at N=20. The resulting gradient is less noisy, and the simpler loss landscape (fewer parameters) makes optimisation more reliable.

This does not mean that backflow is unnecessary at N=20; it means that the current training pipeline cannot supply enough gradient signal to train backflow effectively at this scale. The implication is that scaling collocation to larger systems requires either (i) memory-efficient backflow architectures (e.g. sparse attention, low-rank message passing), or (ii) improved sampling that provides higher ESS at constant computational cost.

7.4 A physics-informed proposal and its limit

The ESS collapse of §7.2 is a limitation of the *proposal*, not of collocation itself, and it can be pushed back a long way. A physics-informed proposal carries Markov-chain-free collocation deep into the Wigner regime; what limits it, and the wall that survives, follow from the same physics.

7.4.1 The origin-centred proposal misses the Wigner shell

Every proposal in the standard pipeline is a mixture of *origin-centred* Gaussians,

$$q(\mathbf{R}) \;=\; \frac{1}{K}\sum_{k} \mathcal{N}\big(\mathbf{R};\, 0,\, \sigma_{f,k}^2/\omega\big)\,. \tag{7.4}$$

But as $\omega \to 0$ the electrons localise into a Wigner molecule on a *shell* far from the origin. The classical $N = 6$ minimum is the $(1, 5)$ configuration: one central electron and five on a ring of radius $1.334\,\omega^{-2/3}$ [2, 3]. In oscillator units that ring sits at 1.96, 2.87 and 4.22 ℓ for $\omega = 0.1$, 0.01 and 0.001, moving outward as the trap weakens and past the reach of a width-2ℓ origin Gaussian. The proposal aims where $|\Psi|^2$ has no mass; hence the collapse.

7.4.2 The Wigner-ring proposal

Before the numbers, one distinction has to be exact, because "Markov-chain-free" is a claim of this thesis. The Wigner geometry enters in three roles and only the third sits inside the training loop. *(i)* A proposal can be *evaluated* against a trusted state obtained by MCMC, which is how Table 7.10 is produced—a measurement, not a training step, and one that uses a chain by design. *(ii)* Its geometry can be *fitted* to $|\Psi_\theta|^2$ samples as a diagnostic. *(iii)* In production it is refitted from the current importance-weighted batch alone, with no chain anywhere; that is the genuinely Markov-chain-free adaptation used in the cascade of §7.4. Throughout, a Metropolis probe still runs outside the gradient loop to select checkpoints and certify the final energy, and no sample it produces ever reaches a parameter update. We replace the origin mixture with a *Wigner-ring* proposal: a radial Gaussian at the shell radius times an angular distribution for the ring electrons, plus a central Gaussian. On trusted states this lifts the effective sample size by 15–32× over the origin mixture (Table 7.10), turning a collapsed sampler into a usable one—without any training. Fitting the ring to $|\Psi|^2$ samples (the simplest adaptive form) doubles the ESS again; a $(1, 5) + (0, 6)$ mixture *hurts* at $N = 6$, since the state is $(1, 5)$ and the extra channel dilutes the proposal mass.

Table 7.10: ESS (of 4096) against a trusted $N = 6, \omega = 0.01$ state. The origin mixture is already collapsed; the ring proposal covers the shell.

proposal	ESS
origin-Gaussian mixture (standard)	1.5
Wigner ring $(1, 5)$, hand-tuned	22.5
Wigner ring $(1, 5)$, fitted to $\|\Psi\|^2$	47.5

7.4.3 Angular crystallisation defeats the uniform-angle ring proposal

Why does the ring work at $\omega = 0.01$ but fail at $\omega = 0.001$? The molecule *crystallises angularly* as the trap weakens. The nearest-neighbour angular-gap standard deviation of the five ring electrons is $40.8°/33.3°/25.5°/6.7°$ at $\omega = 1/0.1/0.01/0.001$: a "ring liquid" at moderate coupling becomes an angular crystal (electrons pinned near 72° apart) at $\omega = 0.001$. A proposal with independent, uniform angles almost never lands on the crystalline configuration—so the angular distribution must crystallise too. This is captured Markov-chain-free by modelling the five gaps (which sum to 2π) as Dirichlet(α): an exact tractable density whose single concentration α interpolates liquid (α small) to crystal (α large), fitted to the measured gap variance. This diagnosis *explains* the collocation wall that is otherwise only observed.

7.4.4 An MCMC-free cascade reaches $\omega = 0.0035$

Combining the ring proposal (radial Gaussian × Dirichlet gaps) with importance oversampling, a *continuous* adaptive re-fit of the proposal to the current importance-weighted batch (every few steps, no MCMC), and a gentle ω-cascade seeded from a converged $\omega = 0.01$ state, Markov-chain-free collocation produces states that lie on the Wigner energy-scaling curve down to $\omega = 0.0035$—four cascade rungs, with heavy-VMC energies within 2% of the scaling estimate, and within 1% on three of the four rungs (Table 7.11), well past where the origin proposal collapses (ESS $\approx$ 1.5 of 4096 at $\omega = 0.01$, Table 7.10).

Three things bound that result. The agreement in Table 7.11 is against a scaling curve fitted to our own $\omega = 0.01$ energy, so it establishes continuity rather than accuracy; two structural checks support it—the radial density peaks at 3.34ℓ against a classical ring at 3.42ℓ, and a sampler initialised on the shell converges to the same peak, not a second basin. No *angular* diagnostic was run at $\omega = 0.0035$: the bond-order and Lindemann machinery of §6.2.5 ran on the production grid, not the cascade rungs, and those states were not retained. So a Wigner molecule at $\omega = 0.0035$ rests here on energy continuity and radial geometry, not on demonstrated angular order. And the Markov-chain-free claim is a claim about the gradient—no Metropolis sample reaches a parameter update at any rung—not about ancestry: each

rung is warm-started from a stronger confinement, and that ladder runs back to a pretrained checkpoint whose residual objective used an external energy as its target. The frontier parameters therefore inherit structure shaped several confinements upstream by reference-guided pretraining, the same inheritance the SR–VMC states carry (§8.1). The essential ingredient is the continuous re-fit: it keeps ESS healthy ($\sim 10^2$) *while* the energy descends, where a static or slowly-updated proposal collapses to ESS ~ 1 as the wavefunction moves away from it.

Table 7.11: Markov-chain-free Wigner-ring cascade, $N = 6$: heavy-VMC energy vs. the fitted Wigner scaling curve $E(\omega) \approx 0.690\,(\omega/0.01)^{0.689}$, whose exponent is close to the classical 2/3 but whose prefactor is anchored at $\omega = 0.01$, where no external reference exists. The agreement is therefore internal scaling consistency, not validation, and it is retained through $\omega = 0.0035$.

ω	E	scaling est.	ratio
0.0077	0.577	0.577	1.00
0.0059	0.481	0.480	1.00
0.0045	0.400	0.398	1.01
0.0035	0.340	0.335	1.02

7.4.5 The remaining limit: the Slater–Jastrow balance

Below $\omega = 0.0035$ the cascade becomes unstable, and—informatively—the instability is *non-monotonic* across proposal variants: wide-exploratory, tempered-tracking, and outward-anchored proposals each converge some rungs and diverge others. That pattern locates the obstacle not in the proposal but in the warm-start, and four transfer diagnostics (taking the converged $\omega = 0.0035$ rung down to $\omega = 0.0027$, a 23% step) pin down its mechanism precisely.

Where the collapse lives. First, the frontier state is a genuinely Wigner-localised one on the evidence available: at its own $\omega = 0.0035$ its radial density peaks at 3.34ℓ against a classical ring at 3.42ℓ, and it occupies a *single* basin—initialising the sampler on the shell converges to the same 3.34ℓ. We stop short of calling it a Wigner molecule in the sense Chapter 6 defines, because that definition requires registered angular order and no angular diagnostic was run on this state. Transferred to $\omega = 0.0027$, however, the wavefunction collapses to $\sim 1\ell$, and two tests show *where* the collapse lives. It is not the sampler: initialising the Metropolis walkers directly on the classical shell (3.57ℓ) still relaxes inward to 1.1ℓ, so the collapsed density is a property of the ansatz, not of sampler initialisation. And it is not the backflow: its displacement is small ($|\Delta\mathbf{x}| \approx 0.13\ell$, radially inward) and essentially *preserved* across the transfer ($0.15\ell \to 0.13\ell$)—it scales correctly with ℓ but is far too small to bridge the 2.4ℓ gap between the Slater core ($\sim 1\ell$) and the Wigner shell.

The mechanism. A delicate cancellation. The bare harmonic-oscillator determinant suppresses the shell by $\exp\!\left(-\omega r^2/2\right) \sim \exp(-5.6)$ in amplitude ($\sim 10^{-5}$ in density); the Wigner peak exists only because the Jastrow amplifies it back by a comparable factor, with the backflow contributing a small inward nudge. A 23% change in ω perturbs both the Slater confinement and the Jastrow correlation length, the fine-tuned balance tips, and the density falls back to the non-interacting shell—from which the collocation re-fit cannot reliably climb out (hence the non-monotonic proposal behaviour). This rules out the natural fix of an ω-robust warm-start by coordinate rescaling: because the ansatz relaxes inward even when handed shell configurations, no coordinate transform preserves the state across the step. It does not rule out representing the $\omega = 0.0027$ state at all—that would require optimising it from scratch there, which we did not do. Reaching $\omega < 0.0035$ would require per-ω re-optimisation against a shell-*amplitude*-anchored objective, or a genuinely ℓ-covariant parametrisation—a distinct problem left to future work. The contribution here is concrete: a physics-informed proposal that carries Markov-chain-free collocation from the origin-proposal wall down to a radially localised Wigner-regime state at $\omega = 0.0035$, and a precise account of the angular-crystallisation and Slater–Jastrow-balance effects that bound it.

Part V

Discussion and Conclusions

Life must be understood backwards. But they forget the other proposition, that it must be lived forwards.

Søren Kierkegaard
Journals

Chapter 8

Discussion

8.1 Geometry between architecture and optimiser

The three parts of this thesis converge on one object. Architecture fixes the shape of the variational tangent space; the optimiser acts on that space through its metric; and the sampling measure decides how well the metric can be estimated. Read in that order, results that arrived separately become a single chain.

Its first link is the one the energy cannot see. Two correlators of matched or deliberately mismatched capacity reach the same reference-quality state at strong confinement, agreeing to four figures, and arrive there through tangent spaces that carry their Fisher weight very differently. Counting directions is the weaker half of that observation—it says how many the correlator uses, it is measured on one of two substitutable components, and it depends on how the network is parametrised and normalised. Which directions is the stronger half, because whether a direction lies inside an operator span does not depend on the parametrisation, although the weight it carries does. The message-passing correlator's leading tangent modes lie inside a four-operator physical dictionary to $R^2 \approx 1$ across confinement and up to N=12, where a separable network of equal or greater capacity leaves close to half its tangent budget outside it. A direction outside a chosen dictionary need not be unphysical, so what the measurement supports is that one tangent space is far more completely accounted for by a small set of collective operators than the other—and that a smaller d_{eff} makes such a dictionary easier to cover cannot explain the gap, since the two effective dimensions converge toward 3.7 at weak confinement while the dictionary gap persists. None of this yet shows that the architecture represents the state with intrinsically fewer degrees of freedom. The comparison that does not depend on parametrisation at all is the local-energy variance, lower for the message-passing correlator at every confinement; it is the one that says the two *states* differ, and not only the two ways of laying out their parameters.

The second link is where the optimiser meets that space. The result there is narrower than the usual one about natural gradients. Under $|\Psi|^2$ sampling the empirical metric is badly conditioned, $\kappa \sim 10^7$–10^{10} on precisely the models where Adam and SR agree to four decimals, so conditioning cannot be the criterion. What we propose separates the regimes is whether a batch resolves the metric at all. Under $|\Psi|^2$ the samples come from the measure that defines S; under a fixed collocation proposal they do not, and preconditioning becomes what makes the optimisation converge; and once the proposal loses the target, inverting the metric amplifies noise instead of signal. The negative half of that is measured. The positive half—that collocation under a fixed proposal occupies an anisotropic but still estimable middle regime—is inferred from the estimator, since κ was never obtained under the collocation measure at matched batch size.

Whether the chain generalises is a fair question and the honest answer is that we cannot settle it here. Message-passing neural quantum states have been adopted for extended fermion systems on the argument that relational structure scales [9, 10]; what this thesis adds is a measurement of *where* that

structure shows up—in the tangent geometry before the energy—and a mechanism for why the effect should grow rather than shrink with correlation strength. The dictionary gap widens with N over the range we can measure. If that continues, the architectures that scale are those maintaining a transported edge representation rather than pooling pairwise features and discarding them. Three sizes is not a scaling study, and the comparison is seeded at N=6.

What agreement with the benchmarks supports. The energies agree closely with the available fixed-node references over the benchmarked grid, and that agreement is neither exactness nor a meeting of two blind calculations. Three separate things limit what it shows. The first is ancestry: where a DMC value exists it is also the target of residual pretraining. The pure-energy SR tail that follows minimises $\langle \hat{H} \rangle$ under $|\Psi_\theta|^2$ and no longer sees the target, which strengthens the comparison considerably, but the final states still descend from reference-guided training. The second is what an energy can certify. The variational bound keeps the reported expectation above E_0 whatever the training aimed at, but it says nothing about how close to E_0 it lies, and a flexible ansatz can reproduce an approximate reference without describing every other property of the exact state. Our evidence beyond the energy is partial: the local-energy variance, which vanishes only for an eigenstate, and a squared overlap of 1.000000 with the exact state where one is available, at N=2. The third concerns the cells where our energies sit below the reference. The reference is variational only with respect to its own imposed nodes, so there is no contradiction, and better nodes in the network would explain it; but so would a convention difference in the Hamiltonian, an extrapolation or sampling error on either side, or an error in the chain of compilations through which the reference reached us, and we cannot tell these apart. A fixed-node DMC calculation that uses the network's nodes would be the cleanest test. The tables accordingly report deviation, not error, and what we claim is close agreement with the available fixed-node references, not a validation of the wavefunctions. Below $\omega = 0.1$ for $N \geq 6$ no external value exists and those states are predictions.

8.2 Physics returned to the solver

The most useful thing this thesis did was not to train a wavefunction but to let the wavefunction correct the method that produced it.

The order matters. A generic three-component Gaussian proposal covers the strong-confinement droplet adequately and collapses in the Wigner regime, where the effective sample size falls to about 1.5 of 4096 and the gradient estimator dies with it. Nothing in the objective or the architecture diagnoses that. What diagnoses it is the structure measured from the trained states themselves: a single ring at N=6 with a well-determined radius, angular order that appears well after radial stiffening, and shell occupancies that are read rather than assumed. Rebuilding the proposal as a radial Gaussian at the measured shell radius times a Dirichlet angular model lifted the effective sample size by 15–32× with no training at all, and carried Markov-chain-free training from $\omega \approx 0.01$ to $\omega = 0.0035$.

That is a loop, and its two halves fail in different ways, which is what makes it informative. Everything up to $\omega = 0.0035$ lies on one causal chain: proposal–target mismatch inflates the weight tail, the tail destroys the gradient estimator, the destroyed estimator makes the optimisation seed-dependent and the metric unreliable, and inverting an unreliable metric amplifies noise. Langevin refinement joins that chain at the first link by breaking the proposal the weights are computed against; CG-SR joins at the last. Even the N=20 reversal, where a Jastrow-only ansatz beats the backflow that should outperform it, reaches the same chain through memory, not confinement. Up to this frontier the failures share a sampling origin, and a better proposal moves the frontier.

Below 0.0035 the obstacle changes character. Metropolis walkers started on the classical shell relax inward regardless; the backflow displacement transfers correctly and is simply too small to bridge the gap; the collapse is non-monotonic across proposal variants in a way a sampling failure would not be. What fails is the warm start. At these confinements the Wigner peak exists only because the Jastrow amplifies an exponentially suppressed determinant back up by a comparable factor, and a 23% change in ω perturbs both sides of that cancellation until the density falls back to the non-interacting shell. The failure is one of covariance in ℓ under a change of trap—of transfer and parameterisation, not of expressivity. We have not shown the ansatz cannot represent the $\omega = 0.0027$ state; the remedy we suggest, per-ω re-optimisation against a shell-amplitude-anchored objective, presumes it can.

The same mechanisms should extend beyond the quantum dot. An importance-sampled scheme whose proposal increasingly misses a strongly correlated target should meet the first chain in that order, and any

second-order method applied to such a system ought to certify that the metric it inverts is statistically resolved at the sample sizes in use.

The staged crossover. The physics that drove the loop stands in a particular relation to what was known. That parabolic dots form Wigner molecules with shell structure is established [1, 2, 4, 11]; that the crossover is *staged*—radial expansion and shell stiffening completing well before angular order appears within the shells—is what the topology-resolved diagnostics here make visible, because they condition on the shell partition before measuring angular correlation instead of averaging over it. The N=6 ring approaches its classical radius $1.334\,\omega^{-2/3}$ monotonically to within 1.7% while its bond order is still at the random-angle null, which is the separation stated quantitatively. Whether the same staging survives a total-spin scan we cannot say.

What "Markov-chain-free" claims. The claim is about the gradient: no configuration entering a parameter update comes from a Markov chain. A small Metropolis probe runs outside the gradient loop to select checkpoints and certify the final energy, and no sample it produces reaches an update. It is not a claim of freedom from Monte Carlo, since importance sampling and resampling are themselves Monte Carlo; nor of freedom from the reference, since where a DMC value exists the residual pretraining anneals toward it and the rollback monitors against it. Nor, at the deep-Wigner rungs where no reference exists, is the state reference-free in ancestry: its warm starts descend from benchmark-guided states at stronger confinement. Against the SR route, collocation is several times less accurate across the benchmarked range—within about a tenth of a percent of DMC at strong confinement and three tenths at N=6, ω=0.1—and not competitive at N=20. The two look complementary rather than rival: collocation for fast, parallel, Markov-chain-free exploration and pretraining; SR for final accuracy where correlation is strong.

8.3 Scope, and what to do next

Five limits bound everything above. All states are the lowest found within a fixed spin-projection sector, and total spin is not scanned; that matters most in the deep-Wigner regime, where spin transitions are plausible and the structural conclusions are strongest. The architecture comparison is seeded at N=6, rests on fewer models at N=12 and on a single production state at N=20, and the error bars on the production energies do not include the variation between independent training runs. The backflow variants are not capacity-matched and cannot be at equal width, so neither the cross-architecture energy gap nor the rank collapse isolates architecture. The tangent-space quantities are properties of the networks as parametrised, not of the states they represent. And one entry in the backflow rank table, at N=20 and $\omega = 0.01$, comes from a re-run whose diagnostics were not retained.

Five things follow that we would do next.

Make the proposal learned, not designed. Up to the frontier the proposal is the bottleneck and not the objective, which argues for a better sampler rather than a better loss: normalising flows trained against $|\Psi|^2$, or the shell-informed construction of §7.4 generalised beyond ring geometries. Past the frontier the target changes: what is needed there is an ℓ-covariant parameterisation, or an objective anchored on the shell amplitude, so that a converged state survives being carried to a weaker trap.

Make rotation equivariance exact. Permutation equivariance holds by construction and the centre of mass is projected out, but the displacement head reads a d-vector from a network that has seen absolute coordinates, so SO(2) equivariance is only encouraged—by relative-vector edge features and by the random in-plane rotation applied to every batch—and never enforced. In a rotationally invariant trap whose weak-confinement physics is a freely rotating molecule, that is the wrong symmetry to leave approximate. The fix is local to one module: build the displacement as $\Delta\mathbf{r}_i = \sum_{j\neq i} g_\beta(\text{invariants}_{ij})\,\mathbf{r}_{ij}$, a learned scalar gain on each relative vector, so the output rotates exactly with the input and the augmentation can be dropped. We did not run it.

Break the quadratic backflow. What stops the method at twenty electrons is the $\mathcal{O}(N^2 h_{\text{bf}})$ edge tensor, a memory limit rather than a physical one. Sparse or low-rank message passing, a local backflow without all-to-all edge features, or the sample-side reductions of minSR and the kernel formulation [44, 45] would extend the reach without giving up the nodal correction that is the reason for having a backflow at all.

Match the backflow capacities. A widened conventional backflow at 182k parameters is the one control that would turn the rank-collapse observation into an architecture ablation, and it is the most important

experiment this thesis leaves undone.

Measure the metric where the hypothesis lives. The estimability account of §7.2 rests on a spectrum measured under $|\Psi|^2$ and inferred under collocation. Measuring the QGT under the collocation measure at matched batch size, with the stability of its leading eigenvalues across independent batches, a bootstrap on the resolvable part, and its sensitivity to batch size and regularisation, would turn that account from a hypothesis into a criterion or refute it.

All things are implicated with one another, and the bond is holy; and there is hardly anything unconnected with any other thing.

Marcus Aurelius
Meditations

Chapter 9

Conclusions

The work formed a feedback cycle between solver and physics. A variational state was constructed and compared against every external reference available, with which it agrees closely; the state was then used as an instrument, and resolved the shell geometry of the Wigner regime: rings at definite occupancies, n-fold angular order becoming large only in the last decade of confinement, a shell radius approaching its classical $\omega^{-2/3}$ scaling. Those are the measurements a sampler needs in order to place its mass correctly. Feeding the measured geometry back into the proposal extended Markov-chain-free training from the failure of the origin-centred sampler at $\omega \approx 0.01$ down to $\omega = 0.0035$, into states the original method could not reach.

Architecture. Message passing supplies a relational channel whose effect appears in the tangent geometry before it appears in the energy (§7.1). Under the parametrisations and normalisation conventions used here, a message-passing correlator reaches a reference-quality solution with its Fisher weight concentrated in half as many effective directions as a separable network of matched or larger capacity, and keeps more of that weight inside a compact operator dictionary. That is a statement about the networks as parametrised, not about how many degrees of freedom the state intrinsically needs. What is independent of parametrisation is that the message-passing correlator carries a lower local-energy variance. In the nodes, toward the Wigner regime, a conventional backflow's displacement field falls spectrally to rank one and loses half a percent of the energy while a larger message-passing field does not; a capacity-matched conventional backflow is the control that would make this an architectural result. A controlled correlator swap moves the collapse boundary, which points to one relational bottleneck rather than two deficits.

Optimiser. Conditioning alone does not decide whether natural-gradient preconditioning earns its cost (§7.2). Under $|\Psi|^2$ sampling the empirical metric is badly conditioned on the very models where Adam and stochastic reconfiguration agree to four decimals. Under a fixed collocation proposal SR is what makes the optimisation work at all; once the sampling that builds the metric collapses, inverting it amplifies noise. We read that pattern as saying that estimability, rather than conditioning, separates the regimes. It is a hypothesis the failures support, not yet a criterion: the metric was never measured under the collocation measure itself.

Paradigm. An accurate wavefunction can be trained with no Markov chain in the gradient, inside a regime whose edges we found by crossing them (§7.3). The claim is about the gradient only: a Metropolis probe still selects checkpoints and certifies the final energy, and the cascades start from reference-guided pretrained states. The weak-form objective comes within about a tenth of a percent of DMC at $N \in \{6, 12\}$ under strong confinement and within three tenths of a percent of our own converged energies well into the correlated regime, as measured by independent Metropolis evaluation of every trained state. It fails through one chain of causes—proposal mismatch, weight-tail inflation, gradient noise, an unestimable metric—which a better proposal moves but does not remove, and then through a second failure of transfer between confinements.

Physics. The trained state serves as an instrument (§6.2). The crossover it resolves is staged: the cloud expands and its shells stiffen radially well before the electrons acquire angular order within them.

The shell geometry that emerges at the weakest traps is resolved topology by topology, and its measured radius approaches the classical value $1.334\,\omega^{-2/3}$ to within 1.7%. The staging itself is better established than where the transition sits: a grid with one point per decade can order the stages but not locate them. And these are the lowest states found at fixed spin projection, in the regime where a change of ground-state spin is most plausible.

9.1 Concluding perspective

All four answers are bounded in the same way. Every state is the lowest found within a fixed spin-projection sector; the architecture comparison is seeded at $N{=}6$ and thinner above it, and its tangent-space measures belong to the networks as parametrised; and the two backflow variants are not capacity-matched, so neither the cross-architecture energy gap nor the rank collapse isolates architecture. §8.3 sets out what would move each of those.

At the widths reached for the well-resolved $N \leq 12$ studies, raw representational capacity was rarely the first bottleneck; alignment, conditioning and sampling were. It was not irrelevant everywhere. At $N{=}20$ the memory cost of the backflow's edge tensor forces a capacity cut that degrades the collocation gradients. But what determined the outcome, across architecture, optimisation and sampling alike, was alignment with the low-dimensional collective structure the Hamiltonian already possesses: breathing, correlation-hole and relative modes at strong confinement, resolving at weak confinement into the shell geometry of the Wigner molecule. That the same alignment matters in all three places is probably a fact about the system rather than about neural networks.

Several of these results were reached only after an initial reading of the same data proved wrong: a rank collapse that survived seeding where a dimension inversion did not, a transfer failure first attributed to the sampler and later to the ansatz, an importance-sampling error whose signature was indistinguishable from an architectural deficiency, and a strong-confinement collocation energy that looked statistically exact until an independent evaluation showed it to be the lucky tail of a best-of-three selection. Each was identified by a diagnostic that could separate the competing explanations. That is the methodological point these chapters have tried to make concrete: a variational calculation is only as trustworthy as the tests that can tell its failure modes apart.

Part VI

Appendices

Chapter A

Laplacian conditioning

A.1 Residual training for Schrödinger

We consider the stationary many-body Schrödinger equation on configuration space $\mathbf{R} = (\mathbf{r}_1, \ldots, \mathbf{r}_N) \in \Omega \subset \mathbb{R}^D$ with $D = dN$ (for our quantum dots, $d = 2$):

$$\hat{H}\Psi(\mathbf{R}) = E\Psi(\mathbf{R}), \qquad \hat{H} = \hat{T} + \hat{V}, \qquad \hat{T} = -\frac{1}{2}\sum_{i=1}^{N} \Delta_i, \tag{A.1}$$

where $\hat{V}$ collects external and interaction potentials.

In *strong residual minimisation* one introduces

$$L := \hat{H} - E \quad \Rightarrow \quad L\Psi = 0, \tag{A.2}$$

and minimises a squared residual loss of the schematic form

$$\mathcal{L}_{\text{res}}(\theta) = \frac{1}{2}\|L\Psi_\theta\|^2_{L^2(\Omega;w)} = \frac{1}{2}\int_\Omega |(L\Psi_\theta)(\mathbf{R})|^2 \, w(\mathbf{R}) \, d\mathbf{R}, \tag{A.3}$$

for a nonnegative weight w induced by the sampling strategy. A key message of the numerical-analysis viewpoint is that strong residual losses can be *severely ill-conditioned*, and training speed is tightly linked to spectral properties of an operator/matrix induced by L [40].

Which of the thesis's objectives this analyses. Three objectives appear in the main text and Eq. (A.3) is exactly one of them. Stage I pretraining (§4.3) minimises $\frac{1}{B}\sum_b (E_L(\tilde{R}_b) - E_{\text{eff}})^2$ over configurations drawn from $|\Psi_\theta|^2$; since $E_L = \hat{H}\Psi/\Psi$, that sample average estimates $\int |\Psi_\theta|^2 \, (\hat{H}\Psi_\theta/\Psi_\theta - E)^2 = \|(\hat{H} - E)\Psi_\theta\|^2_{L^2}$, which is Eq. (A.3) with $w \equiv 1$. Everything in this appendix therefore applies to stage I directly. It does *not* apply in the same way to the other two. Stage II is variational energy minimisation (§4.4), whose objective is linear rather than quadratic in the residual and whose conditioning is governed by the quantum geometric tensor measured in §7.2, not by the Δ^2 stiffness derived below. The collocation route (§4.5) replaces the strong residual with a weak-form score-function objective under an analytic proposal, for the reason developed in §A.4 and Appendix C: the weak form is what removes the second-derivative amplification, and it is the escape from the mechanism analysed here rather than an instance of it. Where the two production objectives inherit anything from this analysis, it is through the single channel they share with stage I—the Laplacian inside E_L—and not through the squared-residual structure.

A.2 Linearized geometry and the matrix A

To expose the mechanism, follow the standard linearization argument used in conditioning analyses [40]. Fix an expansion point θ_0 and define tangent (feature) functions

$$q_j(\mathbf{R}) := \partial_{\theta_j}\Psi_\theta(\mathbf{R})\big|_{\theta=\theta_0}, \qquad j = 1, \ldots, P. \tag{A.4}$$

For small parameter increments $\delta\theta$, we have the first-order model

$$\Psi_{\theta_0+\delta\theta}(\mathbf{R}) \approx \Psi_0(\mathbf{R}) + \sum_{j=1}^{P} q_j(\mathbf{R})\,\delta\theta_j, \qquad \Psi_0 := \Psi_{\theta_0}. \tag{A.5}$$

Applying the residual operator gives

$$(L\Psi_{\theta_0+\delta\theta})(\mathbf{R}) \approx (L\Psi_0)(\mathbf{R}) + \sum_{j=1}^{P} (Lq_j)(\mathbf{R})\,\delta\theta_j. \tag{A.6}$$

Introduce the weighted inner product (complex-valued wavefunctions allowed)

$$\langle f, g\rangle_w := \int_\Omega f(\mathbf{R})^*\, g(\mathbf{R})\, w(\mathbf{R})\, d\mathbf{R}. \tag{A.7}$$

Let the Jacobian J be the linear map whose jth column is Lq_j. Then the loss (A.3) becomes (to second order in $\delta\theta$)

$$\mathcal{L}_{\text{res}}(\theta_0 + \delta\theta) \approx \frac{1}{2}\|r_0 + J\,\delta\theta\|_w^2, \qquad r_0 := L\Psi_0. \tag{A.8}$$

with normal-equation matrix

$$A := J^*J, \qquad A_{ij} = \langle Lq_i,\, Lq_j\rangle_w. \tag{A.9}$$

Here A is Hermitian positive semidefinite; in practice it can be close to singular, so one often effectively works with a damped variant $A + \lambda I$ (implicit or explicit).

Now consider gradient descent on the quadratic model (A.8) (with step size η). The error in parameter space evolves as

$$\delta\theta^{(t+1)} - \delta\theta^\star = (I - \eta A)\left(\delta\theta^{(t)} - \delta\theta^\star\right), \tag{A.10}$$

so each eigen-direction of A contracts by a factor $(1 - \eta\lambda)$. Hence convergence speed is governed by the eigenvalue spread of A, typically summarized by the (effective) condition number

$$\kappa(A) = \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)} \tag{A.11}$$

(over the nonzero/regularised spectrum). Large $\kappa(A)$ implies stiffness: a step size small enough to be stable for $\lambda_{\max}$ yields slow progress along directions associated with $\lambda_{\min}$. This is precisely the conditioning bottleneck emphasized in physics-informed training analyses [40].

A.3 Bi-Laplacian scaling

The defining feature of Schrödinger is that L contains a second-order differential operator. To isolate the kinetic contribution, consider the regime where the residual is dominated by the Laplacian part,

$$L \approx -\frac{1}{2}\Delta_{\mathbf{R}}. \tag{A.12}$$

From L to L^*L. The matrix A in (A.9) is a Gram matrix in the feature space $\{Lq_j\}$. The corresponding operator-level object is the *Hermitian square* L^*L (with adjoint taken in the $L^2(\Omega; w)$ inner product) [40]. For general non-constant w, the adjoint L^* differs from the formal differential adjoint by weight-induced lower-order terms; the cleanest picture is obtained for constant w and standard boundary conditions (periodic/Dirichlet), where the Laplacian is self-adjoint.

In the Laplacian-dominant regime (A.12), one has (up to a constant factor)

$$L \approx -\tfrac{1}{2}\Delta \quad \Rightarrow \quad L^*L \approx \tfrac{1}{4}\,\Delta^2, \tag{A.13}$$

i.e. a bi-Laplacian scaling. The prefactor does not affect conditioning ratios.

Fourier/eigenmode derivation of k^4. On a periodic domain (or, more generally, in a Laplacian eigenbasis) $-\Delta\phi_{\mathbf{k}} = |\mathbf{k}|^2\phi_{\mathbf{k}}$, so $\Delta^2\phi_{\mathbf{k}} = |\mathbf{k}|^4\phi_{\mathbf{k}}$. If an error $\delta\Psi$ expands as $\delta\Psi = \sum_{\mathbf{k}} c_{\mathbf{k}}\phi_{\mathbf{k}}$ and $L \approx -\frac{1}{2}\Delta$, then

$$\|L\,\delta\Psi\|_{L^2}^2 \approx \frac{1}{4}\sum_{\mathbf{k}} |\mathbf{k}|^4 |c_{\mathbf{k}}|^2. \tag{A.14}$$

Thus the squared residual penalizes high-frequency components with a *quartic* weight $|\mathbf{k}|^4$. If the relevant spectrum spans $|\mathbf{k}| \in [k_{\min}, k_{\max}]$, the induced eigenvalue ratio scales like

$$\kappa \sim \left(\frac{k_{\max}}{k_{\min}}\right)^4, \tag{A.15}$$

which is a sharp mathematical expression of "Laplacian-driven stiffness". This quartic growth is consistent with the Laplacian/Poisson toy examples and frequency-dependent conditioning effects emphasized in [40].

Length-scale interpretation. Let ℓ denote the smallest length scale that must be represented. Then $k_{\max} \sim 1/\ell$ and (A.15) implies

$$\kappa \text{ grows rapidly as } \ell \downarrow 0. \tag{A.16}$$

In many-body Schrödinger problems, small ℓ arises from nodal structure, strong localisation, and short-range correlation features; in all such regimes, the Laplacian makes the strong residual extremely curvature-sensitive.

A.4 Barron/spectral perspective: the Laplacian

The conditioning story above can be rephrased in approximation terms. In spectral Barron-type viewpoints [37], function classes are controlled by norms that weight the Fourier transform by powers of frequency (or related spectral weights). A key implication is that applying derivatives systematically increases the importance of high-frequency tails [40]. Since Δ multiplies a Fourier mode by $|\mathbf{k}|^2$, and the squared residual effectively involves L^*L (hence Δ^2 in Laplacian-dominant regimes), the residual can require substantially more "spectral budget" than the wavefunction itself. In practical terms: even if Ψ is approximable in a given network class, *approximating its strong residual* can be much harder, because the loss implicitly demands accurate high-frequency curvature (cf. (A.14)). This is one of the motivations, in numerical analysis of PINNs, for considering weak/variational formulations that reduce derivative order when the strong form is too stiff [40].

A.5 Implications for Stochastic Reconfiguration

Stochastic reconfiguration (SR) updates parameters using a natural-gradient-like preconditioner [24, 26, 57]. Writing $O_i = \partial_{\theta_i} \log \Psi_\theta$, one common SR update is

$$S(\theta)\,\delta\theta = -g(\theta), \qquad S_{ij} = \mathrm{Cov}_{p_\theta}[O_i, O_j], \qquad g_i = 2\,\mathrm{Cov}_{p_\theta}[E_L, O_i], \tag{A.17}$$

where $p_\theta \propto |\Psi_\theta|^2$ and $E_L = (\hat{H}\Psi_\theta)/\Psi_\theta$. SR improves conditioning in parameter space by using the (quantum) information geometry of the variational family [24, 57]. However, the Laplacian-induced stiffness described above is an *operator-level* phenomenon: strong residuals magnify curvature errors via Δ and Δ^2 (cf. (A.14)), which remains a fundamental challenge even when parameter updates are geometrically preconditioned.

Chapter B

Coulomb conditioning

B.1 Coulomb structure and singularity

For our two-dimensional electronic systems ($\mathbf{r}_i \in \mathbb{R}^2$), the interaction contains Coulomb terms of the form

$$\hat{V}_{\mathrm{C}}(\mathbf{R}) = \sum_{1\le i<j\le N} \frac{1}{r_{ij}}, \qquad r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|. \tag{B.1}$$

This is a *singular multiplication operator*: it diverges at particle coalescence $r_{ij} \to 0$. The exact eigenfunction remains finite (or vanishes by antisymmetry) but develops non-smooth short-distance structure (a cusp) so that divergences in kinetic and potential contributions cancel in physically meaningful quantities such as the local energy [26].

> Coulomb does not merely add "another term" to the residual; it introduces *singular short-range physics* that forces delicate cancellations with the Laplacian. Strong residual objectives square these effects and become extremely sensitive to tiny cusp errors.

B.2 2D cusp conditions: antiparallel vs parallel spins

We present the 2D cusp logic in a way that separates the two spin channels. The cleanest statement (especially for practical ansätze) is in terms of the short-distance behaviour of a *Slater–Jastrow* factorization

$$\Psi(\mathbf{R}) = D(\mathbf{R})\, J(\mathbf{R}), \qquad J(\mathbf{R}) = \exp\Big(\sum_{i<j} u(r_{ij})\Big), \tag{B.2}$$

because (i) the Slater part D enforces antisymmetry and determines whether Ψ vanishes at coalescence, and (ii) the Jastrow slope $u'(0)$ is what cancels the $1/r$ Coulomb singularity in the local energy [26].

Fix a pair (i, j) and introduce relative/centre-of-mass coordinates in 2D:

$$\mathbf{R}_{ij} := \frac{\mathbf{r}_i + \mathbf{r}_j}{2}, \qquad \mathbf{r} := \mathbf{r}_i - \mathbf{r}_j, \qquad r := |\mathbf{r}|. \tag{B.3}$$

A direct computation gives, in any dimension,

$$\Delta_i + \Delta_j = \frac{1}{2}\,\Delta_{\mathbf{R}_{ij}} + 2\,\Delta_{\mathbf{r}}, \tag{B.4}$$

hence the singular short-distance kinetic behaviour is controlled by the relative term $-\Delta_{\mathbf{r}}$.

A key 2D identity is

$$\Delta_{\mathbf{r}} r = \frac{1}{r} \qquad (r > 0,\ \mathbf{r} \in \mathbb{R}^2). \tag{B.5}$$

Antiparallel spins (distinguishable in the Slater part). If the two electrons have opposite spin, they enter different determinants in D and generically D does *not* vanish at $\mathbf{r} \to 0$. Then $\Psi(0) \neq 0$ and a direct cusp expansion in 2D is legitimate:

$$\Psi(\mathbf{r}, \ldots) = \Psi(0, \ldots)\Big(1 + a\, r + \mathcal{O}(r^2)\Big), \qquad r \to 0. \tag{B.6}$$

Using (B.5), the singular part of the relative Laplacian is

$$\Delta_{\mathbf{r}}\Psi \approx \Psi(0)\, a\, \Delta_{\mathbf{r}} r = \Psi(0)\, \frac{a}{r}, \qquad \frac{\Delta_{\mathbf{r}}\Psi}{\Psi} \approx \frac{a}{r}. \tag{B.7}$$

Since the pair kinetic contribution contains $-\Delta_{\mathbf{r}}$, the singular part of the local energy is

$$E_L(\mathbf{R}) = \frac{\hat{H}\Psi}{\Psi} \supset \left(-\frac{a}{r} + \frac{1}{r}\right) = \frac{1-a}{r}. \tag{B.8}$$

Thus finiteness of E_L at coalescence requires

$$a = 1, \qquad \text{i.e.} \qquad \partial_r \log \Psi|_{r=0} = 1 \quad \text{(2D, antiparallel spins)}. \tag{B.9}$$

Equivalently, in a Slater–Jastrow form with $D(0) \neq 0$, this is the same as $\partial_r \log J|_0 = 1$ for the ij-pair.

Parallel spins (identical fermions, node at coalescence). For same-spin electrons, antisymmetry forces D (and hence Ψ) to vanish as $\mathbf{r} \to 0$. Generically one has a *linear* zero:

$$D(\mathbf{r}, \ldots) \;=\; \boldsymbol{\eta}(\ldots) \cdot \mathbf{r} \;+\; \mathcal{O}(r^2), \qquad r \to 0, \tag{B.10}$$

with some nonzero vector $\boldsymbol{\eta}$ depending on the other coordinates. In this case, the cancellation of the Coulomb singularity is controlled by the Jastrow factor through cross terms in $-\Delta_{\mathbf{r}}(DJ)/(DJ)$. A standard short-distance analysis (carried out explicitly for 2D quantum dots) shows that the divergent kinetic contribution behaves like

$$-\frac{\Delta_{\mathbf{r}}(DJ)}{DJ} \supset -\frac{(2k+d-1)}{r}\, \partial_r \log J \qquad \text{with} \qquad d = 2, \quad k = 1 \text{ (identical particles)}, \tag{B.11}$$

while the Coulomb potential contributes $+1/r$ to E_L as before. Hence finiteness of the local energy requires

$$\partial_r \log J|_{r=0} = \frac{1}{2k+d-1} = \frac{1}{3} \quad \text{(2D, parallel spins)}. \tag{B.12}$$

This 1 (antiparallel) versus 1/3 (parallel) 2D cusp statement is the standard quantum-dot/Jastrow–Slater cusp condition [58].

What to remember. In 2D:

$$\text{antiparallel spins:} \quad \partial_r \log \Psi|_0 = 1 \iff \partial_r \log J|_0 = 1, \tag{B.13}$$

$$\text{parallel spins:} \quad \Psi(0) = 0 \text{ (linear node)}, \quad \partial_r \log J|_0 = \frac{1}{3}. \tag{B.14}$$

B.3 Why Coulomb is ill-conditioned (2D emphasis)

The cusp conditions above show *how* Coulomb creates stiffness: it forces kinetic and potential contributions to be individually large and singular, while their *sum* is finite only under a short-distance constraint. Strong residual minimisation is therefore a *cancellation problem.*

Sensitivity amplification in the local energy. If the learned antiparallel-spin cusp slope is $a_\theta = 1 + \delta a$, then (B.8) implies

$$E_L(\mathbf{R}) \supset -\frac{\delta a}{r}, \tag{B.15}$$

so an arbitrarily small cusp mismatch produces an arbitrarily large local-energy spike as $r \to 0$.

For parallel spins, if the learned Jastrow slope is $\partial_r \log J|_0 = \frac{1}{3} + \delta b$, then the same short-distance analysis leading to (B.12) implies

$$E_L(\mathbf{R}) \supset -\frac{3\,\delta b}{r}. \tag{B.16}$$

Thus both channels generate $1/r$ spikes unless the cusp is satisfied.

Why squaring is especially dangerous in 2D. A $1/r$ defect in E_L (or in $(\hat{H}-E)\Psi$) is concentrated near coalescence. But in 2D the squared singularity is not integrable—it diverges logarithmically:

$$\int_{r<\varepsilon} \left(\frac{1}{r}\right)^2 d^2\mathbf{r} = \int_0^{\varepsilon} \frac{1}{r^2}\,(2\pi r\,dr) = 2\pi \int_0^{\varepsilon} \frac{dr}{r} = +\infty.$$

Therefore, without effectively resolving the cusp, *strong* squared-residual objectives can be dominated (or even made ill-defined in the continuum limit) by the coalescence region. In Monte Carlo practice, this manifests as heavy-tailed fluctuations and extreme sensitivity to rare close-pair samples.

Residual expansion and cancellation structure. Writing $L = \hat{T} + (\hat{V} - E)$ and expanding the squared residual norm:

$$\|L\Psi\|_w^2 = \|\hat{T}\Psi\|_w^2 + \|(\hat{V}-E)\Psi\|_w^2 + 2\,\mathrm{Re}\langle \hat{T}\Psi,\ (\hat{V}-E)\Psi\rangle_w, \tag{B.17}$$

we see that near coalescence both $\hat{T}\Psi$ and $\hat{V}_{\mathrm{C}}\Psi$ are large. The exact solution relies on cancellation between these contributions (cf. (B.8) and (B.12)), which means (B.17) demands high *relative* accuracy in the cusp region: if either term is off by a small fraction, the net residual can be large in absolute magnitude. This is a classic recipe for ill-conditioning: the loss landscape contains directions where small parameter changes cause huge residual changes in a small region.

Operator-level picture (L^*L). From the same viewpoint as Appendix A, strong residual training induces matrices/operators tied to L^*L [40]. For $L = \hat{T} + (\hat{V} - E)$, one has (formally)

$$L^*L = \hat{T}^2 + (\hat{V}-E)^2 + \hat{T}(\hat{V}-E) + (\hat{V}-E)\hat{T}. \tag{B.18}$$

Coulomb enters here in two destabilizing ways:

1. the positive term $(\hat{V}-E)^2$ involves a $1/r^2$-type singular weight;
2. the mixed terms contain derivatives of V through product rules, e.g. $\hat{T}(V\Psi)$ generates $\nabla V \cdot \nabla\Psi$ and ΔV contributions, which are even more singular/distributional at coalescence.

This formal expansion explains why the Coulomb singularity can severely inflate the effective spectral spread of the training operator, aligning with the operator-conditioning viewpoint [40].

B.4 Barron/spectral perspective: the Coulomb singularity

Coulomb forces the *true* 2D solution to have cusp-like, non-smooth short-range structure, so approximating the *strong residual* requires representing high-frequency content that is not well captured by globally smooth/low-complexity function classes.

In spectral/Barron-style approximation theories [37], approximation quality is governed by spectral decay and norms that weight the Fourier tail [40]. A cusp (or a linear node times a cusp in J) implies slower spectral decay than a globally smooth function. Strong residual training then *further* amplifies the high-frequency tail because it applies Δ (and effectively L^*L), which multiplies Fourier modes by powers of $|\mathbf{k}|$ (Appendix A). Hence, even if Ψ can be approximated reasonably in function value, the strong residual can remain large unless the cusp structure is represented accurately. This is exactly the type of regularity mismatch that motivates weak/variational formulations in numerical analysis of PINN-type methods, since weak forms reduce the derivative order demanded of the approximation [40].

Practically, QMC wavefunctions therefore incorporate explicit short-range correlation structure (e.g. cusp-enforcing Jastrow factors), precisely to remove these singular residual pathologies and reduce the variance of local quantities [26, 58].

B.5 Implications for Stochastic Reconfiguration

SR performs a natural-gradient step using a (quantum) Fisher/metric matrix [24, 26, 57]. In one common formulation,

$$S_{ij} = \mathrm{Cov}_{p_\theta}[O_i, O_j], \qquad g_i = 2\,\mathrm{Cov}_{p_\theta}[E_L, O_i], \qquad S\,\delta\theta = -g, \tag{B.19}$$

with $O_i = \partial_{\theta_i} \log \Psi_\theta$.

Coulomb-induced cusp mismatch produces E_L spikes of the form (B.15)–(B.16), which has two consequences:

1. **Heavy-tailed Monte Carlo forces.** The estimator of $g_i = \mathrm{Cov}[E_L, O_i]$ becomes noisy when E_L has rare but extreme spikes (short-range events). This increases variance and can destabilize the linear solve unless one uses damping/regularisation, as standard in SR practice [26].

2. **Coupling to curvature sensitivity.** Even if SR corrects anisotropy in parameter space (natural gradient) [24, 57], it does not remove the underlying operator-driven stiffness: the spikes originate from imperfect kinetic–potential cancellation enforced by Coulomb cusp physics.

Summary. Laplacian dominance creates stiffness through k^4-type amplification of curvature errors; Coulomb creates stiffness by enforcing cusp constraints (in 2D: 1 vs 1/3 depending on spin channel) and cancellation against the Laplacian. Both mechanisms worsen conditioning of strong residual objectives and increase statistical difficulty of SR through heavy-tailed local-energy fluctuations [26, 40, 58].

Chapter C

The collocation–backflow catch-22

This appendix formalizes a structural difficulty encountered when training backflow transformations under the *strong-form* collocation objective—the one that minimises the variance of the pointwise local energy E_L—and presents extensive numerical evidence for the N=6, ω=0.5 quantum dot. The core observation is a *catch-22*: backflow's learning signal is concentrated near the nodal surface, but the local energy diverges there for any imperfect wavefunction, and every sampling strategy we tried that includes those points produced gradients whose variance we could not control. The result, across every intervention below, is an accuracy floor at roughly 0.3%—set by the Jastrow alone—that is absent under VMC + SR training of the same ansatz family. We call the floor irreducible *within this pipeline*: it is an empirical bound established by exhaustion over the interventions we tested, not a proof that no strong-form scheme can do better.

The scope of this result needs two qualifications. First, it is a property of the *strong form* specifically: the weak-form (score-function / REINFORCE) collocation objective used for the results in the main text never differentiates through the Laplacian and does not inherit this pathology, which is why it trains the same ansatz to far below the 0.3% floor. The value here is therefore a diagnosis of one objective's limits, not a measure of what Markov-chain-free training can achieve. Second, this analysis is what motivated the shift to the weak form, and is reported as a negative result and not as a headline number.

C.1 Setup and notation

We consider a Slater–Jastrow–backflow ansätz of the form

$$\Psi_\theta(\mathbf{R}) = \det \tilde{S}_\uparrow(\mathbf{R};\theta_{\mathrm{BF}})\ \det \tilde{S}_\downarrow(\mathbf{R};\theta_{\mathrm{BF}})\ \exp\big(J(\mathbf{R};\theta_J)\big), \tag{C.1}$$

where $\tilde{S}_\sigma$ is the Slater matrix evaluated at backflow-shifted coordinates $\tilde{\mathbf{r}}_i = \mathbf{r}_i + \Delta\mathbf{x}_i(\mathbf{R};\theta_{\mathrm{BF}})$, J is a neural Jastrow correlator, and $\theta = (\theta_J, \theta_{\mathrm{BF}})$ are trainable parameters. A CTNN message-passing network produces the backflow shifts $\Delta\mathbf{x}_i$.

We train by minimising the *screened collocation* objective

$$\mathcal{L}(\theta) = \mathrm{Var}_{\mathcal{S}}\big[E_L(\mathbf{R};\theta)\big], \qquad E_L = \frac{\hat{H}\Psi_\theta}{\Psi_\theta}, \tag{C.2}$$

where the collocation set $\mathcal{S}$ is obtained by importance-sampling from a Gaussian proposal q and retaining the top-K points ranked by $|\Psi_\theta|^2/q$ ("screened collocation"). This top-K selection naturally suppresses near-node configurations where $|\Psi|^2 \approx 0$.

C.2 The divergence of E_L near nodes

The local energy decomposes as

$$E_L = -\frac{1}{2}\frac{\nabla^2\Psi}{\Psi} + V = -\frac{1}{2}\Big[\nabla^2 J + |\nabla J|^2 + 2\,\nabla J\cdot\frac{\nabla D}{D} + \frac{\nabla^2 D}{D}\Big] + V, \tag{C.3}$$

where $D := \det \tilde{S}_\uparrow \det \tilde{S}_\downarrow$ and the last term $\nabla^2 D/D$ contains second derivatives of the backflow shifts $\nabla^2 \Delta \mathbf{x}_i$ amplified by the inverse determinant.

For the exact wavefunction, E_L is constant. If Ψ is an eigenstate, $E_L = E$ everywhere. The cancellation is a property of the *full* local energy $\hat{H}\Psi/\Psi$ with $\Psi = D\,e^J$, in which the Jastrow cross term $2\nabla J \cdot \nabla D/D$ of Eq. (C.3) participates alongside $\nabla^2 D/D$; we isolate the determinant terms below because they are the ones the backflow parameters move, not because the others are absent. For an exact eigenstate, the potentially singular contributions in the full expression $\nabla^2(De^J)/(De^J)$ cancel at a simple node. This is not a property of $\nabla^2 D/D$ in isolation: for a generic smooth determinant with $D \sim d$ near the nodal surface $\mathcal{N}$, $\nabla^2 D$ may remain $\mathcal{O}(1)$. The exact eigenstate satisfies

$$\frac{\nabla^2(De^J)}{De^J} \text{ finite on } \mathcal{N}, \tag{C.4}$$

through cancellation among the determinant, Jastrow-cross, and remaining Jastrow terms in Eq. (C.3).

For an approximate wavefunction, the cancellation generically fails. During training, the backflow parameters θ_{BF} are imperfect: the *learned* nodal surface $\mathcal{N}_\theta$ (where $D(\cdot;\theta) = 0$) does not coincide with the *true* nodal surface. Let d_θ denote the signed distance from the learned node and d^* that from the true node. Near a point where $d_\theta \approx 0$ but $d^* \neq 0$, the determinant $D \sim d_\theta \to 0$ while the kinetic contributions from the Jastrow and potential terms remain finite:

$$E_L(\mathbf{R}) \supset \frac{\nabla^2 D(\mathbf{R};\theta)}{D(\mathbf{R};\theta)} \sim \frac{c}{d_\theta} \to \pm\infty \qquad \text{as } d_\theta \to 0,\ d^* \neq 0, \tag{C.5}$$

for some configuration-dependent constant $c \neq 0$. Conversely, near a point on the true node ($d^* = 0$) but off the learned node ($d_\theta \neq 0$), $\Psi \sim d^* \to 0$ and the wavefunction assigns vanishing weight. The pathology (C.5) is therefore not an artefact of the nodal structure itself—it is a property of *mismatched* nodes, and is worst precisely when backflow still has something to learn.

C.3 The geometric mismatch argument

We now locate the divergence. It is *not* in the variance of the local energy, which the calculation below shows to be finite for a simple node; it is one derivative later, in the parameter gradient.

Consider the contribution to $\mathrm{Var}[E_L]$ from a neighborhood U_ε of the learned nodal surface:

$$\mathrm{Var}[E_L] \geq \int_{U_\varepsilon} |\Psi(\mathbf{R};\theta)|^2 \left(E_L(\mathbf{R};\theta) - \bar{E}\right)^2 d\mathbf{R}. \tag{C.6}$$

When the node is "simple" (D vanishes linearly), we have

$$|\Psi|^2 \sim d_\theta^2, \qquad\qquad E_L^2 \sim d_\theta^{-2\alpha}, \tag{C.7}$$

where α characterises the severity of the mismatch. For the generic case where the numerator $\nabla^2 D$ does *not* vanish on $\mathcal{N}_\theta$ (because $\mathcal{N}_\theta \neq \mathcal{N}^*$), one has $\alpha = 1$ so that $E_L^2 \sim d_\theta^{-2}$.

In D-dimensional configuration space, the volume element in a tubular neighborhood of a codimension-1 surface scales as $dV \sim d_\theta^0\, d\sigma\, dd_\theta$. Thus the integrand in (C.6) behaves as

$$|\Psi|^2 \cdot E_L^2 \cdot dV \sim d_\theta^2 \cdot d_\theta^{-2} \cdot dd_\theta = dd_\theta, \tag{C.8}$$

which integrates to $\mathcal{O}(\varepsilon)$ and is therefore *finite*: a simple node does not, by itself, make $\mathrm{Var}[E_L]$ diverge under the $|\Psi|^2$ measure. The instability enters one derivative later, in the *parameter gradient*. But the key observation is that this is the *best case*. In practice, the mismatch between $\mathcal{N}_\theta$ and $\mathcal{N}^*$ is not uniform: there exist regions where the surfaces cross at angles, creating pockets where $|d^*|/|d_\theta|$ is large (the true wavefunction is non-negligible while the learned determinant is near-zero), and there the E_L^2 spike is not compensated by $|\Psi|^2$ suppression.

More concretely, the gradient $\partial\mathcal{L}/\partial\theta_{\mathrm{BF}}$ involves terms of the form

$$\frac{\partial E_L}{\partial \theta_{\mathrm{BF}}} \supset \frac{1}{D}\frac{\partial(\nabla^2 D)}{\partial \theta_{\mathrm{BF}}} - \frac{\nabla^2 D}{D^2}\frac{\partial D}{\partial \theta_{\mathrm{BF}}}, \tag{C.9}$$

both of which diverge as $D \to 0$. Even in the marginal case where $\mathrm{Var}[E_L]$ is finite, the *gradient variance* (which involves these higher-order poles) generically diverges, making gradient-based optimisation impossible at points near the learned node.

C.4 The catch-22

The preceding analysis reveals a three-sided dilemma:

1. **Backflow learns at the node.** The purpose of backflow is to reshape the nodal surface of the Slater determinant. The gradient signal $\partial E_L/\partial\theta_{\mathrm{BF}}$ is largest where the backflow shifts actually change which side of the node a configuration point lies on, i.e., where $D \approx 0$.

2. **E_L diverges at the node for imperfect θ.** As shown in §C.2–C.3, the local energy and its parameter gradient contain poles of the form D^{-1} and D^{-2} whenever the learned node does not coincide with the exact node. The resulting gradient variance is unbounded.

3. **Good collocation sampling must avoid the node.** Under screened collocation, the top-K selection ranks points by $|\Psi|^2/q \propto D^2 e^{2J}/q$. Since $D^2 \to 0$ at the node, near-node points are systematically excluded from the training batch. This is *protective*—it prevents gradient explosion—but it also removes the points where backflow's learning signal lives.

A "catch-22" emerges:

> *Including near-node points gives divergent, noisy gradients and backflow shrinks defensively or diverges. Excluding them gives stable training but removes backflow's learning signal. In either regime, collocation cannot effectively train backflow beyond an accuracy floor set by the correlator quality alone.*

C.5 Why VMC + SR escapes

Under VMC, the sampling distribution *is* $|\Psi_\theta|^2$, which assigns vanishing measure to the nodal surface. Near-node configurations are strongly *suppressed*—for a simple node the probability of lying within a distance ϵ scales as ϵ^3—rather than strictly absent, which is enough to keep the D^{-1} singularity out of the gradient estimator in practice. However, VMC *still optimises the node position* through a subtle mechanism:

Indirect nodal information through bulk correlations. The variational energy $\langle E_L\rangle_{|\Psi|^2}$ is sensitive to nodal position because shifting the node changes the wavefunction *everywhere* (through normalisation and through the functional form of D). The SR force vector $g_i = \mathrm{Cov}_{|\Psi|^2}[E_L, O_i]$ (where $O_i = \partial_{\theta_i}\log\Psi$) receives contributions from the *bulk*—regions where both $|\Psi|^2$ and E_L are well-behaved—that are *correlated with* nodal position but do not require evaluating E_L at the node.

SR amplifies the signal. The Fisher metric $S_{ij} = \mathrm{Cov}[O_i, O_j]$ captures how sensitive $\log\Psi$ is to each parameter direction. By solving $(S + \varepsilon I)\,\delta\theta = -g$, SR amplifies exactly those subtle bulk signals that encode nodal information, compensating for the fact that the node itself has zero weight under $|\Psi|^2$.

Core asymmetry. The asymmetry needs stating carefully, because a power count that is too quick proves too much. Under $|\Psi|^2$ the SR force $g_i = \mathrm{Cov}[E_L, O_i]$ has a finite *mean*: both E_L and $O_i = \partial_{\theta_i}\log\Psi$ carry a d_θ^{-1} pole near a mismatched node, their product goes as d_θ^{-2}, and the measure supplies $|\Psi|^2 \sim d_\theta^2$, so the integral converges. Its *variance* does not, by the same counting—so VMC is not immune to near-node pathology either, and in practice this is what the damping and trust region of §4.4 are for.

The real distinction is one derivative higher. VMC and SR never differentiate E_L: the local energy enters only as a multiplicand in a covariance, so the D^{-2} pole of Eq. (C.9)—which comes from ∂_θ acting on $\nabla^2 D/D$—never appears. Strong-form collocation differentiates E_L by construction, and inherits it. That is why the same node is survivable in one scheme and not in the other, and it is the argument

§7.3.1 makes for the weak form as well. None of the sampling, loss, or preconditioning strategies we tested (§C.6) overcame this asymmetry within the strong-form collocation framework; we present it as a structural obstruction of the tested pipeline rather than a theorem that no collocation method can ever avoid it.

C.6 Experimental evidence

We present a systematic study for N=6 electrons in a 2D harmonic trap with ω=0.5 (DMC reference: $E_{\mathrm{DMC}} = 11.78484(6)$ [12]). The ansätz uses a neural Jastrow correlator (∼15,000 parameters) and a CTNN backflow network (∼8,600 parameters) with bf_scale = 0.7. All energies are evaluated via MCMC with 15 000 samples after burn-in.

C.6.1 Baseline: correlator alone vs. correlator + backflow

Table C.1: Baseline collocation results for N=6, ω=0.5. "Correlator only" trains without backflow; "joint" trains both networks from scratch for 300 epochs with bf_scale = 0.7. The backflow provides only a marginal improvement of ∼0.1 percentage points.

Configuration	E (Ha)	σ_E	err (%)
Correlator only (300 ep)	11.834	0.003	0.42
Correlator + BF, joint (300 ep)	11.823	0.003	0.33
Correlator + BF, separate opt	11.822	0.003	0.32
DMC reference	11.78484	—	0.00

The backflow reduces the error from 0.42% to 0.32–0.33%, far short of the <0.01% reached by VMC + SR on the same ansatz *family* (Table 6.2). The two are not matched in size: the networks here total ∼24k parameters against ∼236k for the production model, so the comparison bounds what this objective achieves at this capacity, not isolating the objective alone. The floor is nonetheless a property of the objective, because it does not move under any of the interventions below, including ones that add capacity.

C.6.2 Sampling strategies

We tested multiple strategies to bring near-node information into the collocation batch, hoping to give the backflow a stronger learning signal.

Table C.2: Effect of sampling strategy on collocation accuracy. All experiments use 50-epoch fine-tuning from the best collocation checkpoint (0.35%), except "hard-smooth (scratch)" which trains 300 epochs from random initialisation.

Method	Description	err (%)	Outcome
Top-K baseline	standard screened collocation	0.35	stable
Hard mining + Huber	top-K of residuals	0.41	neutral
Hard mining + smoothness	+ $\|\nabla^2\Delta x\|^2$ penalty	0.38	slight help
SIR (single proposal)	$\|\Psi\|^2$-reweighted resampling	0.47	worse
SIR (mixture)	Gaussian mixture proposal	0.44	worse
Gumbel top-K [59]	stochastic top-K selection	5.00	catastrophic

Every strategy that includes more near-node points (SIR, Gumbel) produced *worse* results—consistent with the catch-22: the additional points carry divergent E_L values that corrupt the gradient. Gumbel top-K (which replaces deterministic selection with stochastic Gumbel-softmax) was catastrophic, with early stopping triggered by rapid energy divergence.

C.6.3 Loss modifications

Table C.3: Effect of loss function modifications. The hybrid loss adds a mean-energy (variational) term; the det filter discards points with small $|\det \tilde{S}|$.

Method	Description	err (%)	Outcome
Variance loss (baseline)	Huber on $E_L - \bar{E}$	0.33	best collocation
Hybrid anneal	$\beta \cdot \mathrm{Var} + (1-\beta) \cdot \mathrm{Mean}$, β:0.7→0.15	2.59	catastrophic
Gumbel + hybrid	stochastic top-K + hybrid	5.00	catastrophic
Hybrid + det filter	β=0.2 fixed + $\lvert\det\rvert$ filter	24.75	broken

The mean-energy gradient $\nabla_\theta \langle E_L \rangle_{\mathcal{S}}$ under biased collocation sampling does *not* equal the VMC gradient—the sampling distribution is not $|\Psi|^2$—so the variational principle does not hold, and the backflow drifts to large, unphysical shifts ($|\Delta \mathbf{x}| \approx 0.83$, versus $\sim$0.23 under variance-only training). The variance loss's implicit constraint on backflow magnitude is *protective*, not pathological.

C.6.4 Gradient regularisation

We attacked the divergence sources in E_L directly:

Table C.4: Regularisation of the E_L singularity (50-epoch fine-tune from 0.35% checkpoint). Each modification smooths or clips one of the divergent contributions to E_L.

Method	Modification	err (%)	Outcome
Soft Coulomb	$1/r \to 1/\sqrt{r^2+\varepsilon^2}$, ε=0.2	1.41	much worse
Regularised det	$\log\lvert D\rvert \to \mathrm{logaddexp}(\log\lvert D\rvert, \log\delta)$	0.49	worse
T clipping	$\max(\lvert T\rvert, T_{\max})$, $T_{\max}$=30	0.43	neutral

Soft Coulomb trains on wrong physics (V is no longer the true potential). Regularising the determinant distorts the $\log\Psi$ landscape. Kinetic energy clipping is neutral because the clipped points are exactly those that carried the (noisy) backflow signal. None of the regularisations improve upon the baseline.

C.6.5 Gradient preconditioning

Table C.5: Gradient preconditioning experiments (300-epoch scratch training unless noted). K-FAC [60] uses block-diagonal Kronecker-factored Fisher on the backflow network only.

Method	err (%)	Outcome
Vanilla Adam	0.33	baseline
K-FAC on CTNN	0.34	no improvement
K-FAC + smoothness	0.51	worse
Variance-minimising pretrain	0.54	worse
Adam fine-tune (30 ep, lr=1e-4)	0.34	no improvement
Adam burst (15 ep, lr=3e-4)	0.56	worse

K-FAC provides a block-diagonal approximation to the Fisher matrix and should, in principle, help with the ill-conditioned parameter-space geometry. In practice, it does not: the conditioning problem is *operator-level* (Appendices A–B), not merely parameter-space anisotropy, and K-FAC cannot address the fundamental divergence in E_L.

C.6.6 Summary of all interventions

The pattern is striking:

- Every intervention that *includes* more near-node points (SIR, Gumbel) *increases* the error—backflow shrinks defensively under gradient noise or diverges entirely.

Table C.6: Complete summary of all collocation-based backflow experiments for N=6, ω=0.5. No intervention breaks the ∼0.32% accuracy floor. The final row is the same ansatz *family* under VMC + SR at production width (Table 6.2), an order of magnitude larger in parameter count; it is included to show the size of the gap, not as a matched comparison.

Experiment	err (%)	Category
Correlator + BF, sep opt (best)	0.32	baseline
Correlator + BF, joint 300 ep	0.33	baseline
K-FAC on CTNN	0.34	preconditioner
Adam fine-tune 30 ep	0.34	optimiser
Top-K baseline (retrained)	0.35	sampling
Hard mining + smoothness	0.38	sampling
Hard mining + Huber	0.41	sampling
Correlator only	0.42	no backflow
T clipping	0.43	regularisation
SIR (mixture)	0.44	sampling
SIR (single)	0.47	sampling
Regularised det	0.49	regularisation
K-FAC + smoothness	0.51	preconditioner
Var-min pretrain	0.54	pretrain
Adam burst lr=3e-4	0.56	optimiser
Soft Coulomb	1.41	regularisation
Hybrid loss (annealed)	2.59	loss
Gumbel top-K + hybrid	5.00	sampling + loss
Hybrid + det filter	24.75	loss + filter
Same ansatz, VMC + SR	**<0.01**	**variational**

- Every intervention that *changes* the loss to tolerate near-node contributions (hybrid mean-energy, det filter) produces catastrophic results because the collocation sampling is not $|\Psi|^2$ and the variational principle does not apply.
- The smoothness penalty $\lambda\|\nabla^2\Delta\mathbf{x}\|^2$ helps modestly (from 0.59% to 0.38% without it vs. with it) by damping the $\nabla^2 D/D$ poles, but cannot eliminate the fundamental divergence.
- Ironically, the only method that trains backflow effectively is MCMC-based VMC + SR—which sidesteps the entire collocation framework.

C.7 Connection to Appendices A and B

The catch-22 is a specific manifestation of the general conditioning difficulties analysed in Appendices A–B:

Laplacian stiffness (Appendix A). The k^4-type amplification of curvature errors (§A.3) is compounded by backflow, because $\nabla^2(\det \tilde{S})$ introduces second derivatives of $\Delta\mathbf{x}$ that are not present in a fixed-node ansatz. These additional high-frequency modes inflate the spectral spread of the training operator L^*L.

Coulomb singularity (Appendix B). The $1/r$ Coulomb potential (§B.1) adds a singular multiplication operator to $\hat{H}$. Near both the coalescence point ($r_{ij} \to 0$) and the node ($D \to 0$), E_L contains products of these two singularities, and the cusp cancellation (§B.2) must be maintained simultaneously with the nodal cancellation (C.4). In the strong residual squared $\|LE_L\|^2$, these compound singularities create an extremely stiff landscape.

The backflow-specific pathology. What distinguishes the backflow catch-22 from the general conditioning problem is the *self-referential* nature of the singularity: the backflow parameters θ_{BF} determine *where* the node is, so updating θ_{BF} moves the singularity in configuration space. Under the collocation loss, the gradient pushes θ_{BF} in a direction determined by E_L values at the current collocation points, but the resulting node shift invalidates those very points. VMC+SR avoids this because it samples from the *current* $|\Psi_\theta|^2$ at every step, so the sampling distribution co-adapts with the node position.

Chapter D

Additional collocation results

This appendix records the collocation results that Chapter 7 refers to but does not tabulate: the N=20 energies, and the variants that were tried and failed. Both bear on the same question—what limits the Markov-chain-free route—from the side of what does not work.

D.1 What was varied

Four axes were explored. The *sampling*: Gaussian and Gaussian-mixture proposals scaled by $1/\sqrt{\omega}$, oversample-then-resample with ESS diagnostics, ESS-adaptive oversampling, weight tempering and log-weight clipping. The *objective*: finite-difference collocation residuals, a hybrid weak form, and pure REINFORCE, which became the default, with the Laplacian kept in the reward path but never differentiated. The *preconditioner*: Adam, diagonal-Fisher natural gradient, and full SR in Woodbury and CG modes with damping anneals and trust-region caps. And the *training style*: single sweeps, wave-based cascades with checkpoint inheritance, and asynchronous campaigns relaunching from the best available checkpoint.

Two results held across the search. Optimisation quality dominated architectural novelty: moderately sized backflow+Jastrow models with stable training beat more ambitious variants with weaker optimisation geometry. And at low ω, regularising the resampling distribution (ESS control, top-mass reduction) helped where hard cusp gating did not—a controlled 2×2×3 matrix over gating, resampling regularisation and three seeds showed no measurable gain from gating and a consistent gain from regularisation.

Warm-starting along the confinement cascade $\omega : 1.0 \to 0.5 \to 0.1 \to 0.01 \to 0.001$ was necessary throughout; cold starts at low ω did not converge. Representative cascade results are N=6 at ω=1.0 ($E = 20.16167 \pm 0.00181$, +0.012%), ω=0.5 ($E = 11.78466 \pm 0.00199$, −0.002%) and ω=0.1 ($E = 3.56525 \pm 0.00046$, +0.32%), and N=12 at ω=1.0 ($E = 65.70699 \pm 0.00595$, +0.010%). These are single end-of-run evaluations of selected runs and were not independently re-evaluated; the selection effect documented in §7.3.4 makes the strong-confinement values among them optimistic by several hundredths of a percent. For N=2 the route covers the whole confinement range: +0.012%, +0.023%, −0.004%, +0.002% and +0.154% at $\omega = 1, 0.5, 0.1, 0.01, 0.001$, zero-variance extrapolated.

D.2 The $N = 20$ collocation energies

At N=20 the backflow's edge-feature tensor, which scales as $\mathcal{O}(N^2 h_{\mathrm{bf}})$, forces the hidden dimension down to h_{bf}=48–64 to fit in memory. The best backflow run at ω=1.0 then reached only ∼+18% error, while a Jastrow-only ansatz—freed of that tensor, and so able to afford a larger collocation budget and higher oversampling—reached +1.32% ($E = 157.934 \pm 0.018$). The more expressive architecture is beaten by the simpler one, which is consistent with the memory-forced capacity cut degrading gradient quality more than its extra flexibility repays.

Polishing the Jastrow checkpoints at reduced learning rate improved the ω=1.0 error from ∼+2.6% to +1.32% over ∼400 epochs, and the ω=0.5 error from ∼+7.0% to +2.74% ($E = 96.447 \pm 0.015$); the best ω=0.1 result was +5.53% ($E = 31.637 \pm 0.008$). The gap to the sub-percent accuracy reached at

$N \in \{6, 12\}$ tracks the capacity cut and the gradient noise it brings. A more memory-frugal backflow is the direct next intervention; these experiments do not isolate the collocation objective as the source of the remaining gap.

D.3 Negative results worth recording

Scalar final energy is not sufficient for method development on this route: ESS, top-mass concentration, rollback counts and the probe-to-final gap are what distinguish a run that converged from one that stopped moving. Several intuitively attractive interventions delivered nothing in the tested range—hard cusp gating most notably—and recording that was necessary to keep the search from circling. Stricter SR trust constraints, heavier damping, ESS-adaptive sampling and rollback controls improved operational stability without closing the accuracy gap at the hardest targets. What eventually closed it at low ω and N=6 was not a stabilisation measure but a change of proposal (§7.4).

Chapter E

The learned representation of one trained solution

This appendix records what the internal representation of *one* high-quality trained solution looks like. It is descriptive, it is measured on a single training run, and—as §7.1.2 sets out—most of the quantities in it are component-level and therefore gauge-like: the correlator and the backflow are partial substitutes, so how the labour is divided between them records the route the training took as much as the state it reached. Nothing in the architectural argument of Chapter 7 rests on anything here. It is reported because the regularities are striking and because they are ones the network was free *not* to adopt, not because they establish a property of the architecture.

Three findings survive that qualification well enough to be worth stating. Across $N \in \{2, 6, 12, 20\}$ and $\omega \in \{1, 0.5, 0.1, 10^{-2}, 10^{-3}\}$ the correlator feature space stays on one to three effective dimensions and never expands, even where the real-space density develops elaborate shell structure; what changes across the crossover is which branch the leading axis is built from, not how many axes there are. The displacement field is the opposite kind of object, occupying nearly all the spatial directions available to it. And the coupling between the two changes character near $\omega \approx 0.1$, the same place the energy partition crosses over (§6.2.3).

Every diagnostic below is evaluated on fresh samples drawn from $|\Psi_\theta|^2$, with the estimators and stability protocol of §5.2.

E.1 Correlator geometry: a low-rank representation that reorganises

We characterise the correlator feature space $Z = \overline{\phi}\|\overline{\psi}\|\mathbf{g}$ by its entropy-based effective rank $r_{\text{eff}}(Z)$, the correlation $|\rho(\text{head}, \text{PC1})|$ between the head output and its first principal component, and the power of PC1 decomposed across architectural branches. Table E.1 reports these quantities across the grid.

The manifold is low-dimensional everywhere. The effective rank stays below 3.1 across the entire grid, and the head output is almost perfectly reconstructed from PC1 alone in all but one analysed cell ($|\rho| \geq 0.94$ except N=20, ω=0.01, where $|\rho| = 0.79$). This holds even at strong correlation where the real-space density develops pronounced shell structure—the correlator does not respond by expanding its latent dimensionality. The modest peak at N=12, ω=0.1 ($r_{\text{eff}} = 3.08$) coincides with the intermediate-correlation regime where neither the shell picture nor the crystalline picture provides a clean single-mode description.

A $\phi \leftrightarrow \psi$ transition in the leading axis. What varies across regimes is *which branch dominates PC1*. At moderate-to-weak correlation ($\omega \geq 0.1$, $N \geq 6$), PC1 is predominantly ψ-loaded: the pair branch carries 60–73% of the variance, meaning the correlator's primary degree of freedom tracks pair correlations. At strong correlation the loading shifts toward ϕ (orbital) or $\mathbf{g}$ (auxiliary) branches for N=6 and N=12. The N=20 run does not follow the same branch-switching pattern: its pair branch remains dominant at both weakest confinements. Given that these are single-run, component-level diagnostics,

Table E.1: Correlator geometry (CTNN). r_{eff}: entropy-based effective rank of Z. $|\rho|$: absolute correlation between head output and PC1 reconstruction. ϕ/ψ/g: fraction of PC1 variance from the orbital, pair, and auxiliary branches. The ψ-dominated entries ($\psi > 0.5$) are shown in bold.

N	ω	r_{eff}	$\lvert\rho\rvert$	ϕ	ψ	g
2	0.001	1.00	1.00	0.00	**1.00**	0.00
2	0.01	1.04	0.66	0.00	**1.00**	0.00
2	0.1	1.31	0.99	0.00	**0.53**	0.46
2	0.5	1.25	1.00	0.00	**0.58**	0.42
2	1.0	1.53	0.98	0.00	0.35	**0.65**
6	0.001	1.52	0.97	0.01	0.12	**0.87**
6	0.01	1.38	0.99	0.05	0.44	**0.50**
6	0.1	2.73	0.94	**0.66**	0.21	0.14
6	0.5	2.69	0.98	0.15	**0.63**	0.22
6	1.0	2.69	0.99	0.03	**0.62**	0.35
12	0.001	1.66	0.96	**0.54**	0.13	0.33
12	0.01	1.54	0.97	0.43	0.09	**0.48**
12	0.1	3.08	0.99	0.12	**0.63**	0.26
12	0.5	2.32	0.99	0.12	**0.69**	0.19
12	1.0	1.76	0.99	0.08	**0.73**	0.19
20	0.001	2.59	0.99	0.18	**0.74**	0.08
20	0.01	1.09	0.79	0.04	**0.94**	0.01
20	0.1	2.20	0.97	0.36	0.44	0.20
20	0.5	2.30	1.00	0.17	**0.70**	0.13
20	1.0	2.32	0.99	0.18	**0.71**	0.11

we do not interpret this discrepancy as a size-dependent physical transition. The transition is sharpest at N=12, where ψ goes from 0.09 at ω=0.01 to 0.63 at ω=0.1.

These loadings track the physics. When interactions are weak relative to the trap ($\omega \gtrsim 0.1$), the non-interacting shell structure provides a good zeroth-order description, and the correlator's primary role is refining pair correlations on top of that. When interactions dominate ($\omega \lesssim 0.01$), the leading correlator coordinate shifts away from pair features toward the single-particle and auxiliary branches. The network reorganises the features through which it represents the state, and that reorganisation—not any growth in dimension—is how it adapts. Nothing in the architecture imposes this behaviour: the N=6 and N=12 regime changes appear as a rotation of a low-dimensional axis across branches rather than as growth in latent dimensionality.

E.2 Backflow: a high-rank field with a low-rank footprint

Table E.2 characterises the message-passing displacement field. We report the normalised displacement magnitude $\|\Delta\mathbf{x}\|/\ell$, the effective rank of $\Delta\mathbf{x}$ (viewed as a $B \times 2N$ matrix of flattened displacements), and the CKA similarity between the correlator features with and without backflow.

Displacement magnitude scales with ω but saturates near $\|\Delta\mathbf{x}\|/\ell \sim 0.1$. At strong correlation ($\omega \leq 0.01$), displacements are small relative to the harmonic length, $\|\Delta\mathbf{x}\|/\ell \sim 0.02$–$0.05$. As the trap tightens and the system enters the moderate-to-weak correlation regime, the magnitude grows and saturates at $\sim$8–10% of ℓ for $\omega \gtrsim 0.1$. The message-passing backflow never produces large displacements—its tanh output bound and copresheaf message-passing structure act as implicit regularisers, in contrast to the conventional backflow, which can exhibit displacements comparable to the system scale in poorly converged regimes (e.g. $\|\Delta\mathbf{x}\|/\ell \sim 0.95$ for N=6, ω=0.001 with the conventional network).

Full-rank displacements, near-unity CKA. The effective rank of the displacement field is consistently $\approx 2N - 2$, saturating the available spatial degrees of freedom modulo centre-of-mass conservation (confirmed by COM shifts of order 10^{-14}–10^{-17}). The backflow transformation is not a low-rank collective mode such as a breathing or rigid rotation; the displacement field spans essentially all COM-free

Table E.2: CTNN backflow geometry. $\|\Delta\mathbf{x}\|/\ell$: mean displacement magnitude normalised by harmonic length. Rank: effective rank of the displacement covariance. $2N-2$: maximum possible rank after centre-of-mass projection. CKA: linear centred kernel alignment between correlator features with and without backflow (1.0 = identical feature manifolds).

N	ω	$\|\Delta\mathbf{x}\|/\ell$	Rank	$2N-2$	CKA
2	0.001	0.003	1.9	2	1.000
2	1.0	0.029	2.0	2	1.000
6	0.001	0.017	10.0	10	0.999
6	0.01	0.035	9.9	10	0.998
6	0.1	0.096	10.0	10	0.995
6	0.5	0.093	10.0	10	0.991
6	1.0	0.089	10.0	10	0.979
12	0.001	0.017	21.9	22	1.000
12	0.01	0.030	21.9	22	0.998
12	0.1	0.096	21.9	22	0.993
12	0.5	0.096	21.9	22	0.993
12	1.0	0.091	21.9	22	0.995
20	0.001	0.023	37.7	38	0.999
20	0.01	0.046	37.4	38	1.000
20	0.1	0.075	37.8	38	0.997
20	0.5	0.099	37.6	38	0.990
20	1.0	0.099	37.6	38	0.991

spatial directions, so each particle receives a correction conditioned on the full configuration. The field stays near that ceiling across the whole grid, including the deepest Wigner points, where one might have expected the correction to become collective. Whether that is a property of *this* architecture or of trained backflows generally is not something the present measurement can decide; Chapter 7 settles it by holding everything else fixed (§7.1.2).

Yet the CKA between correlator features with and without backflow exceeds 0.99 in nearly all cases (and exceeds 0.979 everywhere). The correlator manifold is almost unperturbed by the displacement field. The correlator's feature geometry is therefore almost unaffected by the presence of the displacement field, which is a statement about that geometry and not about the subspace the backflow occupies. Whether the two components are genuinely complementary, rather than merely uncorrelated on this measure, is tested by ablation in §7.1.2.

E.3 Correlator–backflow coupling: from independence to cooperation

The previous subsection established that backflow leaves the correlator manifold intact. But are the correlator and the backflow *functionally* coupled—does the displacement depend on what the correlator computes? We measure this through three diagnostics: (i) the mean cosine similarity $\langle\cos(\nabla_{\mathbf{x}} f,\, \Delta\mathbf{x})\rangle$ between the correlator's gradient in configuration space and the displacement field (alignment implies the backflow pushes along directions the correlator is sensitive to); (ii) the fraction of configurations where $\Delta\mathbf{x}$ is aligned with ∇f; and (iii) $|\delta f|/\sigma_f$, where $\delta f = f(\tilde{R} + \Delta\mathbf{x}) - f(\tilde{R})$ is the change in the correlator output when it is evaluated on the backflow-shifted coordinates, in units of the correlator's own standard deviation σ_f. Two cautions attach to this last quantity. It is a *counterfactual sensitivity*: in the production ansatz the correlator sees the unshifted $\tilde{R}$ (only the determinant sees $\tilde{R} + \Delta\mathbf{x}$; §3.5), so δf measures how much the correlator *would* respond to the displacement, not an actual term in the wavefunction. And it is a typical relative change, not a variance decomposition: $|\delta f|/\sigma_f = 0.7$ means the counterfactual change is 0.7 standard deviations, not that 70% of the variance is attributable to backflow. Table E.3 reports these alongside the fraction of particles pushed toward their nearest neighbour.

Table E.3: Correlator–backflow coupling diagnostics (message-passing ansatz). $\langle\cos\rangle$: mean cosine between ∇f and $\Delta\mathbf{x}$. Aligned: fraction of configs with $\nabla f \cdot \Delta\mathbf{x} > 0$. $|\delta f|/\sigma_f$: backflow-induced change in the correlator output, relative to its standard deviation. →NN: fraction of particles displaced toward their nearest neighbour (0.5 = isotropic). Only $N \geq 6$ shown; N=2 displacements are near-zero for most ω.

N	ω	$\langle\cos\rangle$	Aligned	$\lvert\delta f\rvert/\sigma_f$	→NN
6	0.001	0.14	0.63	0.03	0.43
6	0.01	0.18	0.77	0.05	0.28
6	0.1	0.49	0.99	0.25	0.16
6	0.5	0.62	1.00	0.33	0.19
6	1.0	0.77	1.00	0.41	0.18
12	0.001	−0.48	0.01	0.03	0.61
12	0.01	0.22	0.82	0.05	0.29
12	0.1	0.56	1.00	0.39	0.12
12	0.5	0.65	1.00	0.49	0.20
12	1.0	0.75	1.00	0.55	0.21
20	0.001	0.32	0.97	0.06	0.44
20	0.01	0.22	0.89	0.07	0.45
20	0.1	0.39	1.00	0.25	0.17
20	0.5	0.65	1.00	0.58	0.22
20	1.0	0.78	1.00	0.69	0.25

A transition at $\omega \approx 0.1$. The data reveal a sharp regime change. At $\omega \leq 0.01$ the coupling is weak and inconsistent in sign: $\langle\cos\rangle$ runs from -0.48 to 0.32 across the six cells, the alignment fraction is erratic, and backflow barely affects the correlator output ($|\delta f|/\sigma_f < 0.07$ everywhere). The N=12, ω=0.001 cell is strongly *anti*-aligned ($\langle\cos\rangle = -0.48$, alignment fraction 0.01), which we record without an account of it: at that confinement the displacement field is small enough (Table E.2) that its direction relative to the correlator gradient carries little energy either way, but we cannot say why the sign is systematic rather than scattered. In this regime the backflow operates in a subspace largely orthogonal to the correlator's gradient—it modifies the wavefunction in directions that W_θ is insensitive to, consistent with its role being to adjust the Slater determinant's orbital positions rather than the correlator.

At $\omega \geq 0.1$, the picture changes qualitatively. The gradient cosine jumps to 0.5–0.8, the alignment fraction reaches 1.0 for $N \geq 6$, and the counterfactual output change reaches $|\delta f|/\sigma_f = 0.25$–$0.69$ (i.e. up to $\sim$0.7 standard deviations of the correlator). Here the backflow acts as a cooperative refinement: it pushes configurations along directions the correlator is maximally sensitive to, systematically correcting the correlator's residual errors. The transition is sharpest between $\omega = 0.01$ and $\omega = 0.1$, coinciding with the crossover in $\Gamma = V_{\text{int}}/T$ from $\Gamma > 10$ (interaction-dominated) to $\Gamma \sim 3$–6 (moderate correlation).

Directional structure: correlation-hole deepening. The nearest-neighbour direction fraction provides a complementary signature. At $\omega \geq 0.1$ for $N \geq 6$, only 12–25% of particles are displaced toward their nearest neighbour. The backflow preferentially pushes particles *apart*, actively deepening the correlation hole. This directional bias is strongest at N=12, ω=0.1 (→NN = 0.12), exactly where the correlator effective rank peaks and the $\phi \leftrightarrow \psi$ transition occurs. At strong correlation ($\omega \leq 0.01$, large N), the fraction rises toward 0.5 (isotropic), because the large equilibrium inter-particle separations leave little room for further correlation-hole deepening.

E.4 Limits of component-level interpretation

Taken together these measurements describe two components of very different character. The correlator occupies one to three directions, whose leading axis tracks the dominant global coordinate of the cloud and changes identity rather than growing in number as the physics changes. The displacement field is high-rank and configuration-dependent, and its presence leaves the correlator's feature geometry almost unchanged (CKA > 0.98). The coupling between them is weak and inconsistent in sign at strong correlation, and strong and systematically aligned at moderate coupling, where the displacements follow the

correlator's gradient; the change is sharp and coincides with the crossover in Γ.

These measurements describe one trained solution and do not by themselves establish which properties are architecture-specific. A different training route could divide the same labour differently and arrive at the same energy, which for two partial substitutes is what one should expect. Deciding the question requires holding everything else fixed and varying only the architecture.

Chapter F

Structural diagnostics in full

Chapter 6 reports the two radial quantities the argument turns on—r_{mode} and the width-to-scale ratio γ. The complete diagnostics are collected here: the mean radius, width, FWHM and decile quantiles for N=2, and the ring-angle spread and pair Lindemann ratio for N=6. Estimators and stability protocol are defined in §5.1. All lengths are in Bohr; angles in radians.

F.1 Fitted energy and radius exponents

Section 6.2.3 states two conclusions from the energy partition—that interactions come to dominate, and that the trap–interaction balance approaches its classical form—and defers the fitted exponents that support them to here.

Log–log fits $E(\omega) \propto \omega^{\alpha}$ over the grid $\omega \in \{10^{-3}, 10^{-2}, 10^{-1}, 0.5, 1\}$ give a kinetic exponent $\alpha_T \simeq 1.00$ at every N, an interaction exponent $\alpha_{V_{\mathrm{int}}} \simeq 0.66$, and trap exponents $\alpha_{V_{\mathrm{trap}}} \simeq 0.81, 0.74, 0.72$ for $N = 2, 6, 12$. The kinetic energy therefore scales essentially linearly with ω while both potential terms grow sublinearly with nearly N-independent exponents; the ratio $\Gamma \propto \omega^{\alpha_{V_{\mathrm{int}}} - \alpha_T} \approx \omega^{-0.34}$ predicts a factor $10^{3\times 0.34} \approx 10.5$ across three decades, against the measured 9.3, 10.2 and 11.4.

From $V_{\mathrm{trap}} = \frac{1}{2}\omega^2 \sum_i \langle r_i^2 \rangle$ the rms radius is $r_{\mathrm{rms}} = \sqrt{2V_{\mathrm{trap}}/(N\omega^2)}$, and fitting $r_{\mathrm{rms}} \propto \omega^{\alpha_r}$ gives $\alpha_r \simeq -0.60$ to -0.64, equivalently $\langle r^2 \rangle \propto \omega^{-\beta}$ with $\beta \simeq 1.19$–1.28. The cloud therefore expands faster than the non-interacting ω^{-1} law for $\langle r^2 \rangle$.

These are averaged quantities and should not be confused with the shell-resolved radius of §F.4, which is the one compared against the classical Wigner prefactor; the two differ because r_{rms} and r_{mode} mix the central electron of a $(1, n)$ topology with the ring, and the shell-resolved radius does not.

Table F.1: Radial diagnostics for N=2 across ω (Bohr units).

ω	r_{mode}	$\langle r \rangle$	σ_r	FWHM	q_{10}	q_{50}	q_{90}	γ
1.00	1.435	1.632	0.705	1.743	0.735	1.581	2.581	0.491
0.50	2.303	2.475	1.007	2.466	1.209	2.409	3.829	0.437
0.10	3.601	3.830	1.728	4.449	1.640	3.714	6.127	0.480
0.01	14.191	14.593	5.615	13.602	7.390	14.322	22.038	0.396
0.001	64.292	63.953	18.479	43.977	40.392	63.814	87.714	0.287

Table F.2: Radial and angular diagnostics for N=6 across ω (Bohr; angles in radians).

ω	r_{mode}	σ_r	γ	σ(ring angles)	$\gamma_{r_{ij}}$
1.00	2.0907	0.9249	0.4424	0.7046	0.3538
0.50	3.0702	1.3648	0.4445	0.6943	0.3397
0.10	8.1686	3.5025	0.4288	0.6842	0.3149
0.01	33.1538	13.9094	0.4195	0.6497	0.2889
0.001	113.5542	47.8076	0.4210	0.6368	0.2664

F.2 Shell-resolved structure across the whole grid

Chapter 6 quotes shell occupancies, radii, bond order and Lindemann numbers for individual (N, ω) cells in running prose. Table F.3 collects them for the whole grid from a single analysis pass, so that the trends can be read off instead of assembled from the text. Shell radii are given in trap units $\tilde{r} = \sqrt{\omega}\, r$; multiply by $\ell = \omega^{-1/2}$ for Bohr.

Reading $|\Phi_n|$ against its null. One column of that table needs a baseline to be interpretable, and its absence would make the bond-order numbers read far more favourably than they should. For n particles placed at *independent uniform* angles on a ring, $n\Phi_n = \sum_k e^{in\phi_k}$ is a two-dimensional random walk of n unit steps, so

$$\big\langle |\Phi_n| \big\rangle_{\text{random}} = \frac{1}{n}\int_0^\infty \frac{1 - J_0(t)^n}{t^2}\, dt \;\xrightarrow{\;n\to\infty\;}\; \frac{\sqrt{\pi}}{2\sqrt{n}} \approx \frac{0.886}{\sqrt{n}}, \qquad \big\langle |\Phi_n|^2 \big\rangle_{\text{random}} = \frac{1}{n}. \tag{F.1}$$

The exact values at the occupancies that occur are 0.525, 0.402, 0.366, 0.338, 0.298, 0.269, 0.257 and 0.204 for $n = 3, 5, 6, 7, 9, 11, 12, 19$; the Rayleigh limit undershoots them by 0.4–2.6%, which is why it is not used here. A finite $|\Phi_n|$ is therefore *not* by itself evidence of angular order: a five-particle ring with no order at all returns $|\Phi_5| \approx 0.40$. The ratio $|\Phi_n| / \langle |\Phi_n| \rangle_{\text{random}}$ is the quantity that carries information, and Table F.3 reports it.

Read that way the result is sharper than the raw values suggest, and it needs stating with the right threshold. From $\omega = 1$ down to $\omega = 0.1$ the ratio is 0.99–1.01 at every particle number and on every ring. At $\omega = 0.01$ it lifts, to 1.08 at N=6 and 1.02 at N=12, while N=20 stays at 1.00. Only at $\omega = 10^{-3}$ does it become large: 1.63 on the N=6 ring, 1.16 and 1.55 on the two rings of the N=12 $(3, 9)$ molecule, and 1.18 and 1.46 on the outer two rings of the N=20 $(1, 7, 12)$ molecule.

Whether the small departures are *resolved* is a separate question from whether they are *large*, and here both can be answered. Because the shells are assigned by radial rank alone and no angular cut is applied, Eq. (F.1) is also the null of this particular pipeline; generating it numerically instead—keeping every frame's radii and redrawing all angles uniformly, then running the identical shell assignment—reproduces it within its error in every cell. Block-bootstrap errors on the ratio are 0.002–0.007. On that scale the N=6 ring at strong confinement sits resolvably *below* the null, at 0.989 ± 0.002 for $\omega = 1$ and 0.990 ± 0.002 for $\omega = 0.5$. That is physics rather than artefact. Since $\langle |\Phi_n|^2 \rangle = n^{-1} + n^{-2} \sum_{j \neq k} \langle \cos n(\theta_j - \theta_k) \rangle$, a depletion of close angular pairs—a correlation hole along the ring—makes the cross terms negative: what the ring shows at strong confinement is short-range angular repulsion without n-fold order. At N=12 and N=20 the ratio stays within 0.5% of the null through $\omega = 0.1$.

The conclusion that survives is the one that matters: statistical detectability is not what separates the regimes here, magnitude is. On that criterion n-fold order is absent through $\omega = 0.1$, where the resolved departures are all within about one percent; weakly present at $\omega = 0.01$ and clearly so only at N=6; and unambiguous at 10^{-3}, where every ring exceeds the null by 16–63%.

The Lindemann numbers of the same table fall monotonically throughout, so radial stiffening precedes n-fold angular ordering rather than accompanying it—the separation between radial and angular melting anticipated in §1.4. Note that the values here are for the *modal* topology over all frames; the sector-conditioned values quoted in §6.2.5 are computed on the frames assigned to a given sector and run slightly higher.

Table F.3: Shell-resolved diagnostics at the modal topology, all frames, for every (N, ω). Occupancies are $(n_{\mathrm{in}}, \ldots, n_{\mathrm{out}})$; $\tilde{r}_s$ are mean shell radii in trap units; $w_s = \sigma(r_s)/\langle r_s \rangle$ are per-shell radial relative widths; $|\Phi_{n_s}|$ is the rotation-registered bond order on shell s; and the final column is $|\Phi_{n_s}|$ divided by its exact random-angle expectation, Eq. (F.1). A value of 1.00 in that column means the ring is equal to the independent-angle null within quoted precision; see the text for why departures of about a percent are at the resolution of this comparison. Single-particle inner shells are omitted from the angular columns, where $|\Phi_1| = 1$ trivially.

N	ω	occ.	$\tilde{r}_s$	w_s	$\lvert\Phi_{n_s}\rvert$	ratio to null
6	1.0	$(1, 5)$	0.70, 1.66	0.43, 0.32	0.397	0.99
6	0.5	$(1, 5)$	0.76, 1.78	0.42, 0.31	0.397	0.99
6	0.1	$(1, 5)$	0.93, 2.15	0.41, 0.28	0.405	1.01
6	0.01	$(1, 5)$	1.33, 2.96	0.42, 0.22	0.434	1.08
6	0.001	$(1, 5)$	1.12, 4.29	0.63, 0.15	**0.654**	**1.63**
12	1.0	$(1, 11)$	0.55, 2.00	0.47, 0.34	0.268	1.00
12	0.5	$(1, 11)$	0.60, 2.18	0.47, 0.33	0.268	1.00
12	0.1	$(1, 11)$	0.74, 2.68	0.45, 0.31	0.270	1.00
12	0.01	$(1, 11)$	1.04, 3.77	0.41, 0.28	0.275	1.02
12	0.001	$(3, 9)$	2.52, 6.04	0.61, 0.24	**0.611**, **0.462**	**1.16**, **1.55**
20	1.0	$(1, 19)$	0.55, 2.32	0.44, 0.36	0.204	1.00
20	0.5	$(1, 19)$	0.60, 2.52	0.44, 0.35	0.203	1.00
20	0.1	$(1, 19)$	0.75, 3.14	0.42, 0.34	0.204	1.00
20	0.01	$(1, 19)$	1.05, 4.47	0.49, 0.32	0.205	1.00
20	0.001	$(1, 7, 12)$	1.27, 4.19, 7.83	0.47, 0.24, 0.11	**0.398**, **0.375**	**1.18**, **1.46**

F.3 Shell-model fit to the pair distribution

The reconstruction quoted in §6.2.5 tests whether the measured pair distribution is explained by the detected shell geometry. Two versions of that test exist and they are not equally informative.

The weaker version decomposes the pairs of two-shell frames into inner–inner, inner–outer and outer–outer histograms and recombines them with the combinatorial weights of the ideal topology. Because the sector histograms and the total are accumulated from the same frames, this recombination is close to exact by construction—cosine similarities exceed 0.99999 at every (N, ω)—and it establishes only that the sector bookkeeping is self-consistent.

The stronger version is a *generative* one and is the test reported here. Shell radii and widths are fitted on one half of the frames; each shell is then modelled as a regular n_s-gon carrying a per-frame global rotation and a fitted angular jitter; and the resulting pair-distance density is compared with the empirical density on the *held-out* half. Against it we run an angle-randomised null with the same fitted radial distributions but uniform angles, which isolates what polygonal order contributes over and above shell radii alone. Table F.4 reports both.

The polygon model reproduces the held-out pair distribution to cosine similarity 0.987–0.999 across the whole grid. The null reaches 0.936–0.996, and the gap between them, $\Delta\cos$, grows monotonically as the trap weakens at every particle number—from 0.012 to 0.058 at N=6, from 0.006 to 0.026 at N=12, and from 0.003 to 0.010 at N=20. Shell radii alone account for most of the pair structure at strong confinement; angular order accounts for a growing remainder as the Wigner molecule forms. This is the same transition the bond-order ratios of Table F.3 record, measured on a different quantity and on held-out data.

How firmly $\Delta\cos$ is determined. The analysis code, restored from the notebook that produced this table, reproduces every published $\Delta\cos$ to six decimals, and two further tests were run on it. The first is a block bootstrap: contiguous blocks of both the training and the held-out frames are resampled, the shell model is refitted and a fresh model sample drawn in every replicate, so the interval covers sampling noise in the data, uncertainty in the fitted parameters and Monte Carlo noise in the model. Over 200 replicates the 95% intervals are 0.0002–0.0036 wide and exclude zero everywhere, including N=20 at $\omega = 1$, $[0.0027, 0.0030]$. The growth with $1/\omega$ is resolved: at N=6 all five intervals are disjoint, and at N=12 and N=20 only the two strongest confinements touch.

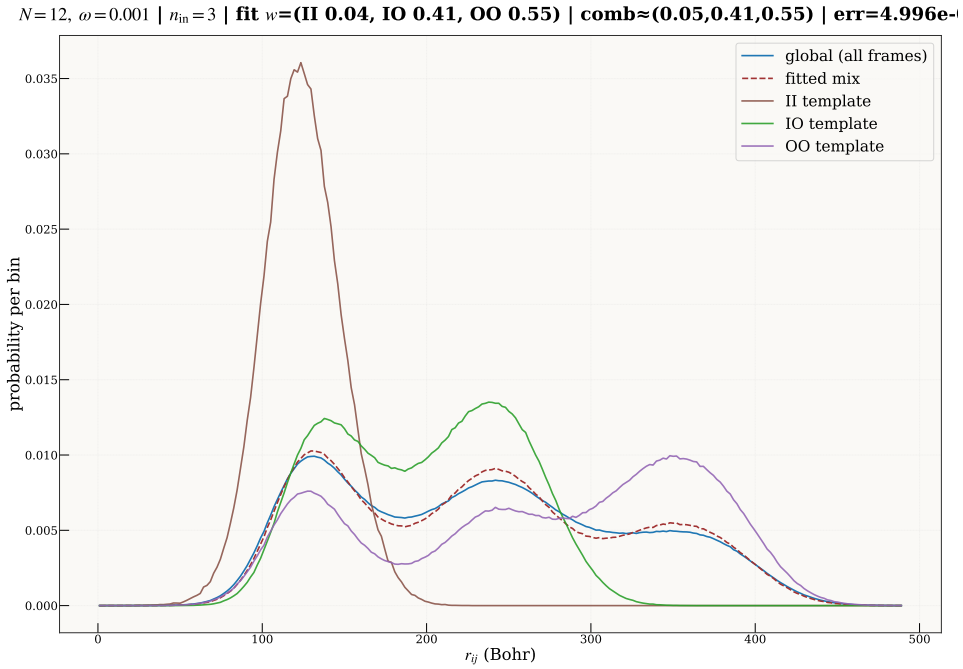


Figure F.1: Shell-resolved reconstruction of $g(r)$ at $\omega = 10^{-3}$ for N=12. The global pair distribution is reproduced by a convex mixture of inner–inner (II), inner–outer (IO) and outer–outer (OO) histograms with weights set by the observed $(n_{\mathrm{in}}, n_{\mathrm{out}})$ statistics.

The second test varies the analysis choices. Halving or doubling the number of histogram bins leaves $\Delta\cos$ unchanged at the fourth decimal, and swapping or interleaving the training and held-out halves moves it by at most 0.0003. The shell-detection threshold matters only where it changes the partition: lowered far enough to split the single N=20 ring into a spurious $(1, 3, 16)$ at $\omega = 0.5$ and 0.1, or to give $(1, 3, 8)$ at N=12, $\omega = 0.01$, it lowers $\Delta\cos$ by 10–16%, and in that variant the N=20 value at $\omega = 0.5$ drops just below the one at $\omega = 1$. The largest dependence is on the angular-jitter model: scaling the fitted jitter by 0.75 or 1.25 moves $\Delta\cos$ by up to +34% and −22%, but in both variants it stays positive and grows monotonically as the trap weakens, at every particle number. The size of $\Delta\cos$ is therefore model-dependent; its sign and its trend are not. The code, the replicates and every variant are in `scripts/wigner_null_bootstrap.py` and `results/analysis/2026-09-12_wigner_null_bootstrap/`.

Table F.4: Shell-model fit to the pair-distance distribution. The empirical $p_{\text{test}}(r_{ij})$ on held-out frames is compared with (i) a *polygon* model, in which each detected shell is a regular n_s-gon with a per-frame global rotation and fitted angular jitter, and (ii) a *uniform-angle* null with the same fitted radial distributions and no polygonal order. $\Delta\cos = \cos_{\text{poly}} - \cos_{\text{uni}}$ is what angular order buys; L^2_{poly} is the polygon model's residual.

N	ω	occ.	$\cos_{\text{poly}}$	$\cos_{\text{uni}}$	$\Delta\cos$	L^2_{poly}
6	1.0	$(1,5)$	0.9988	0.9866	0.0122	4.69×10^{-3}
6	0.5	$(1,5)$	0.9986	0.9836	0.0150	5.09×10^{-3}
6	0.1	$(1,5)$	0.9974	0.9741	0.0234	6.83×10^{-3}
6	0.01	$(1,5)$	0.9922	0.9536	0.0386	1.14×10^{-2}
6	0.001	$(1,5)$	0.9942	0.9358	**0.0585**	1.04×10^{-2}
12	1.0	$(1,11)$	0.9993	0.9937	0.0056	3.36×10^{-3}
12	0.5	$(1,11)$	0.9990	0.9927	0.0063	4.01×10^{-3}
12	0.1	$(1,11)$	0.9980	0.9896	0.0085	5.43×10^{-3}
12	0.01	$(1,11)$	0.9929	0.9805	0.0124	9.92×10^{-3}
12	0.001	$(3,9)$	0.9868	0.9604	**0.0264**	1.34×10^{-2}
20	1.0	$(1,19)$	0.9992	0.9964	0.0028	3.43×10^{-3}
20	0.5	$(1,19)$	0.9990	0.9958	0.0032	3.78×10^{-3}
20	0.1	$(1,19)$	0.9982	0.9942	0.0040	4.95×10^{-3}
20	0.01	$(1,19)$	0.9947	0.9900	0.0047	8.21×10^{-3}
20	0.001	$(1,7,12)$	0.9915	0.9820	**0.0095**	1.02×10^{-2}

F.4 The measured shell radius against the classical prediction

The classical minimum-energy configuration of six charges in a parabolic trap places five of them on a ring of radius $1.334\,\omega^{-2/3}$ [2, 3]. Nothing in the ansatz, the training objective or the sampler knows that number, so the fitted outer-shell radius of Table F.3 is an external check on the whole structural picture—and a sharper one than the exponent fits of §6.2.4, because it tests a prefactor as well as a power.

Table F.5: Fitted outer-shell radius of the N=6 $(1,5)$ molecule against the classical Wigner ring $r_{\text{cl}} = 1.334\,\omega^{-2/3}$. Radii in Bohr.

ω	r_{shell}	r_{cl}	ratio
1.0	1.66	1.33	1.247
0.5	2.52	2.12	1.188
0.1	6.80	6.19	1.098
0.01	29.65	28.74	1.032
0.001	135.67	133.40	1.017

The ratio falls monotonically toward unity, reaching 1.017 at $\omega = 10^{-3}$: the quantum ring sits 1.7% outside the classical one at the weakest trap, and the excess grows smoothly with confinement as zero-point motion pushes the shell outward. Fitting $r_{\text{shell}} \propto \omega^{\alpha}$ gives $\alpha = -0.637$ over the full grid and -0.650 over the weakest three confinements, against the classical $-2/3$.

Two remarks on why this quantity, rather than the r_{mode} of Table 6.3, is the right one to compare. First, r_{mode} is the mode of the *single-particle* radial density, which for a $(1,5)$ topology mixes the central electron with the ring and is therefore not the ring radius; fitting it gives $\alpha \approx -0.58$ at N=6 and -0.54 at N=2 over the full grid, and approaches $-2/3$ only on the weak-trap half. Second, the shell-resolved radius is the quantity the physics-informed proposal of §7.4 actually needs, so the agreement recorded here is what licenses building a sampler out of it.

Chapter G

Code, data and reproducibility

Code. All code used in this thesis—the ansatz, both training routes, the structural and tangent-space diagnostics, the evaluation scripts and the figure scripts—is public at `https://github.com/Im2ql4u/Thesis`, together with the run configurations, the metadata of every saved model and the derived tables from which the numbers in Chapters 6 and 7 are quoted. The text alone, with the figures it includes, is at `https://github.com/Im2ql4u/Thesis_text`.

Models and raw data. Trained weights and the $|\Psi|^2$ sample bundles from which every structural diagnostic is computed are too large for the repository. They are kept on the University of Oslo machine-learning nodes where the calculations ran and are available on request.

Seeds and initialisation. Every run records its seed in its metadata, and seeds are stated in the text wherever a claim rests on seed variation. The collocation runs are warm-started, and two facts limit how far they can be repeated from the current code. The canonical pretrained initialisation behind the continuation chains was lost in a repository rewrite in June 2026, so those runs can be re-evaluated from their saved checkpoints but not retrained bit for bit. And the Jastrow cusp convention changed in May 2026, from a fixed decay length to the oscillator length; checkpoints trained before the change must be evaluated with the convention they were trained with, which the evaluation script infers from the checkpoint metadata.

The importance-sampling error and what was checked after it. In March 2026 the collocation sampler was found to return, for each proposal draw, the log-density of the Gaussian component that generated it rather than the log-density of the whole mixture. The resulting error in the log-weights grows with the dimension Nd and with the spread of the component widths: negligible at N=2, $\omega = 1$, it reached a factor of order e^{12} in the importance weights at N=6, $\omega = 0.01$, which is why its signature—loss of accuracy at larger N and weaker confinement—was indistinguishable from an architectural limit. The error sat in the importance weights and therefore only in the collocation route; the production energies of Chapter 6 are evaluated by Metropolis sampling of $|\Psi|^2$, which does not use those weights. After the correction (26 March 2026) three things were checked. The collocation results reported here come from runs made after it, or, for N=12, from continuation runs whose final training stage used the corrected weights. Every N=6 and N=12 collocation energy in Tables 7.5 and 7.7 was independently re-evaluated for this thesis by Metropolis sampling, with four separately seeded chains per checkpoint and a blocking analysis (`scripts/eval_collocation_blocked.py`). And the structural analysis, whose original code had been lost with a deleted notebook, was recovered and restored (`src/wigner_structure.py`); it reproduces every published $\Delta\cos$ to six decimals.

Known gaps. Two quantities cannot be regenerated from the saved outputs: the cross-shell registration statistic $\tilde{C}_{\mathrm{global}}$ at N=20 (§6.2.5), whose analysis code was not retained, and the N=20, $\omega = 0.01$ entry of the backflow rank table (Table 7.3), whose per-run artefacts were not kept. One inconsistency also remains: the single-particle radial diagnostics of Table 6.3 and Appendix F were computed in an earlier analysis pass, on runs that at the weakest traps are not the same trained states as those behind the shell-resolved diagnostics and the radial-density figures.

Chapter H

Declaration on the Use of Artificial Intelligence

The collocation approach in this thesis was developed over several months of failed experiments. Its central difficulty was not merely that a network did not converge, but that the parts of the Schrödinger residual that made the method physically meaningful also made its gradients unstable near the wavefunction's nodes. During that period, AI systems were used as sounding boards: to test interpretations of unexpected results, to formulate possible explanations, and to decide which experiment should come next. The exchange below is a short sample from one longer conversation at a point where the negative evidence had begun to make abandoning collocation seem plausible.

Author

I am trying to develop a PINN to find ground-state solutions of the Schrödinger equation. I have tried multiple approaches, but I am not succeeding. ... What do you think about this? What should we do from here?

AI response 1

... Anything that injects near-node points makes gradients explode or become uselessly noisy. ...

⋮

AI response 2

... This asymmetry is fundamental and cannot be resolved by sampling tricks, loss modifications, or gradient preconditioning within the collocation framework. ...

⋮

AI response 3

... Stick with SR + energy (or energy+variance) VMC. ...

Artificial-intelligence systems were therefore used for discussion, planning, and review, rather than as a source of results. They helped turn observations into hypotheses that could be checked, suggested alternative diagnostics when a run failed, and reviewed the final prose for clarity and mistakes. The code, the experiments, the scientific judgments, and the text of this thesis were written by the author. In the final revision an AI coding assistant also helped write and run analysis scripts—the independent re-evaluation of the collocation energies, and the null model and bootstrap of the structural diagnostics—whose code and outputs are in the repository. All derivations and every numerical claim in this thesis have been checked by the author; where an AI system suggested an argument or pointed to a possible error, the argument was re-derived and the number recomputed before it was used. In particular, the recommendation shown above was not accepted as a methodological conclusion: it became one claim among several to investigate. The work that followed led to the collocation analysis reported in this thesis.

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